

Default with Policy-Randomness Overestimation

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Abstract

Why do some sovereigns pay high spreads despite moderate debt? I study lenders who overweight the dispersion of the government's policy shocks (policy-randomness overestimation, PRO), a *second-moment* belief distortion. Unlike every first-moment friction (reputation, ambiguity/robust control, diagnostic expectations), which shifts the *level* of the spread–debt schedule, PRO *pivots* its shape: within a single bond-pricing operator it raises the cost of debt in the borrowing region and lowers it near the brink, crossing the rational schedule once at a state-contingent threshold. This pivot, with a single-crossing characterization, is the headline contribution; its sharp, falsifiable signature is a cross-sectional *sign flip*, tightening the friction widens spreads on low-debt instruments and compresses them on high-debt ones, which no level shift can produce. The sovereign deleverages yet defaults at higher debt, so spreads persist as debt falls, a stability illusion. I discipline the friction as an as-if robust-control object via a detection-error probability and prove the pivot invariant for every degree of the friction; at admissible θ the magnitude is small everywhere by design, so the falsifiable content is the *sign* of the cross-sectional response, not its size. A rational-inattention microfoundation endogenizes the friction as the lenders' perceived residual uncertainty, and the framework rationalizes Argentina's decoupling of falling debt from persistent spreads.

Keywords: Sovereign Default, Behavioral Macroeconomics, Policy-Randomness Overestimation (PRO), Sovereign Spreads, Information Frictions, Argentina, Debt Management.

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1 Introduction

Sovereign spreads in some emerging markets stay high even when debt is moderate and fundamentals improve. I trace this to a *second-moment* belief distortion, policy-randomness overestimation (PRO): lenders overweight the dispersion of the government’s policy shocks. This is the paper’s organizing distinction. Every first-moment friction studied in the sovereign-debt literature, reputation, ambiguity and robust control over fundamentals, diagnostic expectations, shifts the spread–debt schedule’s *level*, up or down, more or less uniformly. PRO instead distorts a second moment, and within a single pricing operator it *pivots* the shape of the schedule around a state-contingent threshold $B^*(y)$: it raises the price of debt in the borrowing region and lowers it near the brink. The headline contribution is this pivot, together with a single-crossing characterization (the perceived default margin m^* changes sign exactly once), which is what no level shift can replicate. A rational sovereign facing the pivot deleverages while defaulting at higher debt, so spreads persist as debt falls, a stability illusion: a pricing wedge that lifts spreads in the safe region offsets the composition shift toward lower-spread states, so the average spread need not fall as the sovereign deleverages.

I discipline the friction without reading it off the spread level. I treat the perceived-dispersion inflation θ as an as-if robust-control object in the tradition of [Pouzo and Presno \[2016\]](#) and [Roch and Roldán \[2023\]](#), and bound its magnitude with a detection-error probability rather than asserting it. The data admit only a modest inflation, so at credible θ the friction’s quantitative *magnitude* is small throughout: both in aggregate time-series moments (the mean spread and its volatility are nearly flat in θ) and per state (the cross-sectional spread response is sub-basis-point across the empirically relevant range). That smallness is the point, not a weakness: it follows by design from the detection-error bound, and the per-state premium is in any case offset in any time-series average by the endogenous deleveraging. PRO’s falsifiable content is therefore not a magnitude but a *qualitative, cross-sectional* signature, the *sign* of the schedule’s response that flips across $B^*(y)$, which no level friction can reproduce. The contribution is the mechanism: the pivot, the single-crossing characterization, the higher default threshold, and the deleveraging hold for every $\theta > 1$, independent of magnitude. The large θ used in the figures is an illustrative comparative-statics dial that makes the pivot visible, and the model’s sharp, falsifiable prediction is the non-monotone, sign-flipping cross-section that separates PRO from any level friction.

I then derive the friction from primitives. A rational-inattention microfoundation lets lenders fit a simplified mental model of the sovereign’s payoff and allocate scarce attention between fundamentals and a residual they identify with the unmodeled policy shock. Residual posterior uncertainty inflates the perceived dispersion, which is exactly the PRO friction evaluated at a state-dependent $\theta_{RI}(s) \geq 1$ inside the same operator. The microfoun-

dation maps Argentina’s misreporting episode into a rise in θ_{RI} . Three extensions clarify the levers: even optimal Ramsey transfers cannot undo the pricing wedge, negativity-biased learning makes PRO persistent, and transparency raises welfare.

Literature This paper contributes to the sovereign default literature by proposing a novel behavioral mechanism to explain persistent empirical puzzles: why do emerging economies often face high spreads and excess volatility that seem disconnected from their macroeconomic fundamentals [Tomz and Wright, 2013, Mitchener and Trebesch, 2023]?¹

The dominant paradigm treats default as a purely *strategic decision* where sovereigns rationally weigh repayment costs against temporary market exclusion [Eaton and Gersovitz, 1981, Aguiar and Gopinath, 2007, Arellano, 2008]. While this framework has been successfully extended to incorporate long-term debt and financial frictions [Hatchondo and Martinez, 2009, Chatterjee and Eyigungor, 2012, Mendoza and Yue, 2012],² it struggles to explain why spreads often remain elevated even when fundamentals improve.

A second approach emphasizes *reputational concerns*, where lenders learn about the government’s hidden type [Cole et al., 1995, Phelan, 2006]. The informational channel is broad: Fourakis [2024] shows that deficit and inflation surprises move perceived default probabilities and that reputation loss often occurs *before* a default. The Argentina misreporting episode, analyzed by Morelli and Moretti [2023], is the leading case, a single breach of trust read as a credible signal of a bad type that led to years of market exclusion. Reputation models also explain graduation to a high-trust state [Amador and Phelan, 2021] and why larger haircuts entail greater reputation loss [Amador and Phelan, 2023].³ The key contrast with my mechanism is one of *level versus shape*. Type learning shifts the spread–debt schedule’s *level*: a worse posterior over types lowers the whole price curve through the perceived default probability, but leaves its curvature in B' intact. Canonical reputation models predict that bad types default at *lower* debt thresholds, whereas PRO lenders perceive *higher* default thresholds, because overestimated policy randomness softens their certainty of doom at high debt. A posterior downgrade therefore moves spreads in one direction across instruments; it cannot reverse the sign of $\partial s / \partial \theta$ between low- and high-debt bonds, the cross-sectional signature that no level friction delivers. Reputation models also assume lenders eventually converge to the truth, leaving unexplained why some risk premia appear persistently excessive.

A third strand treats market *sentiment itself* as a fundamental force. Beginning with Calvo et al. [1996] and Cole and Kehoe [2000], this literature shows how pessimistic expect-

¹For comprehensive surveys of the sovereign debt literature, see [Meyer et al., 2022] and [Abbas et al., 2019].

²Empirical evidence on emerging market business cycles and debt restructurings is provided by [Neumeyer and Perri, 2005], [Cruces and Trebesch, 2013], and [Arellano and Ramanarayanan, 2012].

³Other contributions include Cole and Kehoe [1998], D’Erasmus [2011], Egorov and Fabinger [2016], and Dovis and Kirpalani [2020, 2021].

tations become self-fulfilling, with quantitative implementations by [Gennaioli et al. \[2014\]](#) and [Bocola and Dovis \[2019\]](#).⁴ I build on this insight but ask a different question: what if markets systematically *misperceive* the nature of sovereign decision-making?

Information choice and rational inattention. A complementary literature studies how agents allocate attention across signals [[Sims, 2003](#), [Maćkowiak and Wiederholt, 2009](#), [Matějka and McKay, 2015](#)]; investor information acquisition and its equilibrium implications appear in [Veldkamp \[2011\]](#) and [Van Nieuwerburgh and Veldkamp \[2009\]](#). I let lenders allocate attention to second-moment (dispersion) information alongside first-moment fundamentals. The resulting state-dependent tail weight $\theta_{\text{RI}}(y, B')$ stays embedded in the same pricing operator, so the baseline comparative statics carry through.

The PRO friction draws on documented behavioral biases [[Tversky and Kahneman, 1974](#), [Kahneman and Tversky, 1979](#), [De Long et al., 1990](#)] but targets a *second* moment, the perceived dispersion of policy. The behavioral evidence must be read with care. The best-documented second-moment bias is *overprecision*, in which agents hold subjective distributions that are too *narrow* [[Moore and Healy, 2008](#), [Ben-David et al., 2013](#), [Ortoleva and Snowberg, 2015](#), [Augenblick et al., 2025](#)]. This shows that second-moment misperception of comparable magnitude is real, but its sign (too narrow, and about one's *own* beliefs) is the opposite of what PRO requires, namely lenders perceiving *another* agent's policy dispersion as too *wide*. I therefore do not calibrate the PRO wedge from that literature. I treat θ as an *as-if* robust-control object [[Pouzo and Presno, 2016](#), [Roch and Roldán, 2023](#)], a perceived second moment inflated relative to the truth, and discipline its magnitude with a detection-error probability.⁵

My mechanism differs from related belief-distortion approaches. *Ambiguity aversion* based on robust control [[Gilboa and Schmeidler, 1989](#), [Hansen and Sargent, 2008](#)] makes lenders uncertain about the model of *macroeconomic fundamentals*, generating an ambiguity premium that lifts spreads roughly *uniformly* [[Pouzo and Presno, 2016](#), [Roch and Roldán, 2023](#)], a worst-case wedge that raises the schedule's *level* without pivoting its shape; related preference formulations tilt decision weights toward adverse scenarios [[Klibanoff et al., 2005](#), [Maccheroni et al., 2006](#)]. PRO instead targets the perceived volatility of the sovereign's *policy choices*, distorting a *second* moment inside the same pricing operator, so it *pivots* the schedule rather than translating it: spreads widen at low debt and compress near the brink, a sign reversal no uniform premium can produce. And unlike *diagnostic expectations* [[Gennaioli and Shleifer, 2018](#), [Bordalo et al., 2018, 2024](#)], which overreact to news in the *first* moment and leave the perceived *variance* unchanged,

⁴Related work studies financial frictions [[Longstaff et al., 2011](#)], risk aversion [[Lizarazo, 2013](#)], maturity choice [[Stangebye, 2020](#)], and rational learning about fundamentals, as in the rare-disaster mechanism of [Paluszynski \[2023\]](#).

⁵Additional behavioral foundations include investor sentiment [[Baker and Wurgler, 2006](#)], rare disasters [[Gabaix, 2012](#)], disposition effects [[Shefrin and Statman, 1985](#)], crisis psychology [[Kindleberger, 1978](#)], and limits to arbitrage [[Brunnermeier and Nagel, 2004](#), [Barberis and Thaler, 2003](#)].

the time-invariant PRO bias distorts the second moment and produces persistent premia, deleveraging, and a softening of doom at high debt. Diagnostic overreaction moves the schedule’s level up or down as beliefs swing; only a second-moment distortion pivots its shape and flips the sign of $\partial s / \partial \theta$ across the debt cross-section, the discriminator that separates PRO from every level friction.

Roadmap Section 2 presents the Argentine motivation; Section 3 sets up the environment and recursive equilibrium; Section 4 develops the comparative statics; Section 6 gives the rational-inattention microfoundation; Appendix C develops policy and information extensions.

2 Motivation: Argentina’s Inflation–Misreporting Episode

A Puzzle of Persistent Risk Argentina illustrates a core puzzle in international finance: why do emerging economies face borrowing costs that seem disconnected from their fundamentals [Tomz and Wright, 2013, Meyer et al., 2022]? Even models with reputational dynamics struggle to account for the persistence and magnitude of the country-specific risk premium in cases where borrowing costs have appeared divorced from repayment capacity [Morelli and Moretti, 2023].

The Misreporting Episode The puzzle came into relief during the inflation misreporting episode that began in 2007, when the Argentine government intervened in its statistics institute, INDEC, and suppressed the official CPI well below credible private and provincial estimates [Morelli and Moretti, 2023, Cavallo, 2013]. The manipulation drew fines on private consultancies [Grajewski and Raszewski, 2013], led *The Economist* to stop publishing the official figures, and prompted a rare IMF declaration of censure [The Economist, 2012, International Monetary Fund, 2013]. Because inflation-indexed bonds were a large share of public debt, the episode amounted to a *de facto* partial default. Argentina’s EMBI+ spread, previously tracking its Latin American peers, decoupled and widened sharply, and the repricing spilled over to dollar-denominated debt. This is mechanically paradoxical: a policy that lowers the real debt burden should have *decreased* default risk on nominal bonds, which marks the event as an information shock about the government’s future conduct.

Reputational Interpretation and Open Questions The conventional reading [Cole et al., 1995, Morelli and Moretti, 2023] treats the episode as a credible signal of a bad type: lenders are rational but uninformed about a government’s hidden commitment to repay, and the misreporting reveals a strategic type, leading to a persistent downgrade of reputation and higher borrowing costs. This view leaves one question open: if the market merely learned

the type, why did the premium contain an apparently excessive component reflecting not a reassessment of character but a new apprehension about the government's *predictability*? The market seems to have reacted not only to the sovereign's *intent*, but to a perceived increase in its *erratic nature*.

Comparative reading A reputational interpretation views misreporting as a credible signal that raises spreads broadly by shifting the posterior over types. PRO instead emphasizes a perceived increase in policy unpredictability, which implies a *pivoting* price schedule: cheaper terms near the brink, from a perceived chance of repayment against the odds, but costlier terms in normal times. This yields two testable implications: a heterogeneous cross-maturity response (reputation effects lift spreads across instruments, whereas PRO can produce divergent effects across the debt distribution), and a persistent spread that does not fall as the sovereign deleverages, alongside declining volatility, a stability illusion. Table 1 summarizes the differences under canonical implementations.

A Behavioral Hypothesis: Policy-Randomness Overestimation (PRO) I posit that the problem is not only what lenders do not know but what they systematically overestimate: the randomness in sovereign policy choices. Lenders perceive the unobserved shocks driving these choices as larger and more volatile than they truly are, and this wedge distorts their assessment of default risk. An event like the misreporting then reads not only as a signal of bad character but as evidence of an erratic nature. The overestimation is not irrational in a colloquial sense; it is a systematic bias in belief formation [Tversky and Kahneman, 1974, Barberis and Thaler, 2003]. That lenders distort the perceived *dispersion* of policy, not its mean, is a second-moment distortion; I formalize it as an as-if robust-control inflation disciplined by a detection-error probability.

2.1 The empirical signature: Argentina's decoupling

The Argentine episode of the late 2000s and early 2010s already displays, in raw data, the qualitative signature PRO implies. Figure 1 documents two facts over a common window. First, a level *decoupling*: Argentina's EMBI spread averaged roughly 854 bp over 2009Q3–2013Q4, while a flat peer group (Brazil, Chile, Mexico) averaged about 178 bp, a gap of nearly 675 bp that the peers' stable spreads cannot account for. Second, a *paradox*: over the same window Argentina's external debt-to-GDP ratio *fell* by half, from about 51% in 2006 to roughly 26% in 2012. Fundamentals thus moved in the direction a standard model would associate with *lower* spreads, even as the spread stayed persistently elevated relative to peers.

This falling-debt-with-persistent-spread divergence discriminates PRO from a *fundamentals or repayment-capacity* account. Such an account ties the spread to the debt path:

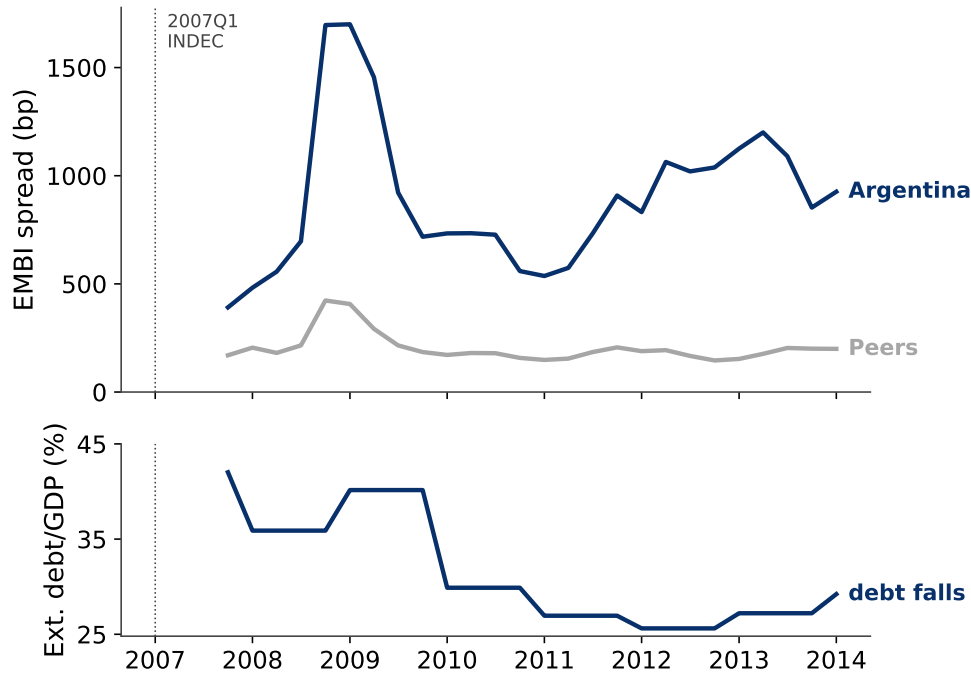


Figure 1: **Argentina’s decoupling and the deleveraging paradox.** Top: Argentina’s EMBI sovereign spread versus a flat peer group (Brazil, Chile, Mexico); Argentina averages roughly 854 bp over 2009Q3–2013Q4 against about 178 bp for the peers. Bottom: Argentina’s external debt-to-GDP ratio falls from about 51% (2006) to roughly 26% (2012). Fundamentals improve while the spread stays elevated.

as debt falls, default risk, and hence the spread, should fall with it. PRO breaks that link, because inflating the perceived dispersion of the policy shock flattens the perceived default-probability schedule and keeps the spread persistent even as the sovereign deleverages. In the language of [Bocola and Dovis \[2019\]](#), the deleveraging is an *orthogonal observable*: it moves opposite to a fundamentals story, so the divergence is the signature of a per-state premium that does not require rising default risk. It does *not* separate PRO from a reputation account [[Morelli and Moretti, 2023](#)] by level decoupling alone, since a reputational penalty need not track the debt level. The separation from reputation rests instead on the cross-maturity prediction (Table 1). Section 5 illustrates this prediction in a cross-maturity event study of Greece’s October 2009 deficit-statistics revision, a textbook predictability shock; the episode is *suggestive but not discriminating*, as explained there. I read this as qualitative consistency rather than a structural test: freely available quarterly EMBI series do not extend reliably before 2007, so a literal before/after regression is not possible, and I return to it as the over-identifying check of Section 5. A scope condition: the deleveraging-with-persistent-spreads pattern applies to credibility episodes like Argentina’s, where high spreads coincide with *falling* debt; cases where high spreads coincide with debt *accumulation* (Lebanon, Venezuela, Sri Lanka) reflect deteriorating fundamentals outside this scope.

A Bridge to the Model I embed this friction into an otherwise standard quantitative default model. The next sections formalize the PRO wedge and show that it pivots the bond price schedule, generating counterintuitive but empirically relevant outcomes: the sovereign deleverages yet faces spreads that persist rather than fall with debt, the risk premium staying elevated, a stability illusion.

3 A Model of Policy-Randomness Overestimation (PRO)

3.1 Environment

Time is discrete and the horizon is infinite, $t = 0, 1, 2, \dots$. The economy receives a stochastic endowment of a single tradable good, y_t . The endowment process is exogenous and follows a stationary, first-order Markov process, which is generated by discretizing the following AR(1) process in logarithms:

$$\ln y' = (1 - \rho_y)\mu_y + \rho_y \ln y + \sigma_y \varepsilon', \quad \varepsilon' \sim \mathcal{N}(0, 1). \quad (1)$$

The transition probabilities are given by the matrix $\Pi(y, y') = \Pr\{y_{t+1} = y' | y_t = y\}$.

In addition to the persistent endowment, the economy is hit each period by a transitory, mean-zero shock that enters the repayment budget additively. I follow [Chatterjee and Eyigungor \[2012\]](#) and assume an i.i.d. component

$$m_t \sim \mathcal{N}(0, \sigma_m^2), \quad m_t \perp m_s \ (t \neq s), \quad m_t \perp y_t, \quad (2)$$

realized at the start of each period, after y_t but before the default/repayment and borrowing decisions are taken. The shock m_t shifts resources available for consumption in repayment only; it plays no role in the default value. As in [Chatterjee and Eyigungor \[2012\]](#), this transitory shock is a smoothing and existence device for long-term debt: by perturbing the repayment budget it convexifies the otherwise discontinuous default decision and restores a well-behaved bond-price schedule. It is the same role played here by the Gumbel taste shock introduced below; the two devices are complementary, with m_t acting on the budget and the taste shock acting on the discrete choice.

The government can borrow from a large number of competitive, risk-neutral international lenders who have access to an international risk-free interest rate r . Debt takes the form of long-term bonds. A bond is a claim to a stream of coupon payments κ in every future period, unless the sovereign defaults. Each period, a fraction $\delta \in (0, 1]$ of outstanding bonds matures, while the remaining fraction $1 - \delta$ carries over to the next period.

The sovereign government has preferences represented by a standard time-separable utility function with a discount factor $\beta \in (0, 1)$. The period utility function is of the CRRA

form:

$$u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}, \quad (3)$$

which is strictly increasing and concave for $\sigma > 0$.

3.2 The Sovereign's Problem

At the beginning of each period t , the state is summarized by the current endowment realization $y \in \mathcal{Y}$ and the stock of outstanding debt B . The sovereign first decides whether to default on its obligations or to repay. This choice is subject to an idiosyncratic preference shock, often referred to as a “taste shock,” which introduces randomness into the decision-making process from the perspective of an outside observer.

Taste Shocks Let $V^D(y)$ be the deterministic component of the value of defaulting, and $V^R(y, B)$ be the deterministic component of the value of repaying. The full, or *ex-post*, value for each choice is the sum of its deterministic part and a random shock:

$$\begin{aligned} \tilde{V}^D(y, \varepsilon_d) &= V^D(y) + \varepsilon_d \\ \tilde{V}^R(y, B, \varepsilon_r) &= V^R(y, B) + \varepsilon_r \end{aligned}$$

The sovereign observes the shocks ε_d and ε_r and chooses the action that yields the highest *ex-post* value. The *ex-ante* value function, from a perspective before the shocks are realized, is the expected maximum of these *ex-post* values:

$$V(y, B) = \mathbb{E}_{\varepsilon_d, \varepsilon_r} \left[\max \left\{ \underbrace{V^D(y) + \varepsilon_d}_{\tilde{V}^D(y, \varepsilon_d)}, \underbrace{V^R(y, B) + \varepsilon_r}_{\tilde{V}^R(y, B, \varepsilon_r)} \right\} \right], \quad (4)$$

where the expectation is taken over the distribution of the shocks.

Following [Dvorkin et al. \[2021\]](#) and [Mihalache \[2024\]](#), I assume that the taste shocks ε_d and ε_r are drawn independently from a Gumbel distribution.⁶ Figure 7 plots the probability density function (PDF) and cumulative distribution function (CDF) for the Gumbel distribution, normalized to have a mean of zero. The scale parameter, η , is pivotal. It governs the variance of the taste shocks, given by $\text{Var}(\varepsilon_i) = \frac{\pi^2 \eta^2}{6}$. A larger η signifies greater dispersion in preferences and introduces more randomness into the sovereign's choice.⁷

⁶The CDF of a Gumbel distribution is given by $F(\varepsilon; \mu_L, \eta) = \exp(-\exp(-(\varepsilon - \mu_L)/\eta))$, where μ_L is the location parameter and $\eta > 0$ is the scale parameter. The mean of this distribution is $\mu_L + \eta\gamma_E$, where $\gamma_E \approx 0.5772$ is the Euler-Mascheroni constant (written γ_E throughout to avoid collision with the re-entry probability γ). For analytical convenience, I choose a specific parameterization, $\text{Gumbel}(-\eta\gamma_E, \eta)$, which makes the mean of the shocks equal to zero: $\mathbb{E}[\varepsilon_i] = -\eta\gamma_E + \eta\gamma_E = 0$.

⁷As $\eta \rightarrow 0$, the shocks' influence diminishes, and the model approaches a deterministic framework where decisions are based solely on $V^D(y)$ and $V^R(y, B)$. Conversely, as $\eta \rightarrow \infty$, the deterministic value components become negligible, and the choice becomes almost entirely random.

Economically, these taste shocks are a reduced-form representation of unmodeled factors that influence policy: political pressures, implementation errors, and policymakers' private information. Modeling them as random draws acknowledges a degree of inherent unpredictability in government behavior (see Figure 7 in Appendix A).

Ex-Ante Value Function The ex-ante value function can be derived using fundamental properties of the Gumbel distribution. The lemma provided in Appendix A provides the closed-form expression for the expected maximum of Gumbel-distributed random variables, which forms the basis for my analytical solution. First, applying Lemma 2 to my setting with $V_1 = V^D(y)$ and $V_2 = V^R(y, B)$, the ex-ante value function is:

$$V(y, B) = \mathbb{E}[\max\{\tilde{V}^d, \tilde{V}^r\}] = \eta \ln \left(\exp \frac{V^D(y)}{\eta} + \exp \frac{V^R(y, B)}{\eta} \right). \quad (5)$$

Discrete Choice The choice probabilities in my model follow directly from another fundamental property of the Gumbel distribution. Using Lemma 3 in Appendix A, the probability of default is:

$$\Pr\{d = 1|y, B\} = \frac{\exp \frac{V^D(y)}{\eta}}{\exp \frac{V^D(y)}{\eta} + \exp \frac{V^R(y, B)}{\eta}}. \quad (6)$$

Value of Default If the sovereign defaults, it is excluded from international credit markets. During exclusion, it bears an output cost and consumes a fraction of its endowment, $c = h(y) \leq y$, where the output cost function is specified similar to Chatterjee and Eyigungor [2012]:

$$h(y) = y - \max\{0, \lambda_0 y + \lambda_1 y^2\}. \quad (7)$$

In each period of exclusion, there is a constant probability $\gamma \in (0, 1)$ that the country regains market access. Upon re-entry, all past debts are forgiven, so it starts with $B = 0$. The value of being in default is therefore:

$$V^D(y) = u(h(y)) + \beta \mathbb{E}_{y'|y} [\gamma V(y', 0) + (1 - \gamma) V^D(y')]. \quad (8)$$

Value of Repayment If the sovereign honors its debt, it pays the coupon κB and retains market access. It can then choose a new level of debt for the next period, B' . This choice is also subject to taste shocks. Given the transitory shock m from (2), the *ex-ante* value of choosing a specific debt level B' , given the state (y, B, m) , is:

$$W(y, B, m, B') = u(y + m - \kappa B + [B' - (1 - \delta)B] q(y, B')) + \beta \mathbb{E}_{y'|y} [V(y', B')], \quad (9)$$

where $q(y, B')$ is the price at which it can issue new bonds and m enters the budget additively. The borrowing choice is subject to i.i.d. Gumbel shocks, denoted by $\{\varepsilon_{B'}\}$, for each

possible debt level B' . Each shock is distributed as $\text{Gumbel}(-\rho\gamma_E, \rho)$. Applying Lemma 4 in Appendix A with $V_i = W(y, B, m, B_i)$ and $\sigma = \rho$, the conditional value of repaying at shock realization m is the log-sum-exp aggregator, and the value of repayment integrates over m :

$$V^R(y, B) = \mathbb{E}_m \left[\rho \ln \left(\sum_{B' \in \mathcal{B}} \exp \frac{W(y, B, m, B')}{\rho} \right) \right], \quad (10)$$

where $\mathcal{B}$ is the discrete set of possible debt levels and the expectation is over $m \sim \mathcal{N}(0, \sigma_m^2)$. The default value $V^D(y)$ in (8) is m -free. The probability of choosing a specific level B' , likewise integrated over m , is:

$$\Pr\{B'|y, B\} = \mathbb{E}_m \left[\frac{\exp \frac{W(y, B, m, B')}{\rho}}{\sum_{B_j \in \mathcal{B}} \exp \frac{W(y, B, m, B_j)}{\rho}} \right]. \quad (11)$$

The taste shock turns the choice of B' from a deterministic point into a smooth distribution over $\mathcal{B}$, peaked at the preferred choice but assigning positive probability to others through the scale ρ . This captures *unobserved heterogeneity* in the sovereign's decision-making.⁸

3.3 International Lenders and Bond Pricing

I depart from the standard model by assuming a *wedge* between the sovereign's true behavior and lenders' perception of it. Lenders in this model are competitive and risk-neutral, pricing bonds to make zero expected profit according to their beliefs. However, their beliefs are systematically biased in a specific way.

Policy-Randomness Overestimation (PRO) The key assumption is that lenders perceive the sovereign's policy environment to be more random than it truly is. Specifically, lenders overestimate the dispersion of the transitory policy shock m : while the sovereign's *true* shock has dispersion σ_m , lenders price the default decision *as if* $m \sim \mathcal{N}(0, (\theta\sigma_m)^2)$, with $\theta \geq 1$. The parameter θ captures the degree of *policy-randomness overestimation (PRO)*; $\theta = 1$ is the rational benchmark. This belief distortion is conceptually distinct from parameter uncertainty or ambiguity aversion: lenders agree on the deterministic value components and on the true taste scale η , but they inflate the residual variance of the policy shock that perturbs the repayment budget.

Consequently, the lenders' perceived probability of default at a future state (y', B') , which I denote $\tilde{P}_\theta(y', B')$, is the m -integrated logit evaluated at the perceived dispersion

⁸This probabilistic approach is also crucial for the numerical stability of the model, as it replaces the non-differentiable "max" operator with a smooth, analytical expression, which is similar to the idea of Chatterjee and Eyigungor [2012].

$\theta\sigma_m$, using the *true* taste scale η :

$$\tilde{P}_\theta(y', B') = \mathbb{E}_{m \sim \mathcal{N}(0, (\theta\sigma_m)^2)} \left[L\left(-\frac{\Delta V(y', B', m)}{\eta}\right) \right], \quad L(z) = \frac{1}{1 + e^{-z}}, \quad (12)$$

where $\Delta V(y', B', m) = V^R(y', B', m) - V^D(y')$ is the repayment–default net value at shock realization m , with $V^R(y', B', m) = \rho \ln(\sum_{B''} \exp[W(y', B', m, B'')/\rho])$ the conditional repayment value and $V^D(y')$ the m -free default value. Note that η enters with no factor θ : the PRO channel acts *only* through the perceived dispersion of m .

Raising θ widens the perceived distribution of the policy shock and *flattens* the perceived default-probability schedule in debt: a wider m -distribution spreads mass into states where, at the true dispersion, default would be nearly certain (high debt) or nearly impossible (low debt). The perceived schedule thus pivots, with *more* default risk in safe, low-debt states and *less* certainty of default in high-debt states. This is the same pivot the temperature formulation delivers, but here it arises from an overestimated residual variance rather than an inflated taste scale, faithful both to the name policy-randomness overestimation and to the rational-inattention object.

Bond Price The equilibrium bond price $q(y, B')$ must satisfy the no-arbitrage condition based on this overestimation. The price equals the discounted expected payoff, where the probability of repayment is assessed using $\tilde{P}_\theta(y', B')$:

$$\begin{aligned} q(y, B') &= \frac{1}{1+r} \mathbb{E}_{y'|y} \left[(1 - \tilde{P}_\theta(y', B')) (\kappa + (1 - \delta) \mathbb{E}_{B''|y', B'} [q(y', B'')]) \right] \\ &= \frac{1}{1+r} \mathbb{E}_{y'|y} \left[(1 - \tilde{P}_\theta(y', B')) \left(\kappa + (1 - \delta) \sum_{B'' \in \mathcal{B}} \Pr\{B''|y', B'\} \cdot q(y', B'') \right) \right] \end{aligned} \quad (13)$$

The inner expectation $\mathbb{E}_{B''|y', B'}$ is taken over the sovereign's *true* borrowing policy $\Pr\{B''|y', B'\}$, governed by the true parameter ρ : lenders understand the sovereign's borrowing behavior but misperceive its propensity to default. This generates a credit spread containing a PRO premium that reflects the lenders' biased beliefs about stability.

3.4 Equilibrium

As is standard in sovereign default literature, the solution concept is a Recursive Markov Perfect Equilibrium, defined as follows:

Definition 1. A Recursive Markov Perfect Equilibrium consists of a set of functions: *value functions for the sovereign* ($V : \mathcal{Y} \times \mathcal{B} \rightarrow \mathbb{R}$, $V^R : \mathcal{Y} \times \mathcal{B} \rightarrow \mathbb{R}$, $V^D : \mathcal{Y} \rightarrow \mathbb{R}$), *policy probabilities for its choices* ($\Pr\{d = 1|\cdot\}$, $\Pr\{B'|\cdot\}$), and a *bond price function* ($q : \mathcal{Y} \times \mathcal{B} \rightarrow \mathbb{R}$), such that for all states (y, B) :

1. **Sovereign Optimality:** Taking the price function q as given, the sovereign's value functions and policy probabilities solve the dynamic programming problem defined by equations (5), (8), and (10). The choices are governed by the true taste shock parameters η and ρ .
2. **Lender Pricing:** The bond price function q satisfies the zero-expected-profit condition for lenders, as specified in (13), which is based on their perceived default probability $\tilde{P}_\theta$ from (12).

Lemma 1 (Threshold characterization of the perceived default probability). *Fix a future state (y', B') and recall $\Delta V(y', B', m) = V^R(y', B', m) - V^D(y')$, where $V^D(y')$ is m -free and $V^R(y', B', m) = \rho \ln(\sum_{B''} \exp[W(y', B', m, B'')/\rho])$ is the logit-aggregated repayment value. Since m enters the repayment budget as $c = y' + m - \kappa B' + q(y', B'')[B'' - (1 - \delta)B']$ and $u^R > 0$, the choice-specific value $W(y', B', m, B'')$ is strictly increasing in m for every B'' ; hence V^R , and therefore $\Delta V(y', B', \cdot)$, is strictly increasing and continuous in m . Define the default threshold*

$$m^*(y', B') := \{m : \Delta V(y', B', m) = 0\}, \quad (14)$$

the unique root when it exists; set $m^ = -\infty$ if $\Delta V > 0$ for all m (always repay) and $m^* = +\infty$ if $\Delta V < 0$ for all m (always default). By monotonicity, $\Delta V(y', B', m) < 0 \iff m < m^*(y', B')$, so the lender perceives default exactly on $\{m < m^*\}$. Then, to leading order in the small taste scale η :*

1. **(Leading-order form.)** *As $\eta \downarrow 0$, $L(-\Delta V(y', B', m)/\eta) \rightarrow \mathbf{1}\{\Delta V(y', B', m) < 0\} = \mathbf{1}\{m < m^*(y', B')\}$ for every $m \neq m^*$. Since $0 \leq L \leq 1$ is dominated by the integrable constant 1 under $\mathcal{N}(0, (\theta\sigma_m)^2)$, dominated convergence gives*

$$\tilde{P}_\theta(y', B') = \mathbb{E}_{m \sim \mathcal{N}(0, (\theta\sigma_m)^2)} \left[L\left(-\frac{\Delta V(y', B', m)}{\eta}\right) \right] \xrightarrow{\eta \downarrow 0} \Phi\left(\frac{m^*(y', B')}{\theta\sigma_m}\right), \quad (15)$$

where Φ is the standard normal CDF. For $\eta > 0$ this is a smooth, strictly-decreasing-in- ΔV mollification of the indicator; all comparative-static signs below are those of (15) and hold for the smoothed object to leading order in η (the $O(\eta)$ qualifier applies throughout).

2. **(Single crossing in B' .)** *On the no-default region, $V^R(y', B', m)$ is strictly decreasing in B' (additional debt service and dilution lower the repayment value), so $\Delta V(y', B', m)$ is strictly decreasing in B' . Differentiating the identity $\Delta V(y', B', m^*(y', B')) = 0$ and using $\partial_m \Delta V > 0$,*

$$\partial_{B'} m^*(y', B') = -\frac{\partial_{B'} \Delta V}{\partial_m \Delta V} \Big|_{m=m^*} > 0, \quad (16)$$

so $m^(y', \cdot)$ is strictly increasing in B' . Provided the zero is interior to the feasible debt range, there is then a unique sovereign-indifference debt level $\hat{B}(y')$ solving*

$m^*(y', \hat{B}(y')) = 0$, with $m^* < 0$ (perceived safe at the median shock) for $B' < \hat{B}(y')$ and $m^* > 0$ (perceived risky) for $B' > \hat{B}(y')$; if m^* does not change sign on the feasible range, $\hat{B}(y')$ is the corresponding boundary point.

3. **(Flattening.)** From (15), $\partial_{B'} \tilde{P}_\theta = \phi(m^*/(\theta\sigma_m)) \partial_{B'} m^*/(\theta\sigma_m) > 0$, and wherever $\phi(m^*/(\theta\sigma_m))/(\theta\sigma_m)$ is decreasing in θ (in particular near the pivot $m^* = 0$) the slope $|\partial_{B'} \tilde{P}_\theta|$ falls in θ : raising θ widens the perceived dispersion and flattens the perceived default-probability schedule, pushing $\tilde{P}_\theta$ toward $\Phi(0) = \frac{1}{2}$.

Proof. Strict monotonicity of ΔV in m (from $u' > 0$ acting through c) gives existence/uniqueness of m^* and the indicator limit; dominated convergence by the constant majorant 1 delivers (15). Strict monotonicity of V^R in B' and the implicit-function identity give (16) and the unique interior crossing $\hat{B}(y')$. Part 3 is the chain rule applied to (15) with $\Phi' = \phi$, sign-tracked through $\partial_{B'} m^* > 0$. $\square$

With the taste shock (logit aggregator) in place, the recursive equilibrium is well behaved.

Proposition 1. Let $\mathcal{Y}$ and $\mathcal{B}$ be compact, $u \in C^1$ strictly increasing and concave with u' bounded on the feasible consumption set, and parameters satisfy $\beta \in (0, 1)$, $r > 0$, $\delta \in [0, 1)$, $\eta > 0$, $\rho > 0$, $\kappa > 0$. Let prices be determined by the pricing operator in (18) with logistic default rule and repayment/default values as defined in the model, and let the Bellman aggregator be the log-sum-exp with taste-shock scale ρ . If the slope condition

$$L_{Jq} L_{TV} < (1 - \beta) \left(1 - \frac{1 - \delta}{1 + r}\right) \quad (17)$$

holds (constants defined below), then the Recursive Markov Perfect Equilibrium as in Definition 1 exists and is unique.

Proof. See Appendix B.4. $\square$

Unified Operators and Notation I adopt an operator view of the recursive equilibrium that anchors all of the following theory. Let J_ρ denote the sovereign's Bellman aggregator (a log-sum-exp with borrowing-choice scale ρ) and $\mathcal{T}_\theta$ the lenders' pricing operator induced by their perceived default probability under “policy-randomness overestimation” θ ; the equilibrium bond price q_θ is the unique fixed point of $\mathcal{T}_\theta$ in (18). Every result below is a comparative static in θ within this same equilibrium of Definition 1: primitives, debt structure, and J_ρ are held fixed, and only the belief parameter θ (or information and policy entering through $\mathcal{T}_\theta$) varies. I do not repeat this caveat at each proposition.

Definition 2 (Pricing operator). *For any belief parameter $\theta \geq 1$, define the lenders' pricing operator $\mathcal{T}_\theta$ acting on a bounded price function q as*

$$(\mathcal{T}_\theta q)(B', y) = \frac{1}{1+r} \mathbb{E}_{y'|y} \left[(1 - P_\theta(y', B')) \left(\kappa + (1 - \delta) \mathbb{E}_{B''|y', B'} [q(y', B'')] \right) \right], \quad (18)$$

where the perceived default probability is the m -integrated logit in the repayment–default net value difference, evaluated at the perceived shock dispersion $\theta\sigma_m$ with the true taste scale η :

$$P_\theta(y', B') = \mathbb{E}_{m \sim \mathcal{N}(0, (\theta\sigma_m)^2)} \left[L\left(-\frac{\Delta V(y', B', m)}{\eta}\right) \right], \quad L(z) = \frac{1}{1 + e^{-z}}, \quad (19)$$

$$\Delta V(y', B', m) \equiv V^R(y', B', m) - V^D(y').$$

Here $V^R(y', B', m)$ is the conditional repayment value at shock realization m and $V^D(y')$ is the m -free default value; the parameter $\theta \geq 1$ enters only through the perceived dispersion of m . Given primitives and θ , the equilibrium bond price q_θ is the unique fixed point of $\mathcal{T}_\theta$.

Notation is collected in Appendix E.

4 Theoretical Analysis

This section develops the comparative statics of PRO. The centerpiece is the *pivot*: because PRO distorts a second moment, it rotates the shape of the spread–debt schedule rather than shifting its level, the qualitative break from every first-moment friction. The single-crossing Lemma 1 (the perceived default margin m^* changes sign exactly once in debt) is what delivers the pivot (Proposition 2) and its falsifiable sign-flip corollary (Corollary 2); the threshold, deleveraging, and welfare results then follow as corollaries of the pivot. The single exogenous change is the belief parameter θ ; the information structure governing its evolution and the planner instruments of the extensions act only *through* $\mathcal{T}_\theta$, leaving primitives untouched. Unless stated otherwise, the comparative-statics signs below are established to leading order in the taste scale η (the $\eta \rightarrow 0$ limit of Lemma 1), which the finite- η objects inherit as approximations.

What is proved I separate three tiers and flag each qualification at the result rather than burying it. *Unconditional* (for every $\theta > 1$ at the modeled taste scales): the price pivot (Proposition 2), the spread pivot (Corollary 1), and the higher default threshold (Proposition 4). *Leading order in η with m^* single-crossing*: the cross-sectional sign flip (Corollary 2). *Conditional, with numerical confirmation*: strict deleveraging holds in the borrowing region $B' < B^*(y)$ in the $\rho \rightarrow 0$ limit, and the welfare loss is established for issuance in that region, with the global statement verified numerically.

Toolbox (in brief) Three ingredients from the appendix recur: (i) the pricing operator $\mathcal{T}_\theta$ is a positive contraction, so $(I - \mathcal{T}_\theta)^{-1}$ exists and is positive; (ii) Bellman and pricing operators are monotone, so orderings propagate to their fixed points; and (iii) fixed-point differentiation signs $\partial_\theta q_\theta$ by an implicit-function argument. Formal statements are in Appendix B.12.

Bond Price Pivot PRO does not uniformly depress bond prices; it pivots the price schedule relative to the rational benchmark, an effect distinct from the monotonic downward shift one associates with reputational deterioration.

Proposition 2. *Consider an economy with PRO lenders ($\theta > 1$) and the baseline economy with rational lenders ($\theta = 1$), both having a small true taste shock parameter $\eta > 0$. Let $q_1(B', y)$ and $q_\theta(B', y)$ be the respective equilibrium bond price functions, each defined as the unique fixed point of the same pricing operator $\mathcal{T}_\theta$ in (18) evaluated at $\theta = 1$ and $\theta > 1$, respectively. For a given endowment level y , there exists a debt threshold $B^*(y)$ such that the price difference $\Delta q(B', y) \equiv q_\theta(B', y) - q_1(B', y)$ satisfies:*

- For levels of future debt $B' < B^*(y)$, $\Delta q(B', y) < 0$ (PRO lenders offer lower prices).
- For levels of future debt $B' > B^*(y)$, $\Delta q(B', y) > 0$ (PRO lenders offer higher prices).

Proof. See Appendix B.5. □

Figure 2 illustrates the pivot: the two opposing belief errors balance at $B^*(y)$. For low debt ($B < B^*(y)$), where a rational lender sees default as remote, the PRO lender overstates how erratic policy is and prices in a non-negligible risk of an out-of-the-blue default, a positive PRO premium that lowers the bond price. For high debt ($B > B^*(y)$), where a rational lender sees default as near-certain, the same overstated dispersion makes the PRO lender *less* certain of doom, raising the price. Because the higher cost falls on the states where the sovereign normally borrows, it deleverages (Proposition 5); yet the spread does not collapse as debt falls, because the PRO premium keeps it elevated relative to the now-lower debt while the flattened default-probability schedule compresses its movement across states. The result is a stability illusion: a spread that stays high even as fundamentals improve, paired with reduced volatility.

Pivot Threshold This pivot point is not static; it responds to the sovereign's economic condition. The next proposition shows that as the sovereign's fortunes improve, the pivot point shifts to higher levels of debt.

Proposition 3. *The debt threshold $B^*(y)$ defined in Proposition 2, at which the baseline and PRO price schedules cross, is monotonically increasing in the endowment level y . That is, $\frac{dB^*(y)}{dy} > 0$.*

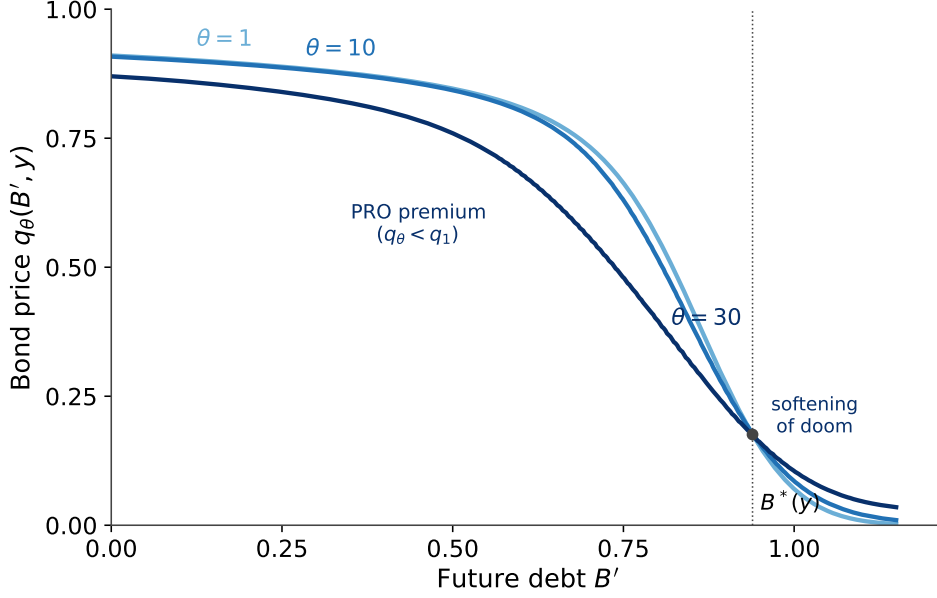


Figure 2: Pivoting Bond Price Schedules

Note: This figure illustrates the pivoting effect described in Proposition 2 for a normal output level. The bond price schedules for three different levels of PRO ($\theta = 1, 10, 30$) cross at the threshold $B^*(y)$ marked by the red dot. To the left of $B^*(y)$, PRO lenders impose a “PRO premium,” offering lower prices than rational lenders. To the right of $B^*(y)$, the “softening of doom” effect emerges, where PRO lenders offer paradoxically higher prices as they perceive less certainty about default in high-debt scenarios.

Proof. See Appendix B.6. □

The intuition is the differential response of the two markets to good news. A higher y improves repayment capacity and shifts both price schedules outward, but the rational market (q_1) responds more strongly to the fundamentals signal than the PRO market (q_θ), whose pricing stays partly anchored by its skeptical prior. Because q_1 shifts further right, its crossing with q_θ at $B^*(y)$ also moves right: a stronger economy sustains more debt before the PRO premium in the low-debt region is outweighed by the softening of doom at high debt.

Spread The credit spread $s(y, B')$ is the yield premium over the risk-free rate r , an inverse mapping from the price $q(y, B')$:

$$s(y, B') = \frac{\kappa}{q(y, B')} - \delta - r. \quad (20)$$

This inverse relationship restates Proposition 2 for credit spreads.

Corollary 1. Let $s_1(B', y)$ and $s_\theta(B', y)$ be the equilibrium credit spreads in the baseline ($\theta = 1$) and PRO ($\theta > 1$) economies, respectively. The spread difference $\Delta s(B', y) \equiv s_\theta(B', y) - s_1(B', y)$ satisfies the opposite relationship to the price difference at the same threshold $B^*(y)$ defined in Proposition 2:

- For low levels of future debt $B' < B^*(y)$, $\Delta s(B', y) > 0$.
- For high levels of future debt $B' > B^*(y)$, $\Delta s(B', y) < 0$.

Proof. See Appendix B.7. □

The discriminating prediction: a cross-sectional sign flip The pivot yields a sharp, falsifiable signature. Consider a marginal increase in the friction θ and trace its effect on prices and spreads state by state.

Corollary 2 (Sign flip of the friction response). *At each endowment y , to leading order in the taste scale η and provided $m^*(y, \cdot)$ is monotone in B' (Lemma 1), the response of the equilibrium price to a marginal increase in θ flips sign across the debt distribution at the pivot $B^*(y)$:*

$$\text{sign}\left(\frac{\partial q_\theta(y, B')}{\partial \theta}\right) = -\text{sign}(m^*(y, B')) = \begin{cases} -, & B' < B^*(y) \quad (\text{prices fall, spreads rise}), \\ +, & B' > B^*(y) \quad (\text{prices rise, spreads fall}), \end{cases}$$

and $\partial s_\theta / \partial \theta$ carries the opposite sign by (20). The cross-sectional pattern is therefore non-monotone: tightening the friction raises spreads on low-debt instruments and lowers them on high-debt instruments.

Proof. To leading order in η the price response is governed by its local source, $\partial_\theta q_\theta \propto -(I - \mathcal{T}_\theta)^{-1} \partial_\theta \tilde{P}_\theta$ with $\text{sign}(\partial_\theta \tilde{P}_\theta) = -\text{sign } m^*$ (from (23)–(20)). Since $m^*(y, \cdot)$ is monotone in B' (Lemma 1), $\partial_\theta \tilde{P}_\theta$ changes sign once at $B^*(y)$, and the positive resolvent $(I - \mathcal{T}_\theta)^{-1}$ preserves the dominant local sign, so $\partial_\theta q_\theta$ inherits a single sign change. The nonlocal smoothing by $(I - \mathcal{T}_\theta)^{-1}$ can in principle displace the crossing from $B^*(y)$; we verify numerically that a single, sharp sign change survives at the calibrated parameters (Section 5). □

This non-monotone, sign-flipping response is the model’s central *discriminating* prediction. Every first-moment friction shifts the schedule’s *level*: ambiguity/robust-control over fundamentals lifts spreads *uniformly* [Pouzo and Presno, 2016, Roch and Roldán, 2023], reputation downgrades raise them broadly across instruments, and diagnostic overreaction moves them in one direction at a time. None of these *first-moment* distortions, whose marginal effect on the perceived default probability keeps a constant sign across B' , can flip the sign of $\partial s / \partial \theta$ within the cross-section: a positive resolvent maps a sign-definite source to a sign-definite response. PRO, a second-moment distortion, instead makes that marginal effect itself change sign in B' (it equals $-\text{sign } m^*$), and so pivots the *shape*, widening spreads on short or low-notional (low- B') instruments while compressing them on long or high-notional (high- B') ones. The prediction is testable across the maturity curve or the cross-section of a sovereign’s outstanding instruments after a predictability shock such as Argentina’s misreporting or Greece’s October 2009

deficit-statistics revision: a divergent, pivot-shaped response separates PRO from any *first-moment* (level) account (Table 1). I illustrate the Greek episode in Section 5; it documents a non-parallel cross-maturity response (the short end moves more than the long end), which is inconsistent with a uniform level shift, but it does not exhibit the distinctive long-end compression and so remains suggestive rather than discriminating. It does *not* separate PRO from a state-dependent robust-control specification or any model that directly inflates the second moment of the policy shock; those share PRO's pivoting mechanism, a scope condition I state rather than overclaim.

Default Threshold The remaining results are corollaries of the pivot. The softening of doom above $B^*(y)$ (better terms at high debt) raises the option value of repaying, so PRO paradoxically makes the sovereign more resilient to debt, pushing its default threshold higher.⁹ This contrasts with standard reputation models, where a sovereign with a worse reputation (i.e., a higher perceived probability of being a 'strategic' type) would typically be expected to default at a lower debt level.

Proposition 4. *Consider economies with PRO lenders ($\theta > 1$) and rational lenders ($\theta = 1$). Let $B_{D,i}^*(y)$ be the sovereign's default threshold for economy $i \in \{1, \theta\}$, defined as the debt level B that satisfies the indifference condition:*

$$V_i^R(B_{D,i}^*(y), y) = V^D(y) \quad \text{for } i \in \{1, \theta\}. \quad (21)$$

For any given endowment level y , the default threshold is higher in the economy with PRO lenders:

$$B_{D,\theta}^*(y) > B_{D,1}^*(y).$$

Proof. See Appendix B.8. □

Default trades the immediate benefit of ceasing payments against the cost of losing market access, whose value depends on future borrowing terms. By Proposition 2, when debt is already high, PRO lenders offer *better* prices (the softening of doom). A sovereign in the PRO economy foresees these more favorable terms upon repayment, which raises the option value of repaying and rolling over debt.

Borrowing Policy The mirror-image corollary of the pivot is on day-to-day borrowing: the PRO premium below $B^*(y)$ (worse terms in the normal borrowing region) disciplines

⁹This result differs from the negative duration effect in Chatterjee and Eyigungor [2012]. In their model, longer maturity reduces the sovereign's incentive to default because it dilutes existing bondholders, creating a debt dilution channel that works through the *quantity* of debt issued. In contrast, this mechanism operates through lender *beliefs* about the sovereign's decision-making process: PRO lenders offer better terms in high-debt states due to their perception of greater randomness in sovereign choices, making continued market access more valuable and pushing out the default threshold through a behavioral pricing channel rather than a debt structure effect.

the sovereign into a more conservative debt policy.

Proposition 5. *Consider economies with PRO lenders ($\theta > 1$) and rational lenders ($\theta = 1$). Let $\mathbb{E}_i[B'|y, B]$ be the expected next-period debt level chosen by the sovereign in economy $i \in \{1, \theta\}$ from a given state (y, B) . For states (y, B) where the sovereign chooses **not to default**, and in the small borrowing-taste limit $\rho \rightarrow 0$ where the borrowing policy concentrates on its mode, expected borrowing is systematically more conservative under PRO:*

$$\mathbb{E}_\theta[B'|y, B] < \mathbb{E}_1[B'|y, B].$$

Proof. See Appendix B.9. □

In the region where the sovereign normally borrows, PRO lenders offer lower prices for new debt, raising the cost of capital, so the sovereign's optimal response is to borrow less. The PRO market disciplines the sovereign into a more conservative fiscal policy through its pricing alone, an endogenous deleveraging by which market beliefs shape real outcomes.

Welfare A direct corollary of the pivot is that the sovereign is worse off: the borrowing-region tax outweighs the remote softening of doom.

Corollary 3 (Welfare loss). *Let $V_i(y, B)$ denote the sovereign's ex-ante equilibrium value function in economy $i \in \{1, \theta\}$, with $\theta > 1$ for PRO lenders and $\theta = 1$ for rational lenders. Then*

$$V_\theta(y, B) \leq V_1(y, B) \quad \text{for all } (y, B),$$

with strict inequality whenever the equilibrium repayment policy places positive probability on the borrowing region $B' < B^(y)$, where PRO lowers the price and raises the cost of debt. (The empirically relevant case has issuance in $B' < B^*(y)$ with probability one; the global statement is verified numerically, Section 5.)*

Proof. See Appendix B.10. □

The sovereign smooths consumption on terms set by the price schedule $q(y, B')$, the price of intertemporal trade. By lowering q in the borrowing ($B' < B^*(y)$) region, PRO acts as a tax on this trade: for any given borrowing the sovereign receives fewer resources today, curtailing consumption. It adjusts (borrowing less, tolerating more debt before default) but cannot escape the loss, since the better prices in the high-debt ($B' > B^*(y)$) region are too remote to compensate for worse prices in the normal course of borrowing.

The Causal Chain of PRO The results follow a clear causal chain from the single shift in beliefs ($\theta > 1$). PRO first pivots the bond price schedule (Proposition 2), and the inverse pivot of the spread schedule (Corollary 1) is an algebraic consequence. The sovereign

then adjusts: it uses the softening of doom at high debt to endure a greater burden before defaulting (Proposition 4), and it reacts to the PRO premium in the primary borrowing region by deleveraging (Proposition 5). These constrained-optimal adjustments cannot overcome the handicap of transacting with a fearful market, culminating in a welfare loss (Corollary 3).¹⁰

Figure 3 illustrates this causal mechanism, showing how a single behavioral friction propagates through the economy's equilibrium relationships. To crystallize the novel con-

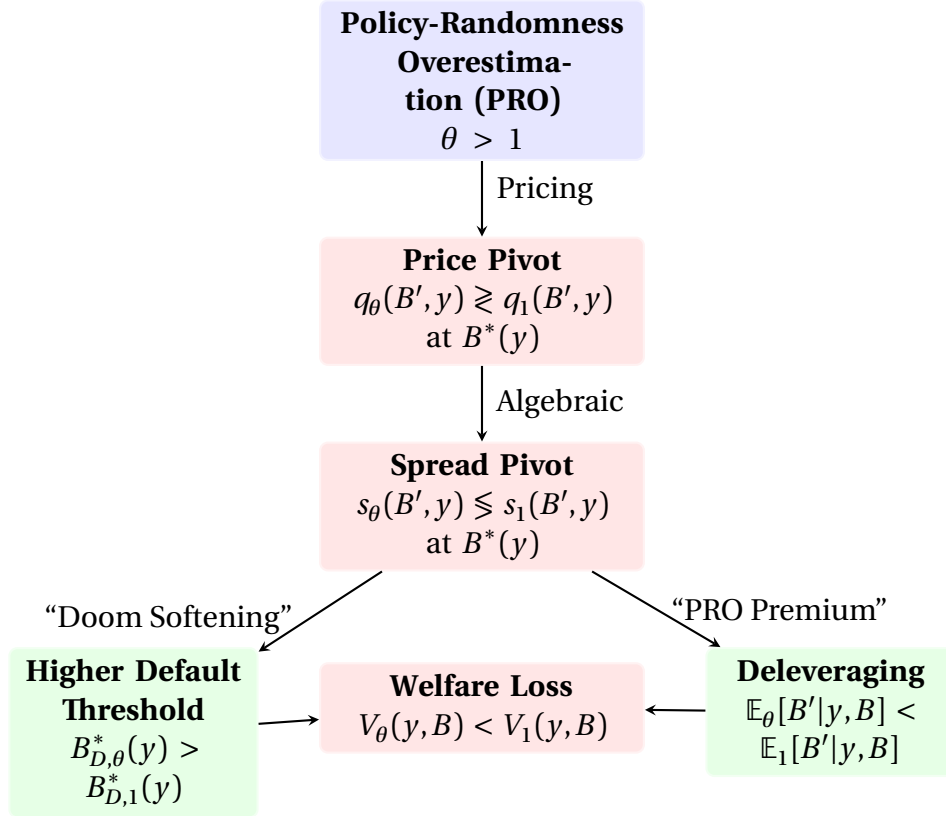


Figure 3: The Causal Chain of PRO

Note: This diagram illustrates how a single behavioral friction ($\theta > 1$) propagates through the economy. PRO first alters market pricing, creating a pivoting bond price schedule. The sovereign optimally responds to this new environment, but cannot fully escape the welfare costs of dealing with biased lenders.

tributions of this behavioral channel, Table 1 explicitly contrasts the key predictions of the PRO model with those of a standard reputation model.

¹⁰This welfare loss cannot be eliminated by optimal lump-sum fiscal policy whenever issuance occurs in the borrowing region (Assumption (A2) of Proposition 7). Appendix C develops a formal Ramsey planning extension (Proposition 7) showing that the PRO-induced distortion of intertemporal prices creates deadweight losses that persist beyond what lump-sum transfers can correct.

Table 1: Comparative Predictions: Reputation and PRO

Prediction Dimension	Reputation (canonical) Cole et al. [1995] , Morelli and Moretti [2023]	Policy-Randomness Overestimation (PRO)
Price Curve, q	Shifts <i>level</i> (monotonic in canonical setups)	<i>Pivots</i> (shape) around baseline (q_1)
Sign of $\partial s / \partial \theta$ in cross-section	<i>Uniform</i> (one sign across instruments)	<i>Flips</i> at $B^*(y)$: + low debt, – high debt
Default Threshold, B_D^*	Tends to be <i>lower</i>	<i>Higher</i> : $B_{D,\theta}^* > B_{D,1}^*$
Expected Borrowing, $\mathbb{E}[B']$	Often <i>lower</i> (constraint-driven)	<i>Lower</i> (price-driven in this model)
Average Spread, $\mathbb{E}[s]$ (time series)	<i>Higher</i>	Nearly <i>flat</i> at admissible θ (offset by deleveraging)
Cross-maturity response after misreporting	Tends to increase across instruments	Divergent across debt distribution (pivot-consistent)
Haircut-size effect on spreads	Strengthens with larger haircuts Amador and Phelan [2023]	May weaken if larger haircuts reduce perceived randomness

Note: This table compares predictions under canonical implementations of reputation models with the behavioral mechanism in this paper. The PRO column reflects this model's implementation assumptions.

5 Quantitative Analysis

In this section, I describe the quantitative implementation of the model. I first outline the calibration of the model parameters and the numerical strategy used to solve for the equilibrium. Then, I present the business cycle properties generated by the baseline model and show that it successfully replicates key features of emerging market economies.

5.1 Calibration and Numerical Solution

Calibration The model is calibrated quarterly to Argentina following the [Chatterjee and Eyigungor \[2012\]](#) (CE-2012) parameterization. One group of parameters is *set externally* to standard literature values; the other is *calibrated internally* to match Argentine business-cycle and default moments (Table 2).

The externally set parameters are standard. The risk aversion coefficient σ is set to 2 and the quarterly risk-free rate to $r = 1.04^{0.25} - 1$. The logarithmic endowment process is an AR(1) with persistence $\rho_y = 0.948503$ and innovation standard deviation $\sigma_\epsilon = 0.027092$,

the CE-2012 values for Argentina. The debt structure delivers a Macaulay duration of five years through a quarterly principal decay rate of $\delta = 0.05$, with the coupon κ set so that a risk-free bond prices at one. The re-entry probability after default is $\gamma = 0.0385$. Following [Chatterjee and Eyigungor \[2012\]](#), the repayment budget carries an i.i.d. transitory shock $m \sim N(0, \sigma_m^2)$ with $\sigma_m = 0.003$, which enters consumption only in repayment, $c = y + m - \kappa B + q(y, B')(B' - (1 - \delta)B)$, so that an adverse draw of m can trigger default; the default value $V^D(y)$ is m -free, and the repayment and pricing objects integrate over m . The Gumbel taste-shock scales are held fixed at $\eta = 5 \times 10^{-4}$ for the default decision and $\rho = 1 \times 10^{-5}$ for borrowing; these are smoothing devices kept small so that decisions are driven by fundamentals.

The internally calibrated parameters are the discount factor $\beta = 0.965$ and the two parameters of the nonlinear output cost of default, $\lambda_0 = -0.19$ and $\lambda_1 = 0.235$. These are chosen jointly so that the rational economy ($\theta = 1$) reproduces Argentina's mean debt-to-GDP ratio, mean and standard deviation of the spread, and default frequency.

The PRO friction. Policy-Randomness Overestimation is recast as an overestimation of the *dispersion of the transitory policy shock* m . Lenders price the default decision as if $m \sim N(0, (\theta \sigma_m)^2)$ with $\theta \geq 1$, while the sovereign's true shock has dispersion σ_m . The perceived default probability is the m -integrated logit

$$\tilde{P}_\theta(y', B') = \mathbb{E}_{m \sim N(0, (\theta \sigma_m)^2)} \left[L(-\Delta V(y', B', m)/\eta) \right], \quad L(z) = \frac{1}{1 + e^{-z}},$$

where $\Delta V(y', B', m) = V^R(y', B', m) - V^D(y')$ and η is the *true* taste scale (no longer scaled by θ). A larger θ widens the perceived m -distribution and flattens the perceived default-probability schedule (more perceived default in safe, low-debt states and less certainty of default in high-debt states), producing the same pricing pivot as the temperature formulation while remaining faithful to the rational-inattention residual-variance interpretation. The rational case is $\theta = 1$. Because only the product $\theta \sigma_m$ enters pricing, results are invariant to the normalization of η ; σ_m is *fixed externally* at the CE-2012 value $\sigma_m = 0.003$, not pinned down by the rational moments.

Identifying and disciplining θ . I treat θ as an *as-if* robust-control object in the tradition of [Pouzo and Presno \[2016\]](#) and [Roch and Roldán \[2023\]](#): lenders price under a perceived second moment inflated relative to the truth, exactly as a worst-case density inflates dispersion in a Hansen–Sargent multiplier model. I therefore do *not* read θ off the spread level, since the rational ingredients ($\beta, \lambda_0, \lambda_1, \sigma_m$) already over-determine it, as [Pouzo and Presno \[2016\]](#) stress; I discipline θ in three complementary ways instead.

Magnitude. I report the detection-error probability (DEP) for the perceived shock distribution: the probability that an econometrician with a sample of the relevant length could

not distinguish $N(0, (\theta\sigma_m)^2)$ from the true $N(0, \sigma_m^2)$ [Hansen and Sargent, 2008]. I take the effective sample length to be $T \approx 74$ quarters, the span of the Argentine sample used by Pouzo and Presno [2016] and Roch and Roldán [2023]. Because the policy shock is latent and inferred only from sparse, noisy signals, the admissible inflation depends on how directly it is observed (Figure 4). Even under the most generous assumption the data tolerate only a *modest* inflation (roughly $\theta \leq 2$), while a literal quarterly-observed inflation becomes detectable already near $\theta \approx 1.15$. This is why θ is best read as an as-if device rather than a literal belief, and why I do not headline a large value.

Invariance. The mechanism does not hinge on the magnitude: the pivot, threshold, and deleveraging results of Section 4 hold for *every* $\theta > 1$ (Propositions 2 and 5). I therefore present the quantitative experiments as a comparative-statics *dial*, a θ -sweep that traces the direction and shape of the mechanism, rather than as a single estimated point. The larger values ($\theta = 10, 30$) shown in the figures are illustrative magnifications that make the qualitative pivot visible, not claims about Argentina’s literal θ .

Consistency. Finally, I ask whether the mechanism’s signature can be told apart from a fundamentals or reputation account using an orthogonal observable, in the spirit of Bocola and Doviš [2019]. The discriminating fact is Argentina’s *falling debt alongside a persistent spread* after 2007 (Section 2): a fundamentals model maps falling debt into falling spreads, whereas a per-state premium that does not require rising default risk reproduces the divergence. This is an over-identifying consistency check, not a point estimate, and I label it as such.

What the quantitative object is: a cross-sectional shape, not a time-series moment.

PRO’s falsifiable content is *cross-sectional and qualitative*: the *sign* of the per-state response, which flips across the debt distribution at $B^*(y)$ (Corollary 2), visible at the illustrative θ in Figure 2. Its *magnitude* is small at data-admissible θ , like the aggregate moments below: at $\theta = 1.15$ the per-state spread response is under a basis point across the empirically relevant range, rising to a few basis points only on deep, near-default instruments. Aggregate time-series moments, the mean spread, the spread volatility, are by contrast nearly *unchanged* at data-admissible θ , and this is by design rather than a weakness: the borrowing-region PRO premium that lifts the per-state spread is offset, in the aggregate, by the endogenous deleveraging that shifts mass toward lower-spread states (the decomposition (22)). The two effects roughly cancel in any single time-series average, so the mean and volatility are flat (indeed non-monotone) in θ at admissible levels. The right object to read the model against the data is therefore the *shape* of the per-state schedule, the pivot and its sign flip, not the level or dispersion of the simulated spread series. The headline quantitative object is the pivot figure (Figure 2); the aggregate moment table below is reported mainly to document that the time-series moments do not move, which is the point. The empirical test of this cross-sectional object is

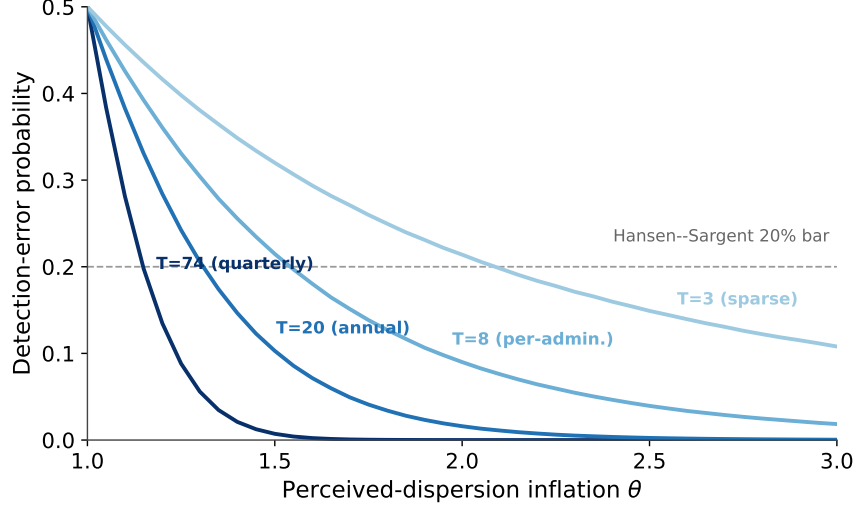


Figure 4: **The data discipline θ to a modest range.** Each curve gives the detection-error probability, the chance an econometrician could *not* distinguish the PRO-perceived distribution $N(0, (\theta\sigma_m)^2)$ from the true $N(0, \sigma_m^2)$, as a function of the inflation θ , for different effective sample sizes T (how directly the latent policy shock is observed). Below the Hansen–Sargent 20% bar the distortion is statistically detectable. Even under the most generous (sparse-observation) assumption the data tolerate only $\theta \lesssim 2$, motivating the as-if robust-control reading of θ and the focus on the θ -invariant mechanism.

correspondingly cross-sectional, not a time-series fit: across the maturity curve, or across a sovereign’s outstanding instruments after a predictability shock, PRO predicts a sign flip in the response to the friction, widening spreads on short or low-notional instruments while compressing them on long or high-notional ones (the discriminating prediction of Corollary 2, Table 1). A level friction moves all instruments the same way, so this cross-maturity sign flip, not the mean or volatility of the aggregate spread series, is what would separate PRO from any level account in the data. The Greek event study below (Section 5.3) provides a first, suggestive look: it rejects a perfectly uniform level shift but does not yet deliver the decisive sign flip, which requires instruments that straddle $B^*(y)$ simultaneously. The discriminating test therefore remains a stated, falsifiable prediction.

Numerical Solution I solve the model by value function iteration on a discretized state space of endowment y and debt B . The endowment process in (1) is discretized into $N_y = 201$ states by Tauchen’s method, and debt onto a uniform grid of $N_B = 600$ points on $[0, 1.5]$. The transitory shock m is integrated over an $N_m = 11$ -point Gauss–Hermite quadrature on $N(0, \sigma_m^2)$ (and, in the lenders’ pricing problem, on $N(0, (\theta\sigma_m)^2)$). The iteration on (V, V^D, V^R) and q runs to a joint fixed point. The log-sum-exp formulation replaces the non-differentiable max operator with a smooth expression, improving stability and speed; stabilized implementations prevent floating-point overflow and underflow from the small taste-shock parameters (Appendix D.4). The algorithm is implemented in

Table 2: Calibration (Quarterly, CE-2012 / Argentina)

Parameter	Value	Description
<i>Externally set</i>		
σ	2.0	CRRA coefficient of relative risk aversion
r	$1.04^{0.25} - 1$	Quarterly risk-free rate (4% ann.)
ρ_y	0.948503	Persistence of log endowment AR(1)
σ_ε	0.027092	Std. dev. of endowment innovations
δ	0.05	Principal decay rate (5-year Macaulay)
κ	$\delta + r$	Coupon (risk-free bond prices at 1)
γ	0.0385	Re-entry probability after default
σ_m	0.003	Std. dev. of transitory repayment shock m
η	5×10^{-4}	Scale of default taste shock
ρ	1×10^{-5}	Scale of borrowing taste shock
<i>Internally calibrated</i>		
β	0.965	Sovereign's discount factor
λ_0	-0.19	Output cost (level)
λ_1	0.235	Output cost (curvature)
<i>Friction</i>		
θ	1.0	PRO coefficient (rational benchmark)

Note: Externally set parameters follow standard values in the sovereign default literature [Chatterjee and Eyigungor, 2012]. The discount factor β and the output-cost parameters λ_0, λ_1 are calibrated internally so that the rational economy ($\theta = 1$) matches Argentina's mean debt-to-GDP ratio, mean and standard deviation of the spread, and default frequency. The PRO coefficient θ governs lenders' overestimation of the dispersion of the transitory policy shock m .

Fortran and parallelized with OpenMP.

5.2 Business Cycle Implications of PRO

I simulate the model under three scenarios: the rational benchmark ($\theta = 1$), a moderate-PRO case ($\theta = 10$), and a high-PRO case ($\theta = 30$). The larger θ values are the comparative-statics dial of Section 5: illustrative magnifications, not estimates of Argentina's literal friction. Table 3 reports the key business cycle moments against their Argentine data counterparts.

The Rational Benchmark The rational economy ($\theta = 1$) matches Argentina's fundamentals closely. The mean debt-to-GDP ratio is 16.1% (data: 17.5%), the mean annualized spread is 6.7% (data: 8.15%), a modest shortfall in the *level* that is the norm for risk-neutral default models [Arellano, 2008, Mendoza and Yue, 2012], while the standard deviation of the spread is 4.44%, essentially identical to the 4.43% in the data. The model also reproduces an annual default frequency of 3.8% (data: $\approx 3\%$), strongly counter-cyclical spreads (correlation -0.73 with GDP, data -0.79), and consumption only slightly more volatile than output ($\sigma_c/\sigma_y = 1.08$, data 1.09). These rational moments jointly pin down $\beta, \lambda_0, \lambda_1$,

and the policy-shock dispersion σ_m , and confirm that the model is a credible benchmark against which to evaluate PRO.

Deleveraging with Persistent Spreads The introduction of PRO reshapes these outcomes along the dimensions predicted by the theory. As θ rises from 1 to 30, the sovereign *deleverages* sharply: the mean debt-to-GDP ratio falls from 16.1% to 10.8%. This is the market discipline of Proposition 5: facing worse prices in the primary borrowing region, the sovereign optimally reduces its issuance. This retreat does *not* buy cheaper credit. The mean spread does not fall. It is *non-monotone* in θ , and at the illustrative $\theta = 30$ it stands at 7.4%, above the rational 6.7%, even as debt drops by a third: the headline is not this modest jump in the average spread but the *persistence* of a high spread on a much safer balance sheet, a borrowing cost far above what the lower debt would command in a rational economy. This is the PRO premium of Corollary 1, a penalty the sovereign cannot offset by reducing debt, because lenders demand compensation for an out-of-the-blue default that its true, safer fundamentals do not warrant.

The Collapse of Default Alongside deleveraging, PRO produces a near-total *collapse in the default frequency*: the annual default rate falls from 3.8% under rationality to essentially zero (0.02%) at $\theta = 30$. Two forces combine. The sovereign's forced deleveraging keeps it far from any debt level at which default is tempting, and the perceived "softening of doom" at high debt (the flatter perceived default schedule, Proposition 4) raises the value of maintaining market access. The economy thus becomes one in which observed defaults are vanishingly rare yet spreads stay elevated: defaults priced but not realized. This near-total collapse is, however, a feature of the *large- θ* illustrations: at the modest values the detection-error probability admits (Figure 4), the default frequency remains close to its rational 3–4% level, so the disciplined region of the model retains a realistic default rate for a serial defaulter while still pivoting the price schedule.

An Illusion of Stability At the illustrative $\theta = 30$, PRO also lowers the *volatility* of financial variables: the standard deviation of the spread falls from 4.44% under rationality to 3.31%. This is not genuine stabilization but an *illusion of stability*. By holding a lower average debt level, the sovereign operates further from its (higher) default threshold, so episodes of soaring spreads become rarer and overall spread volatility falls, even though the average spread remains high. The cyclical volatility of spreads, if anything, intensifies, with $\text{Corr}(\text{spread}, \text{GDP})$ moving from -0.73 to -0.85 . The impact on consumption volatility is minimal (σ_c/σ_y stays near 1.06–1.08): the sovereign adapts to the harsher borrowing environment by reducing its reliance on foreign debt for consumption smoothing, trading away the benefits of financial integration for a quieter but more expensive life.

The admissible inflation depends on how directly the latent policy shock is observed

Table 3: Data vs. Model: Business Cycle and Default Moments

Moment	Data (Arg.)	Rational ($\theta=1$)	$\theta=1.5$	$\theta=10$	PRO ($\theta=30$)
Mean Debt/GDP (%)	17.50	16.13	16.00	15.17	10.81
Mean Spread (%)	8.15	6.72	6.85	6.49	7.36
Std Spread (%)	4.43	4.44	4.64	4.05	3.31
Default freq. (% , ann.)	3.00	3.79	3.78	2.68	0.02
Corr(Spread, GDP)	-0.79	-0.73	-0.73	-0.73	-0.85
σ_c/σ_y	1.09	1.08	1.07	1.08	1.06

Note: Data moments are for Argentina at quarterly frequency; spreads and default frequencies are annualized. Model moments are computed from a 100,000-period simulation under four values of the PRO coefficient: the rational benchmark ($\theta = 1$), a modestly inflated case ($\theta = 1.5$, admissible only under sparse observation; the quarterly-observed bound is $\theta \lesssim 1.15$, Figure 4), a moderate case ($\theta = 10$), and a high case ($\theta = 30$). The rational column ($\theta = 1$) is targeted to the data; the $\theta = 10, 30$ columns are illustrative magnifications along the comparative-statics dial (holding all other parameters fixed), shown to make the θ -invariant mechanism visible, not estimates of Argentina's literal θ , which the detection-error probability (Figure 4) bounds to a modest range. At the illustrative $\theta = 10, 30$, debt and the default frequency fall sharply and the spread volatility eventually declines; but both the mean spread (6.72, 6.85, 6.49, 7.36) and the volatility (4.44, 4.64, 4.05, 3.31) are *non-monotone* in θ , and at the modest $\theta = 1.5$ neither moves appreciably. The robust, θ -invariant object is the per-state price pivot of Proposition 2, which holds for every $\theta > 1$, not any aggregate moment. Results depend only on the product $\theta \sigma_m$ and the rational moments, and are robust to the normalization of the taste scale η .

(Figure 4): at the quarterly sample frequency ($T \approx 74$) it is $\theta \lesssim 1.15$, reaching $\theta \leq 2$ only under sparse observation. At any such modest inflation, illustrated by the $\theta = 1.5$ column, the aggregate moments are essentially unchanged from the rational benchmark (Table 3): debt, the default frequency, the mean spread, and the spread volatility all move negligibly, and the volatility, like the mean, is non-monotone in θ . The deleveraging, the default collapse, and the volatility decline appear only at the illustrative $\theta = 10, 30$. The quantitative bite of PRO at data-admissible θ is therefore small *everywhere*, not only in the aggregate but state by state: at $\theta = 1.15$ the per-state spread response is under a basis point across the empirically relevant range, reaching a few basis points only on deep near-default instruments. The contribution is accordingly the qualitative per-state *mechanism*, invariant for all $\theta > 1$, and its discriminating signature is the *sign* of the cross-sectional response, not its size or any monotone aggregate moment. Figure 9 (Appendix) makes the aggregate level fit of the rational benchmark explicit; note the model matches the location but not the fat right tail, the standard modest-mean-spread feature.

5.3 A first cross-maturity look: the Greek 2009 revision

The discriminating object of Section 5 is cross-sectional, so the natural first place to look is a single sovereign's maturity curve around a predictability shock. The cleanest external analogue to Argentina's 2007 misreporting episode (Section 2) is Greece's October 2009 deficit-

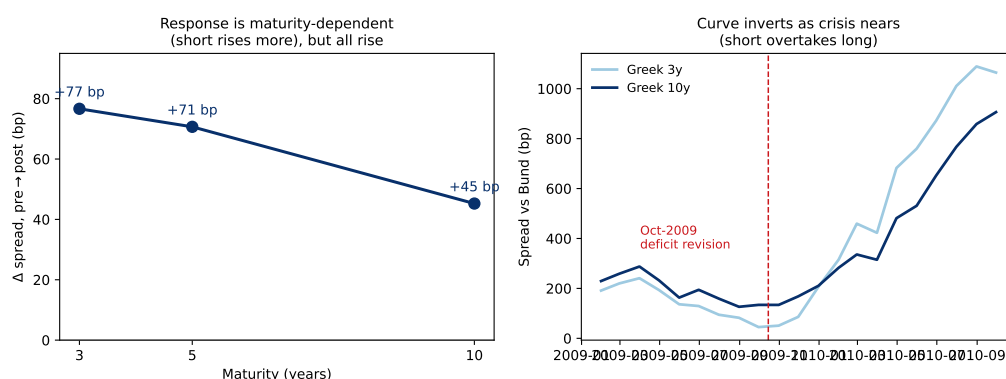


Figure 5: **Cross-maturity spread response to the Greek October 2009 revision.** *Left:* the change in the Greek-minus-German benchmark spread from the pre-window (2009-06–2009-09) to the post-window (2009-10–2010-01), by tenor: +77 bp at 3 years, +71 bp at 5 years, and +45 bp at 10 years. The response is monotonically decreasing in maturity, so the short end widened about 1.7 times as much as the long end. *Right:* the Greek 3-year and 10-year spread levels over 2008–2012; the dashed line marks the October 2009 revision, after which the curve inverts (the 10-year falls below the 3-year). The non-parallel response rejects a perfectly uniform level shift, but every tenor widened, so this is not the distinctive long-end compression that PRO predicts above the pivot. Spreads in basis points; data from FRED/ECB/stooq.

statistics revision, when the reported 2009 deficit was abruptly restated from roughly 6% to roughly 12.7% of GDP. Like INDEC, this is a textbook PRO shock: not new fundamentals but a sudden revelation that policy is less reliable than markets had believed. I assemble monthly benchmark yields at the 3-, 5-, and 10-year tenors for Greece and Germany (FRED/ECB/stooq, 2008–2012) and define the spread at each tenor as the Greek yield minus the German yield, in basis points.

The cross-maturity response is non-parallel. Comparing the pre-window (2009-06–2009-09) with the post-window (2009-10–2010-01), the 3-year spread rose by 77 bp, the 5-year by 71 bp, and the 10-year by 45 bp (Figure 5, left panel). All tenors widened, but the short end moved about 1.7 times as much as the long end, and the credit curve subsequently inverted: the 10-year-minus-3-year slope went from +89 bp in September 2009 to –32 bp in January 2010 and –220 bp by June 2010 (right panel).

I read this episode honestly, as suggestive rather than discriminating. On the positive side, the response is maturity-dependent, not a parallel shift: a naive level account, in which a reputation downgrade lifts every instrument by the same amount, is rejected by the data, and rejected in the direction PRO permits. On the negative side, the long end did not compress; it rose by 45 bp. The distinctive PRO signature is a long-end *softening of doom*, an opposite-signed or falling response on the near-default instrument (Corollary 2), and that signature is absent here. The observed pattern, in which the short end rises more and the curve inverts, is equally consistent with PRO's premium region and with the textbook imminent-default credit-curve inversion, so it does not separate the two accounts.

The reason the episode cannot discriminate is structural, and it is exactly the reason Section 5 insists the test be cross-sectional in the right place. In October 2009 Greece was still below the default brink, in PRO's premium region $B' < B^*(y)$, where the model predicts that *all* tenors' spreads rise and only their *size* is maturity-graded (the larger short-end move is fully consistent with PRO, but does not isolate it). The discriminating sign flip appears only across the pivot $B^*(y)$, on instruments at near-certain default, a region Greece reached only in 2011–2012, by which time bailout and PSI dynamics dominate the pricing and no longer isolate the friction. A clean test therefore needs instruments that *straddle* $B^*(y)$ simultaneously, which a single sovereign curve early in its crisis cannot provide; it requires a bond-level cross-section with variation in notional or seniority (for example Bloomberg issue-level data). The Greek revision is thus an honest illustration that the cross-maturity response is non-uniform in the direction PRO allows, not a confirmation of the sign flip itself. The discriminating test of Corollary 2 and Table 1 remains a stated, falsifiable prediction awaiting straddling cross-sectional data.

5.4 The Mechanics of PRO: Distributions and Policy Functions

Long-Run Outcomes: A Shift in Distributions The aggregate statistics in Table 3 reflect shifts in the sovereign's stationary distributions. Figure 8 plots the simulated histograms for debt and spreads. Panel (a) shows the deleveraging of Proposition 5: the entire debt-to-GDP distribution shifts left, the sovereign's optimal retreat from the punitive pricing it faces. Yet, as Panel (b) shows, this retreat does not earn better terms: the spread distribution is pushed right. This is the PRO premium of Corollary 1, a low-debt trap in which a fundamentally safer sovereign faces a persistently higher cost of capital.

State-Contingent Policies: The Pivoting Effect The long-run distributional shifts trace back to the state-by-state price pivot of Figure 2, analyzed in Section 4: worse terms in the normal borrowing region induce deleveraging (Proposition 5) while better terms at high debt push out the default threshold (Proposition 4). Both responses surface in the stationary distributions of Figure 8: the debt distribution shifts left while spreads do not fall. The corresponding welfare loss is the “tax on intertemporal trade” of Corollary 3, verified numerically in the solved model by a uniformly lower value function under higher PRO.

Why Spreads Persist Despite Deleveraging Let $s_i(y, B')$ denote the spread defined in (20) with price q_i for economy $i \in \{1, \theta\}$, where q_i is the equilibrium bond price under rational lenders ($i = 1$) and PRO lenders ($i = \theta$). Denote by μ_i the stationary distribution over non-default states and optimal choices (including the sovereign's optimal B') in

economy i . The difference in average spreads can be written as the decomposition

$$\bar{s}_\theta - \bar{s}_1 = \underbrace{\kappa \mathbb{E}_{\mu_\theta} \left[\frac{1}{q_\theta} - \frac{1}{q_1} \right]}_{\text{price wedge at PRO weights}} + \underbrace{\kappa \left(\mathbb{E}_{\mu_\theta} \left[\frac{1}{q_1} \right] - \mathbb{E}_{\mu_1} \left[\frac{1}{q_1} \right] \right)}_{\text{composition (policy) effect}}. \quad (22)$$

The pivot result (Proposition 2) implies that for each y there exists $B^*(y)$ such that $q_\theta(y, B') < q_1(y, B')$ (equivalently, $1/q_\theta > 1/q_1$) for $B' < B^*(y)$, and the opposite inequality holds for $B' > B^*(y)$. In the empirically relevant set where new issuance occurs (the primary borrowing region), one typically has $B' < B^*(y)$ and $V^R(y, B') > V^D(y)$.

Two implications follow.

- *Local comparative statics.* Differentiating (13) with respect to θ yields the linear equation

$$(I - \mathcal{T}_\theta) \frac{\partial q_\theta(\cdot)}{\partial \theta}(y, B') = -\frac{1}{1+r} \mathbb{E}_{y'|y} \left[(\partial_\theta \tilde{P}(y', B')) \Lambda(y', B') \right], \quad (23)$$

where $\Lambda(y', B') \equiv \kappa + (1-\delta) \sum_{B'' \in \mathcal{B}} \Pr\{B''|y', B'\} q_\theta(y', B'')$ and $\mathcal{T}_\theta$ is the positive linear operator defined in (18). Because $\|\mathcal{T}_\theta\| < 1$ under discounting and $(1-\delta) < 1$, $(I - \mathcal{T}_\theta)^{-1}$ exists and is positive. Using (12) with $\Delta V(y', B') \equiv V^R(y', B') - V^D(y')$, and using the leading-order form $\tilde{P}_\theta(y', B') = \Phi(m^*(y', B')/(\theta\sigma_m))$ of Lemma 1 (so m^* does not depend on θ), gives

$$\partial_\theta \tilde{P}_\theta(y', B') = \phi\left(\frac{m^*(y', B')}{\theta\sigma_m}\right) \frac{\partial}{\partial \theta} \left[\frac{m^*(y', B')}{\theta\sigma_m} \right] = -\phi\left(\frac{m^*(y', B')}{\theta\sigma_m}\right) \frac{m^*(y', B')}{\theta^2\sigma_m},$$

so $\text{sign}(\partial_\theta \tilde{P}_\theta) = -\text{sign } m^*(y', B')$ ($\phi = \Phi' > 0$ the standard normal density). Hence, in the perceived-safe region where $m^*(y', B') < 0$ (repayment dominates at the median shock; typically $B' < B^*(y)$), the right-hand side of (23) is negative, and since $(I - \mathcal{T}_\theta)^{-1}$ exists and is positive, $\partial q_\theta / \partial \theta < 0$, and therefore $\partial s_\theta / \partial \theta > 0$ by (20); the sign flips where $m^* > 0$.

- *Average spread dominance.* By (22), the first term (price wedge at PRO weights) is strictly positive and *strengthened* by deleveraging, because mass shifts toward $B' < B^*(y)$ where $1/q_\theta - 1/q_1 > 0$. The second term (composition effect at baseline prices) is weakly negative since $1/q_1$ is lower at smaller B' . Under mild regularity (small taste-shock scale η and monotone price/borrowing policies), the first term offsets or outweighs the second, so the average spread need not fall as the sovereign deleverages and may rise modestly. Quantitatively (Table 3) the rise in the *mean* spread is small; the salient and robust effects are the *persistence* of the spread relative to the now-lower debt (the spread is far higher than the reduced leverage alone would warrant) and the decline in its *volatility*, the illusion of stability.

6 Microfoundations for PRO: Mental Model and Residual Uncertainty

The baseline takes the perceived-dispersion inflation $\theta > 1$ as a primitive; this section *derives* it through a rational-inattention (RI) microfoundation. Lenders fit a simplified mental model of the sovereign's net payoff difference and acknowledge its imperfection via a residual, which they identify with the unmodeled transitory policy shock m . They allocate limited attention between learning the model parameters and learning the residual's dispersion. The residual-uncertainty channel inflates the perceived dispersion of the policy shock, so a lender prices as if the shock had dispersion $\theta_{\text{RI}}\sigma_m \geq \sigma_m$, which is exactly the PRO friction of Definition 2, with $\theta_{\text{RI}} = 1$ recovered only under unlimited attention. The challenge is to distinguish known unknowns (model-parameter uncertainty) from unknown unknowns (the dispersion of the residual m).

Same-operator discipline Let $s \equiv (y, B')$ denote the state. As in the baseline, lenders price bonds using the operator

$$(\mathcal{T}_\theta q)(y, B') = \frac{1}{1+r} \mathbb{E}_{y'|y} \left[(1 - \tilde{P}_\theta(y', B')) (\kappa + (1 - \delta) \mathbb{E}_{B''|y', B'} [q(y', B'')]) \right],$$

with the m -integrated logistic default of Definition 2,

$$\tilde{P}_\theta(y', B') = \mathbb{E}_{m \sim \mathcal{N}(0, (\theta\sigma_m)^2)} \left[L\left(-\frac{\Delta V(y', B', m)}{\eta}\right) \right], \quad L(z) \equiv (1 + e^{-z})^{-1}, \quad \eta > 0,$$

where $\theta \geq 1$ is the ratio of the lenders' perceived to the sovereign's true policy-shock dispersion. The sovereign's problem and all primitives are unchanged. The microfoundation endogenizes the state-dependent evaluation parameter $\theta_{\text{RI}}(s) \geq 1$; the operator itself is the same.

6.1 Mental model with residual

Lenders believe the net value difference can be decomposed as

$$\Delta V(s) = f(s; \beta) + \xi, \quad s = (y, B'). \quad (24)$$

Here f is a known functional form with an unknown parameter vector $\beta \in \mathbb{R}^k$ and ξ is the unmodeled transitory policy residual, identified with the shock m that perturbs the repayment budget. Its true dispersion is σ_m , but lenders do not know it and must learn it.

Priors and data are:

$$\beta \sim \mathcal{N}(\bar{\beta}, \Sigma_\beta), \quad s_\beta = \beta + \varepsilon_\beta, \quad \varepsilon_\beta \sim \mathcal{N}(0, (\psi_\beta a_\beta)^{-1} I_k), \quad (25)$$

$$\xi | \sigma^2 \sim \mathcal{N}(0, \sigma^2), \quad \ell_\sigma \equiv \log \sigma^2 \sim \mathcal{N}(\bar{\ell}, \tau^2), \quad s_\ell = \ell_\sigma + \varepsilon_\sigma, \quad \varepsilon_\sigma \sim \mathcal{N}(0, (\psi_\sigma a_\sigma)^{-1}), \quad (26)$$

where σ^2 is the lenders' *perceived* policy-shock variance (true value σ_m^2). The attention levels $a_\beta, a_\sigma \geq 0$ scale effective signal precisions via $\psi_\beta, \psi_\sigma \in (0, 1]$. Attention is costly:

$$\Psi(a_\beta, a_\sigma) = \frac{\kappa_\beta}{2} a_\beta^2 + \frac{\kappa_\sigma}{2} a_\sigma^2, \quad \kappa_\beta, \kappa_\sigma > 0. \quad (27)$$

Posteriors Given (25), the posterior for β is Gaussian with

$$\hat{\Sigma}_\beta = (\Sigma_\beta^{-1} + \psi_\beta a_\beta I_k)^{-1}, \quad \hat{\beta} = \hat{\Sigma}_\beta (\Sigma_\beta^{-1} \bar{\beta} + \psi_\beta a_\beta s_\beta). \quad (28)$$

For the residual variance, using a log-normal prior and Gaussian signal for ℓ_σ yields a Gaussian posterior for ℓ_σ with precision increasing in $\psi_\sigma a_\sigma$; define the posterior mean

$$\widehat{\sigma^2} \equiv \mathbb{E}[\sigma^2 | s_\ell] = \mathbb{E}[\exp(\ell_\sigma) | s_\ell], \quad (29)$$

which is well-defined and continuous in the sufficient statistic s_ℓ and strictly increasing in its posterior mean for ℓ_σ .

6.2 From posterior uncertainty to a perceived policy-shock dispersion

Linearize f at $\hat{\beta}$ and define

$$Z(s) := -\frac{1}{\eta} \left(f(s; \hat{\beta}) + \nabla_\beta f(s; \hat{\beta})^\top (\beta - \hat{\beta}) + \xi \right), \quad z_0(s) := -\frac{f(s; \hat{\beta})}{\eta}. \quad (30)$$

Conditional on posteriors, the residual entering the logit is mean-shifted Gaussian, and the lenders' effective *perceived* policy-residual variance combines the posterior variance of fundamentals along ∇f with the posterior mean residual variance,

$$\widehat{\sigma_m^2}(s) := \nabla f^\top \hat{\Sigma}_\beta \nabla f + \widehat{\sigma^2}, \quad \nabla f := \nabla_\beta f(s; \hat{\beta}). \quad (31)$$

Pricing the default decision under this perceived variance, that is, as if $m \sim \mathcal{N}(0, \widehat{\sigma_m^2}(s))$, is, by construction, the dispersion-form pricing of the friction evaluated at $\theta = \widehat{\sigma_m^2}/\sigma_m$: the two integrate the same m -logit against the same Gaussian. Given the perceived variance, this mapping to θ uses the same m -integral and involves *no logit-temperature approximation*; the only approximation is the linearization of f at $\hat{\beta}$ that defines $\widehat{\sigma_m^2}$ in the first place (Appendix B.13). The lenders' perceived default probability

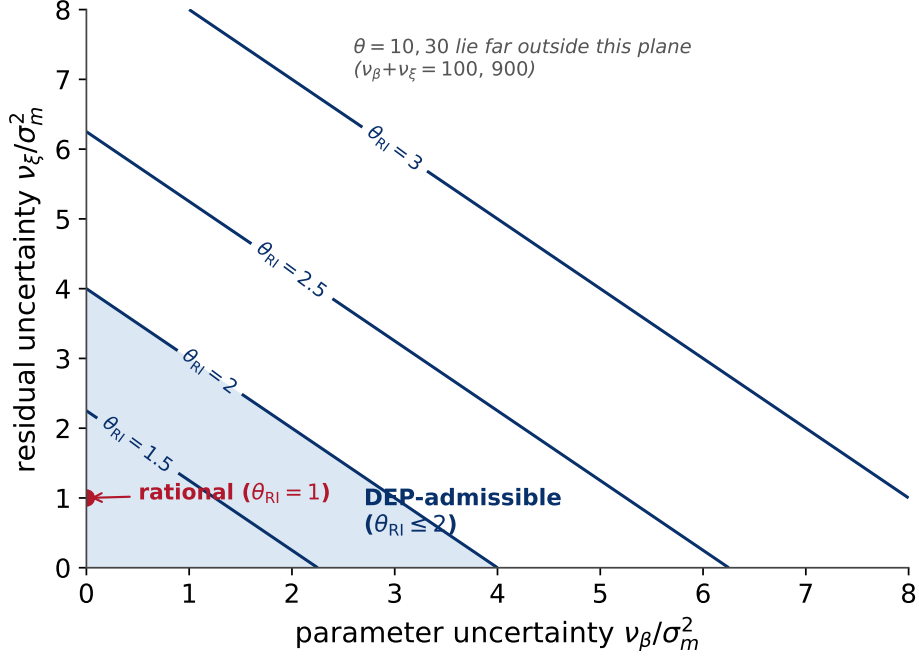


Figure 6: **From rational-inattention primitives to θ .** Contours of the implied PRO coefficient $\theta_{\text{RI}} = \sqrt{(v_\beta + v_\xi)/\sigma_m^2}$ over the plane of parameter uncertainty v_β and residual uncertainty v_ξ (both in units of σ_m^2). The shaded region is detection-error admissible ($\theta_{\text{RI}} \leq 2$); the rational benchmark is the red point. The illustrative $\theta = 10, 30$ used in the figures lie far outside this plane, requiring $v_\beta + v_\xi = 100, 900$.

$\tilde{P}_\theta$ is the one so obtained. Writing this perceived variance as $(\theta_{\text{RI}}(s)\sigma_m)^2$ identifies $\theta_{\text{RI}}(s)$ with the ratio of perceived to true policy dispersion,

$$\theta_{\text{RI}}(s) = \sqrt{\frac{\widehat{\sigma_m^2}(s)}{\sigma_m^2}} = \sqrt{\frac{\nabla f^\top \hat{\Sigma}_\beta \nabla f + \widehat{\sigma}^2}{\sigma_m^2}} \geq 1, \quad (32)$$

where $\theta_{\text{RI}} = 1$ when the perceived residual variance coincides with the true policy-shock variance σ_m^2 . Assuming the variance term is bounded on compact subsets of the state space, we can restrict to $\theta_{\text{RI}}(s) \in [1, \bar{\theta}]$ with finite $\bar{\theta} > 1$.

Equation (32) is the dictionary between the rational-inattention objects and the pricing primitives: the perceived variance $\widehat{\sigma_m^2}(s)$ plays the role of $(\theta\sigma_m)^2$, $\theta_{\text{RI}}(s)$ is the PRO coefficient θ (now state-dependent), and $\mathcal{T}_{\theta_{\text{RI}}}$ is the same operator $\mathcal{T}_\theta$. Figure 6 maps the two underlying uncertainties (parameter uncertainty v_β and residual uncertainty v_ξ) into the implied coefficient θ_{RI} , showing that the detection-error-admissible range occupies only a small corner of this plane: at the quarterly sample frequency the admissible inflation is $\theta \lesssim 1.15$, reaching $\theta \leq 2$ only under sparse observation (Figure 4).

6.3 Decomposing θ_{RI} and attention sensitivities

Decompose the perceived policy-shock variance in (31) into a parameter and a residual source,

$$\widehat{\sigma}_m^2(s) = \underbrace{\nabla f^\top \hat{\Sigma}_\beta \nabla f}_{v_\beta(s)} + \underbrace{\widehat{\sigma}^2}_{v_\xi(s)}, \quad \theta_{\text{RI}}(s) = \sqrt{\frac{\widehat{\sigma}_m^2(s)}{\sigma_m^2}}. \quad (33)$$

Thus $\theta_{\text{RI}}(s)^2 - 1$ is, in units of the true policy variance σ_m^2 , the sum of a parameter component $v_\beta(s)$ and a residual component $v_\xi(s)$: lenders overestimate m 's dispersion exactly to the extent that residual posterior uncertainty exceeds σ_m^2 . Local attention sensitivities (holding ∇f fixed) follow by the chain rule $\partial \theta_{\text{RI}} = \partial \widehat{\sigma}_m^2 / (2\sigma_m^2 \theta_{\text{RI}})$,

$$\frac{\partial \theta}{\partial a_\beta}(s) = -\frac{\psi_\beta}{2\sigma_m^2 \theta_{\text{RI}}} \nabla f^\top \hat{\Sigma}_\beta^2 \nabla f \leq 0, \quad \frac{\partial \theta}{\partial a_\sigma}(s) = \frac{1}{2\sigma_m^2 \theta_{\text{RI}}} \frac{\partial \widehat{\sigma}^2}{\partial a_\sigma}(s). \quad (34)$$

With the objective in (38), the state-weighted first-order conditions compactly read

$$\kappa_\beta a_\beta = \mathbb{E} \left[w(s) \psi_\beta \nabla f^\top \hat{\Sigma}_\beta^2 \nabla f \right], \quad \kappa_\sigma a_\sigma = -\mathbb{E} \left[w(s) \frac{\partial \widehat{\sigma}^2}{\partial a_\sigma}(s) \right], \quad (35)$$

where $w(s) \propto L'(-\Delta V(s)/\eta)$ concentrates mass near the default boundary.¹¹

Posterior mechanics for $\widehat{\sigma}^2$ Under the Gaussian prior/likelihood for $\ell_\sigma \equiv \log \sigma^2$ in (28)–(29), the posterior is $\ell_\sigma | s_\ell \sim \mathcal{N}(\hat{\ell}, \hat{\tau}^2)$ with $\hat{\tau}^2 = (\tau^{-2} + \psi_\sigma a_\sigma)^{-1}$ and $\hat{\ell} = (\tau^{-2} \bar{\ell} + \psi_\sigma a_\sigma s_\ell) / (\tau^{-2} + \psi_\sigma a_\sigma)$. Hence

$$\widehat{\sigma}^2(a_\sigma, s_\ell) = \exp \left(\hat{\ell}(a_\sigma, s_\ell) + \frac{1}{2} \hat{\tau}^2(a_\sigma) \right), \quad (36)$$

and a direct calculation (Appendix B.14) yields the local sensitivity

$$\frac{\partial \widehat{\sigma}^2}{\partial a_\sigma} = \widehat{\sigma}^2 \psi_\sigma \hat{\tau}^4 \left(\frac{s_\ell - \bar{\ell}}{\tau^2} - \frac{1}{2} \right). \quad (37)$$

Equation (37) makes explicit when dispersion attention “inflates” (raises $\widehat{\sigma}^2$, and thus the perceived dispersion of m and θ) or “deflates” it: large instability signals (s_ℓ sufficiently above $\bar{\ell}$) overturn the certainty effect ($-\frac{1}{2}$ term) and increase $\widehat{\sigma}^2$; otherwise, posterior concentration dominates and $\widehat{\sigma}^2$ falls toward the true σ_m^2 .

¹¹In (35), under symmetric innovations and stability-biased priors, attention typically reduces the posterior mean of the residual variance, so $\mathbb{E}[-\partial \widehat{\sigma}^2 / \partial a_\sigma] > 0$. With adverse (instability) signals, $\partial \widehat{\sigma}^2 / \partial a_\sigma$ can locally turn positive, which the sign-robust form in (35) accommodates.

Economic intuition $\theta_{\text{RI}}(s)$ is the ratio of the lenders' (state-wise) perceived policy-shock dispersion to its true value σ_m . The perceived variance has two sources: uncertainty about fundamentals along the most relevant direction ∇f , and uncertainty about the residual policy variance σ^2 . Accordingly, a_β compresses posterior variance precisely along the direction that most shifts the default margin (typically lowering the perceived dispersion and hence θ), whereas a_σ diagnoses instability: if attention reveals stability it lowers the perceived dispersion (and θ); if it reveals larger residual policy risk (bad news) it inflates the perceived dispersion (and θ). The kernel $w(s) \propto L'(-\Delta V/\eta)$ concentrates attention near the threshold, ensuring scarce attention is spent where prices are most sensitive.

Pricing with state-dependent θ_{RI} Define $(\mathcal{T}^{\text{RI}} q)(s) := (\mathcal{T}_{\theta_{\text{RI}}(s)} q)(s)$. Since $\mathcal{T}_\theta$ is positive and a contraction uniformly in $\theta \in [1, \bar{\theta}]$ (Appendix B.12), continuity of θ_{RI} implies $\mathcal{T}^{\text{RI}}$ is a contraction with the same modulus. Hence there exists a unique price fixed point q^* solving $q^* = \mathcal{T}^{\text{RI}} q^*$. All operator-based comparative statics from the baseline therefore carry over with θ replaced by $\theta_{\text{RI}}(\cdot)$.

6.4 Attention choice and substitution

Lenders choose (a_β, a_σ) taking primitives and the operator mapping as given. To keep the same-operator discipline, attention affects prices only through $\theta_{\text{RI}}(\cdot)$ via (28)–(32). A convenient objective is to minimize a weighted posterior dispersion around the decision margin subject to attention costs:

$$\max_{a_\beta, a_\sigma \geq 0} \mathcal{J}(a_\beta, a_\sigma) := -\mathbb{E}[\widehat{\sigma}_m^2(s; a_\beta, a_\sigma) w(s)] - \Psi(a_\beta, a_\sigma), \quad w(s) \propto L'\left(-\frac{\Delta V(s)}{\eta}\right), \quad (38)$$

where $w(s)$ weights states near the default boundary. Under (28)–(29) and convex (27), the program admits a unique solution (a_β^*, a_σ^*) .

Proposition 6 (Mean-quality shock induces attention substitution). *Under Assumptions (A1)–(A4) stated in Appendix B.10, suppose ψ_β falls while ψ_σ and other primitives are fixed. Then at any interior optimum of (38), a_β^* weakly decreases and a_σ^* weakly increases. Consequently, $\widehat{\sigma}^2$ weakly rises in expectation and $\theta_{\text{RI}}(s)$ weakly increases pointwise via (32).*

Proof. See Appendix B.11. □

Event mapping: two channels Argentina's inflation misreporting can be read through two distinct channels:

- *Credibility collapse (quality channel)*: a persistent fall in ψ_β lowers the marginal value of fundamentals signals. By Proposition 6, attention substitutes toward dispersion

$(a_\beta^* \downarrow, a_\sigma^* \uparrow)$, which raises the perceived policy-shock variance $\widehat{\sigma}_m^2(s)$ and hence $\theta_{\text{RI}}(s)$ via (32).

- *Instability shock (signal channel)*: the act of misreporting is itself a large instability signal $s_\ell \gg \bar{\ell}$. By (37), dispersion attention amplifies $\widehat{\sigma}^2$ on impact, inflating the perceived dispersion of m even holding ψ_β fixed.

Both channels operate through the same pricing operator $\mathcal{T}_\theta$ evaluated at a higher $\theta_{\text{RI}}(\cdot)$, i.e. a more overestimated policy-shock dispersion, generating steeper price pivots, higher default thresholds with deleveraging, and spreads that persist even as debt falls.

7 Conclusion

I unify a behavioral sovereign-risk mechanism, policy-randomness overestimation (PRO), with standard default environments through a single pricing operator. PRO tilts pricing around a state-contingent threshold, so governments borrow less while tolerating higher debt at default; spreads then persist as debt declines, a stability illusion. A rational-inattention microfoundation endogenizes the state-dependent perceived dispersion θ_{RI} inside that same operator (Section 6), tying the mechanism to the Argentina event. Policy and information extensions clarify the levers: transfers cannot undo the pricing wedge, negativity-biased learning makes the distortion persistent, and transparency improves welfare by curbing tail tilting (Appendix C). The framework yields compact, testable implications (the pivot, threshold shifts, and decoupling) consistent with event evidence such as Argentina's misreporting, and offers a disciplined path for calibration and event-study validation.

Appendices for “Default with Policy-Randomness Overestimation”

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This appendix collects auxiliary results, proofs, and computational details. Section A details the Gumbel distribution and the logit identities used in the text. Section B provides proofs for the lemmas and propositions. Section B.12 states contraction, positivity/order, and differentiability results for the pricing operator (including Assumption RI.1 and a uniform convergence argument). Section D documents the computational algorithm and the numerical stability techniques for the discrete-choice implementation.

A Gumbel Distribution in Default Models

In this section, I provide some useful results about the Gumbel distribution to help formulate the close form solution presented in Section 3.

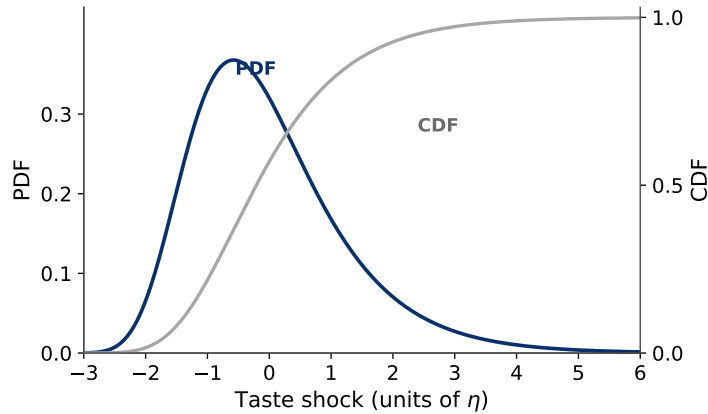


Figure 7: The mean-zero Gumbel distribution ($\eta = 5 \times 10^{-4}$).

Note: The Gumbel distribution is used to model the taste shocks in the sovereign’s default and borrowing decisions. The scale parameter η controls the variance of these shocks. The left y axis is the PDF, the right y axis is the CDF.

Lemma 2. Let $\varepsilon_1, \varepsilon_2$ be independent random variables distributed as $\text{Gumbel}(-\eta\gamma_E, \eta)$, where γ_E is the Euler-Mascheroni constant. Let $V_1, V_2 \in \mathbb{R}$ be deterministic constants. Then:

$$\mathbb{E}[\max\{V_1 + \varepsilon_1, V_2 + \varepsilon_2\}] = \eta \ln \left(\exp \frac{V_1}{\eta} + \exp \frac{V_2}{\eta} \right). \quad (\text{A.1})$$

Proof. See Appendix B.1. □

Lemma 3. Let $\varepsilon_1, \varepsilon_2$ be independent $\text{Gumbel}(-\eta\gamma_E, \eta)$ random variables, and let $V_1, V_2 \in \mathbb{R}$ be deterministic values. Then:

$$\Pr\{V_1 + \varepsilon_1 > V_2 + \varepsilon_2\} = \frac{\exp \frac{V_1}{\eta}}{\exp \frac{V_1}{\eta} + \exp \frac{V_2}{\eta}}. \quad (\text{A.2})$$

Proof. See Appendix B.2. □

Lemma 4. Let $\{V_i\}_{i=1}^n$ be deterministic values and $\{\varepsilon_i\}_{i=1}^n$ be independent $\text{Gumbel}(-\sigma\gamma_E, \sigma)$ random variables. Then:

$$\mathbb{E} \left[\max_{i \in \{1, \dots, n\}} \{V_i + \varepsilon_i\} \right] = \sigma \ln \left(\sum_{i=1}^n \exp \frac{V_i}{\sigma} \right), \quad (\text{A.3})$$

$$\Pr \left\{ \arg \max_{i \in \{1, \dots, n\}} \{V_i + \varepsilon_i\} = j \right\} = \frac{\exp \frac{V_j}{\sigma}}{\sum_{i=1}^n \exp \frac{V_i}{\sigma}}. \quad (\text{A.4})$$

Proof. See Appendix B.3. □

B Proofs

This section contains the proofs for the lemmas and propositions in the paper.

B.1 Proof of Lemma 2

Proof. Let $X_1 = V_1 + \varepsilon_1$ and $X_2 = V_2 + \varepsilon_2$, where $\varepsilon_1, \varepsilon_2$ are independent $\text{Gumbel}(-\eta\gamma_E, \eta)$ random variables. I derive the expected value $\mathbb{E}[\max\{X_1, X_2\}]$. The CDF of a $\text{Gumbel}(\mu, \sigma)$ random variable is $F(x; \mu, \sigma) = \exp(-\exp(-(x - \mu)/\sigma))$. For the parametrization $\text{Gumbel}(-\eta\gamma_E, \eta)$:

$$F_\varepsilon(x) = \exp \left(-\exp \left(-\frac{x}{\eta} - \gamma_E \right) \right).$$

The CDF of $X_i = V_i + \varepsilon_i$ is obtained by translation:

$$F_{X_i}(x) = F_\varepsilon(x - V_i) = \exp \left(-\exp \left(-\frac{x - V_i}{\eta} - \gamma_E \right) \right).$$

The CDF of $\max\{X_1, X_2\}$ is:

$$\begin{aligned}
F_{\max}(x) &= \Pr\{\max\{X_1, X_2\} \leq x\} = \Pr\{X_1 \leq x, X_2 \leq x\} \\
&= F_{X_1}(x) \cdot F_{X_2}(x) \\
&= \exp\left(-\exp\left(-\frac{x - V_1}{\eta} - \gamma_E\right)\right) \cdot \exp\left(-\exp\left(-\frac{x - V_2}{\eta} - \gamma_E\right)\right) \\
&= \exp\left(-\exp\left(-\frac{x - V_1}{\eta} - \gamma_E\right) - \exp\left(-\frac{x - V_2}{\eta} - \gamma_E\right)\right) \\
&= \exp\left(-e^{-\gamma_E} \left(\exp\left(-\frac{x - V_1}{\eta}\right) + \exp\left(-\frac{x - V_2}{\eta}\right)\right)\right).
\end{aligned}$$

Let $\tilde{\mu} = \eta \ln\left(\exp \frac{V_1}{\eta} + \exp \frac{V_2}{\eta}\right)$. I can rewrite:

$$\begin{aligned}
&\exp\left(-\frac{x - V_1}{\eta}\right) + \exp\left(-\frac{x - V_2}{\eta}\right) \\
&= \exp\left(-\frac{x}{\eta}\right) \left(\exp \frac{V_1}{\eta} + \exp \frac{V_2}{\eta}\right) \\
&= \exp\left(-\frac{x}{\eta}\right) \exp \frac{\tilde{\mu}}{\eta} = \exp\left(-\frac{x - \tilde{\mu}}{\eta}\right).
\end{aligned}$$

Therefore,

$$F_{\max}(x) = \exp\left(-e^{-\gamma_E} \exp\left(-\frac{x - \tilde{\mu}}{\eta}\right)\right) = \exp\left(-\exp\left(-\frac{x - \tilde{\mu}}{\eta} - \gamma_E\right)\right).$$

Thus $\max\{X_1, X_2\}$ follows a Gumbel($\tilde{\mu} - \eta\gamma_E, \eta$) distribution. Since the expectation of a Gumbel(μ, σ) random variable is $\mu + \sigma\gamma_E$:

$$\begin{aligned}
\mathbb{E}[\max\{X_1, X_2\}] &= (\tilde{\mu} - \eta\gamma_E) + \eta\gamma_E = \tilde{\mu} \\
&= \eta \ln\left(\exp \frac{V_1}{\eta} + \exp \frac{V_2}{\eta}\right).
\end{aligned}$$

□

B.2 Proof of Lemma 3

Proof. Let $X_1 = V_1 + \varepsilon_1$ and $X_2 = V_2 + \varepsilon_2$ as before. I compute $\Pr\{X_1 > X_2\} = \Pr\{\varepsilon_1 - \varepsilon_2 > V_2 - V_1\}$. I first establish that for independent Gumbel(μ, σ) random variables ε_1 and ε_2 , the difference $\varepsilon_1 - \varepsilon_2$ follows a logistic distribution. The CDF of ε_i is $F(x) = \exp(-\exp(-(x -$

$\mu)/\sigma))$. For $Z = \varepsilon_1 - \varepsilon_2$, I compute:

$$\begin{aligned} F_Z(z) &= \Pr\{\varepsilon_1 - \varepsilon_2 \leq z\} = \Pr\{\varepsilon_1 \leq z + \varepsilon_2\} \\ &= \int_{-\infty}^{\infty} F_{\varepsilon_1}(z + u) f_{\varepsilon_2}(u) du \end{aligned}$$

where $f_{\varepsilon_2}(u) = \sigma^{-1} \exp(-(u - \mu)/\sigma) \exp(-\exp(-(u - \mu)/\sigma))$ is the PDF of ε_2 . Using the substitution $v = \exp(-(u - \mu)/\sigma)$ and the fact that both variables have the same parameters, the integral evaluates to:

$$F_Z(z) = \frac{1}{1 + \exp(-z/\sigma)}$$

This is the CDF of a logistic distribution with location 0 and scale σ . With $\sigma = \eta$ and the parametrization:

$$\begin{aligned} \Pr\{X_1 > X_2\} &= \Pr\{\varepsilon_1 - \varepsilon_2 > V_2 - V_1\} \\ &= 1 - F_Z(V_2 - V_1; 0, \eta) \\ &= 1 - \frac{1}{1 + \exp(-(V_2 - V_1)/\eta)} \\ &= 1 - \frac{1}{1 + \exp((V_1 - V_2)/\eta)} \\ &= \frac{1 + \exp((V_1 - V_2)/\eta) - 1}{1 + \exp((V_1 - V_2)/\eta)} \\ &= \frac{\exp((V_1 - V_2)/\eta)}{1 + \exp((V_1 - V_2)/\eta)}. \end{aligned}$$

Multiplying by $\exp(V_2/\eta)$ yields:

$$\Pr\{X_1 > X_2\} = \frac{\exp \frac{V_1}{\eta}}{\exp \frac{V_1}{\eta} + \exp \frac{V_2}{\eta}}.$$

□

B.3 Proof of Lemma 4

Proof. The proof proceeds by induction on n .

Base case: For $n = 2$, the results follow from Lemmas 2 and 3.

Inductive step: Assume the results hold for $n \geq 2$. For $n + 1$ alternatives, let $Y_n = \max_{1 \leq i \leq n} \{V_i + \varepsilon_i\}$ and $X_{n+1} = V_{n+1} + \varepsilon_{n+1}$. Then:

$$\max_{1 \leq i \leq n} \{V_i + \varepsilon_i\} = \max\{Y_n, X_{n+1}\}.$$

By the inductive hypothesis, Y_n is distributed as $\text{Gumbel}(\sigma \ln(\sum_{i=1}^n \exp(V_i/\sigma)) - \sigma \gamma_E, \sigma)$.

Since X_{n+1} is Gumbel($-\sigma\gamma_E, \sigma$), applying Lemma 2:

$$\begin{aligned}
& \mathbb{E} \left[\max_{1 \leq i \leq n+1} \{V_i + \varepsilon_i\} \right] \\
&= \sigma \ln \left(\exp \frac{\sigma \ln \left(\sum_{i=1}^n \exp \frac{V_i}{\sigma} \right)}{\sigma} + \exp \frac{V_{n+1}}{\sigma} \right) \\
&= \sigma \ln \left(\sum_{i=1}^n \exp \frac{V_i}{\sigma} + \exp \frac{V_{n+1}}{\sigma} \right) \\
&= \sigma \ln \left(\sum_{i=1}^{n+1} \exp \frac{V_i}{\sigma} \right).
\end{aligned}$$

For the choice probabilities, Lemma 3 yields:

$$\begin{aligned}
& \Pr \left\{ \arg \max_{1 \leq i \leq n+1} \{V_i + \varepsilon_i\} = j \right\} \\
&= \begin{cases} \Pr \{Y_n > X_{n+1}\} \cdot \Pr \{\arg \max_{1 \leq i \leq n} \{V_i + \varepsilon_i\} = j\} & \text{if } j \leq n \\ \Pr \{X_{n+1} > Y_n\} & \text{if } j = n+1 \end{cases}
\end{aligned}$$

By the inductive hypothesis and preceding lemmas:

$$\Pr \left\{ \arg \max_{1 \leq i \leq n+1} \{V_i + \varepsilon_i\} = j \right\} = \frac{\exp \frac{V_j}{\sigma}}{\sum_{i=1}^{n+1} \exp \frac{V_i}{\sigma}}.$$

□

B.4 Proof of Proposition 1

Proof. Spaces and operator. Let $\mathcal{S} = \mathcal{Y} \times \mathcal{B}$ (compact) and $\mathcal{C}(\mathcal{S})$ be bounded continuous functions on $\mathcal{S}$ with the sup-norm $\|\cdot\|_\infty$. Define the product space $\mathbf{X} = \mathcal{C}(\mathcal{S}) \times \mathcal{C}(\mathcal{S})$ and, for a weight $\lambda > 0$, the norm

$$\|(V, q)\|_\lambda = \max\{\|V\|_\infty, \lambda \|q\|_\infty\}.$$

$(\mathbf{X}, \|\cdot\|_\lambda)$ is complete. Define $\mathcal{T} : \mathbf{X} \rightarrow \mathbf{X}$ by

$$\mathcal{T}(V, q) = (J(V, q), T(V, q)),$$

where J is the Bellman operator (log-sum-exp/taste shocks) and T is the pricing operator (18) (default probability given by the logistic rule).

Step 1: $\mathcal{T}$ maps $\mathbf{X}$ into itself. By compactness of (y, B) and the discrete (or compact) choice set for B' , continuity of the primitives, and the log-sum-exp aggregator with $\rho > 0$,

$J(V, q)$ is bounded and continuous whenever (V, q) is. For T , note that for any (y, B') ,

$$0 \leq q(y, B') \leq \frac{1}{1+r} (\kappa + (1-\delta) \|q\|_\infty),$$

hence taking suprema gives the uniform bound

$$\|q\|_\infty \leq \frac{\kappa}{r} =: \bar{q}. \quad (\text{B.5})$$

Thus $T(V, q)$ is bounded and continuous, so $\mathcal{T}(\mathbf{X}) \subseteq \mathbf{X}$.

Step 2: Lipschitz bounds (decoupled). I record the following constants (finite by compactness and (B.5)):

(i) J with respect to V . The continuation part of J is multiplied by β and the derivative of log-sum-exp is a probability in $[0, 1]$, so

$$\|J(V_1, q) - J(V_2, q)\|_\infty \leq \beta \|V_1 - V_2\|_\infty.$$

(ii) J with respect to q . Consumption is $c(B') = y - \kappa B + [B' - (1-\delta)B] q(y, B')$. Hence for some bound

$$M_B := \sup_{(y, B) \in \mathcal{S}} \sup_{B' \in \mathcal{B}} |B' - (1-\delta)B| < \infty,$$

and with $\bar{u}' := \sup u'(c)$ on the feasible set, the envelope theorem and the $[0, 1]$ -valued logit weights imply

$$\|J(V, q_1) - J(V, q_2)\|_\infty \leq L_{Jq} \|q_1 - q_2\|_\infty, \quad L_{Jq} \leq \bar{u}' M_B. \quad (\text{B.6})$$

(iii) T with respect to q . In (18), q enters only via the resale term $(1-\delta)\mathbb{E}q$ with one-period discount $1/(1+r)$, hence

$$\|T(V, q_1) - T(V, q_2)\|_\infty \leq m_q \|q_1 - q_2\|_\infty, \quad m_q := \frac{1-\delta}{1+r} < 1. \quad (\text{B.7})$$

(iv) T with respect to V . Here $P_\theta(y', B') = \mathbb{E}_{m \sim \mathcal{N}(0, (\theta\sigma_m)^2)} [L(-\Delta V(y', B', m)/\eta)]$ is the m -integrated logit at the *true* scale η , with $L'(z) = L(z)(1-L(z)) \leq \frac{1}{4}$. Let $\Pi := \kappa + (1-\delta)\bar{q}$ bound the one-step payoff in (18). The mapping $V \mapsto \Delta V(\cdot, m)$ is, for each m , linear in V through continuation values and default recursion; its operator norm is bounded uniformly in m by

$$\|\Delta V(V_1) - \Delta V(V_2)\|_\infty \leq C_\Delta \|V_1 - V_2\|_\infty, \quad C_\Delta \leq \frac{\beta}{1 - \beta(1-\gamma)}.$$

Differentiating P_θ in a uniform shift of ΔV and using $L' \leq \frac{1}{4}$, the exact modulus is

$\frac{1}{\eta} \mathbb{E}_m[L'(-\Delta V/\eta)] \leq \frac{1}{4\eta}$ for every θ . Therefore

$$\|T(V_1, q) - T(V_2, q)\|_\infty \leq \frac{1}{1+r} \cdot \frac{1}{4\eta} \cdot \Pi \cdot C_\Delta \|V_1 - V_2\|_\infty =: L_{TV} \|V_1 - V_2\|_\infty, \quad (\text{B.8})$$

which is the rigorous Lipschitz bound at the true $\eta > 0$ and is *independent of θ* , so the slope condition (17) is unaffected by PRO. To leading order in η this exact modulus has the sharper closed form obtained from the threshold displacement of Lemma 1: a uniform perturbation of V^R by ε shifts the default threshold by $|\varepsilon|/(\partial_m V^R)|_{m^*}$ (implicit-function theorem on $\Delta V = 0$), so by (15) the sensitivity of $P_\theta = \Phi(m^*/(\theta\sigma_m))$ is

$$\frac{\phi(m^*(y', B')/(\theta\sigma_m))}{\theta\sigma_m(\partial_m V^R)|_{m^*}} \leq L_P := \frac{\phi(0)}{\theta\sigma_m \underline{u}'} < \infty, \quad \underline{u}' := \inf u' > 0, \quad (\text{B.9})$$

where finiteness uses $\phi \leq \phi(0) = (2\pi)^{-1/2}$ and $\partial_m V^R \geq \underline{u}' > 0$ on the compact, strictly-interior consumption set. The dominating bound $L_P = \phi(0)/(\theta\sigma_m \underline{u}')$ is *strictly decreasing in θ* , so PRO does not loosen the contraction (the per-state sensitivity is not monotone in θ far from the pivot, but it is dominated by this decreasing envelope). This dispersion-form modulus is the analogue of the prior slope bound rather than numerically equal to it: the temperature bound $1/(4\eta)$ diverges as $\eta \downarrow 0$, whereas L_P is $O(1)$ in η , so for small η the dispersion model is no harder (and generically easier) to make a contraction.

Step 3: Joint contraction under a weighted norm. For any $(V_1, q_1), (V_2, q_2)$, combine the bounds above to get

$$\|J(V_1, q_1) - J(V_2, q_2)\|_\infty \leq \beta \|V_1 - V_2\|_\infty + L_{Jq} \|q_1 - q_2\|_\infty,$$

$$\|T(V_1, q_1) - T(V_2, q_2)\|_\infty \leq L_{TV} \|V_1 - V_2\|_\infty + m_q \|q_1 - q_2\|_\infty.$$

Multiplying the second inequality by λ and taking the max of the two gives

$$\|\mathcal{T}(V_1, q_1) - \mathcal{T}(V_2, q_2)\|_\lambda \leq \alpha(\lambda) \|(V_1, q_1) - (V_2, q_2)\|_\lambda,$$

where

$$\alpha(\lambda) = \max\left\{\beta + \frac{L_{Jq}}{\lambda}, m_q + \lambda L_{TV}\right\}.$$

If there exists $\lambda > 0$ such that

$$\beta + \frac{L_{Jq}}{\lambda} < 1 \quad \text{and} \quad m_q + \lambda L_{TV} < 1,$$

then $\alpha(\lambda) < 1$ and $\mathcal{T}$ is a contraction. These inequalities can be satisfied if and only if

$$L_{Jq} L_{TV} < (1 - \beta)(1 - m_q),$$

which is exactly the slope condition (17). In that case one admissible choice is

$$\lambda^* = \frac{(\beta - m_q) + \sqrt{(\beta - m_q)^2 + 4L_{Jq}L_{TV}}}{2L_{TV}},$$

for which $\alpha(\lambda^*) < 1$.

Step 4: Conclusion. In the weighted complete normed space $(\mathbf{X}, \|\cdot\|_{\lambda^*})$, $\mathcal{T}$ is a contraction mapping. By the Contraction Mapping Theorem, it has a unique fixed point (V^*, q^*) , which by construction satisfies the model's recursive equilibrium conditions. Therefore, the Recursive Markov Perfect Equilibrium exists and is unique. $\square$

B.5 Proof of Proposition 2

Proof. The proof relies on a standard monotone-operator result showing that pointwise orderings of operators are preserved by their fixed points.

Lemma 5. Let $(X, \|\cdot\|)$ be a Banach space and $T_1, T_2 : X \rightarrow X$ satisfy:

1. **Monotonicity:** $f \geq g \Rightarrow T_i(f) \geq T_i(g)$ for $i \in \{1, 2\}$.
2. **Discounting:** $\|T_i(f + c\mathbf{1}) - T_i(f)\| \leq \beta c$ for some $\beta \in (0, 1)$ and constant function $\mathbf{1}$.

If $T_1(f) \geq T_2(f)$ pointwise for all $f \in X$, then their unique fixed points satisfy $f_1^* \geq f_2^*$ where $T_i(f_i^*) = f_i^*$.

Proof of Lemma 5. Let $f_1 = T_1(f_1)$ and $f_2 = T_2(f_2)$. Start from $f_0 = f_2$ and iterate $f_{n+1} = T_1(f_n)$. By contraction, $f_n \rightarrow f_1$. Using $T_1(f_2) \geq T_2(f_2) = f_2$ and monotonicity of T_1 , the sequence is nondecreasing and bounded below by f_2 ; taking limits yields $f_1 \geq f_2$. $\square$

Applied to the pricing operators $T_i = \mathcal{T}_{\theta_i}$, the payoff term $R_i(y', B') \equiv \kappa + (1 - \delta) \mathbb{E}_{B''|y', B'}[q(y', B'')]$ multiplying $(1 - P_i)$ in (18) satisfies $R_i > 0$ for all $q \geq 0$; since the operator ordering in q is uniform, the sign of $(T_\theta q - T_1 q)$ is governed entirely by the probability ordering $P_1 - P_\theta$, which is what Step 1 establishes.

Let $q_i(B', y)$, $i \in \{1, \theta\}$ with $\theta_1 = 1$ and $\theta > 1$, be the fixed point of the pricing operator in (18), with perceived default probabilities given by (19).

Step 1: Probability ordering (correct sign). Fix (y, B') and let $m^* \equiv m^*(y, B')$ be the default threshold of Lemma 1, assumed finite and nonzero. To leading order in η , (15) gives

$$P_\theta(y, B') = \Phi\left(\frac{m^*}{\theta \sigma_m}\right), \quad P_1(y, B') = \Phi\left(\frac{m^*}{\sigma_m}\right).$$

Because $\theta > 1$ shrinks $|m^*/(\theta\sigma_m)|$ toward 0 while preserving its sign, and Φ is strictly increasing with $\Phi(0) = \frac{1}{2}$,

$$P_\theta(y, B') - P_1(y, B') = \Phi\left(\frac{m^*}{\theta\sigma_m}\right) - \Phi\left(\frac{m^*}{\sigma_m}\right), \quad \text{sign}(P_\theta - P_1) = -\text{sign } m^*. \quad (\text{B.10})$$

By Lemma 1, $m^*(y, B') < 0$ for $B' < \hat{B}(y)$ and $m^*(y, B') > 0$ for $B' > \hat{B}(y)$, where $\hat{B}(y)$ is the sovereign-indifference debt level. Hence

$$m^*(y, B') < 0 \Rightarrow P_\theta(y, B') > P_1(y, B'), \quad (\text{B.11})$$

$$m^*(y, B') > 0 \Rightarrow P_\theta(y, B') < P_1(y, B'). \quad (\text{B.12})$$

with equality at $m^* = 0$, and $P_\theta \rightarrow \frac{1}{2}$ as $\theta \rightarrow \infty$. Intuitively: in perceived-safe states ($m^* < 0$, repayment dominates at the median shock) PRO lenders assign a *higher* default probability (the “PRO premium”); in perceived-risky states ($m^* > 0$) they assign a *lower* default probability (the “softening of doom”). This level ordering is rigorous *globally* in m^* ; it does not rely on any slope comparison.

Given the strictly positive payoff term in (18), I have

$$\text{sign}((T_\theta q - T_1 q)(B', y)) = \text{sign}(\mathbb{E}_{y'|y}[P_1(y', B') - P_\theta(y', B')]). \quad (\text{B.13})$$

Step 2: Safe region ($m^* < 0$). Define the safe set at income y by

$$\underline{B}(y) \equiv \inf_{y' \in \mathcal{Y}} \hat{B}(y') = \sup\{b : m^*(y', b) < 0 \ \forall y' \in \mathcal{Y}\},$$

where $\hat{B}(y')$ is the sovereign-indifference level of Lemma 1. For any $B' < \underline{B}(y)$ we have $m^*(y', B') < 0$ for all y' , so (B.11) gives $P_\theta > P_1$ for all y' ; hence

$$\mathbb{E}_{y'|y}[P_1 - P_\theta] < 0 \Rightarrow (T_\theta q)(B', y) < (T_1 q)(B', y) \Rightarrow q_\theta(B', y) < q_1(B', y).$$

Step 3: Risky region ($m^* > 0$). Define the risky set at income y by

$$\overline{B}(y) \equiv \sup_{y' \in \mathcal{Y}} \hat{B}(y') = \inf\{b : m^*(y', b) > 0 \ \forall y' \in \mathcal{Y}\}.$$

For any $B' > \overline{B}(y)$ we have $m^*(y', B') > 0$ for all y' , so (B.12) yields $P_\theta < P_1$ for all y' ; thus

$$\mathbb{E}_{y'|y}[P_1 - P_\theta] > 0 \Rightarrow (T_\theta q)(B', y) > (T_1 q)(B', y) \Rightarrow q_\theta(B', y) > q_1(B', y).$$

Step 4: Pivot existence. Let $\Delta q(B', y) \equiv q_\theta(B', y) - q_1(B', y)$. By continuity of q_i in B' (Proposition 1), $\Delta q(\cdot, y)$ is continuous. Steps 2–3 imply

$$\Delta q(B', y) < 0 \quad \text{for } B' < \underline{B}(y), \quad \Delta q(B', y) > 0 \quad \text{for } B' > \bar{B}(y).$$

By the Intermediate Value Theorem there exists at least one $B^*(y) \in [\underline{B}(y), \bar{B}(y)]$ such that

$$\Delta q(B^*(y), y) = 0,$$

and the local sign pattern is

$$\text{sign}(\Delta q(B', y)) = \begin{cases} -, & B' < B^*(y), \\ 0, & B' = B^*(y), \\ +, & B' > B^*(y). \end{cases}$$

This establishes the pivoting of q_θ around q_1 at $B^*(y)$. □

B.6 Proof of Proposition 3

Proof. Define $F(B', y) \equiv q_\theta(B', y) - q_1(B', y)$ and let $B^*(y)$ be implicitly defined by $F(B^*(y), y) = 0$. I verify the conditions of the Implicit Function Theorem at any $(B^*(y), y)$ where the crossing is interior.

Regularity. (i) Smoothness. Under the primitives assumed in the main text (CRRA utility, log-sum-exp smoothing in the sovereign's choices, compact state grids, and the pricing operator with strictly positive payoff term), the pricing operator is C^1 in (B', y) . Hence each fixed-point price function $q_i(B', y)$ is C^1 in both arguments.¹²

(ii) Non-degeneracy. I show $F_{B'}(B^*(y), y) \neq 0$ and in fact $F_{B'}(B^*(y), y) > 0$. Recall the lenders' perceived default probability in its leading-order form (Lemma 1),

$$P_i(y', B') = \Phi\left(\frac{m^*(y', B')}{\theta_i \sigma_m}\right), \quad \Delta V_i \equiv V_i^R - V_i^D, \quad m^*(y', B') \text{ solving } \Delta V = 0.$$

Since $\Phi' = \phi > 0$ and $\partial_{B'} m^* > 0$ by (16) (higher future debt raises the shock needed for indifference),

$$\frac{\partial P_i(y', B')}{\partial B'} = \frac{1}{\theta_i \sigma_m} \phi\left(\frac{m^*(y', B')}{\theta_i \sigma_m}\right) \frac{\partial m^*}{\partial B'} > 0. \quad (\text{B.14})$$

The non-degeneracy is evaluated at the pivot $B^*(y)$, which the Intermediate-Value construction of Proposition 2 brackets at the sign change $m^*(\cdot, B^*(y)) = 0$ in expectation; there

¹²This follows from differentiability of the Bellman/pricing operators and standard fixed-point differentiation arguments on compact spaces.

the slope ratio is exactly $1/\theta$, so

$$0 < \frac{\partial P_\theta}{\partial B'} \Big|_{m^*=0} = \frac{1}{\theta} \frac{\partial P_1}{\partial B'} \Big|_{m^*=0} < \frac{\partial P_1}{\partial B'} \Big|_{m^*=0}. \quad (\text{B.15})$$

(Away from the pivot the θ -ordering of slopes need not hold, since $\phi(m^*/(\theta\sigma_m))$ can rise faster than $\theta\sigma_m$; only the value at $m^* = 0$ is used below.)

Write the pricing operator as

$$(T_i q)(B', y) = \frac{1}{1+r} \mathbb{E}_{y'|y} \left[(1 - P_i(y', B')) R_i(y', B') \right], \quad R_i(y', B') \equiv \kappa + (1-\delta) \mathbb{E}_{B''|y', B'} [q_i(y', B'')].$$

Since $R_i > 0$, the operator is pointwise *decreasing* in P_i . Hence, as B' increases and P_i rises by (B.14), the price $q_i(B', y)$ strictly *decreases* in B' . Furthermore, because $\partial P_\theta / \partial B'$ is smaller than $\partial P_1 / \partial B'$ by (B.15), the magnitude of the induced price decline is smaller under θ than under 1. Therefore,

$$\frac{\partial q_\theta}{\partial B'}(B^*(y), y) > \frac{\partial q_1}{\partial B'}(B^*(y), y), \quad \text{so} \quad F_{B'}(B^*(y), y) = \frac{\partial q_\theta}{\partial B'} - \frac{\partial q_1}{\partial B'} > 0. \quad (\text{B.16})$$

Sign of the numerator. Income persistence implies that higher y raises expected future endowments, increases ΔV_i , and lowers the indifference shock m^* (formally $\partial_y m^* = -\partial_y \Delta V / \partial_m \Delta V < 0$, as in the auxiliary identity below). Hence higher y lowers $P_i = \Phi(m^*/(\theta_i \sigma_m))$ and raises prices, $\partial q_i / \partial y > 0$. Evaluated at the pivot ($m^* \approx 0$),

$$\frac{\partial P_i}{\partial y} \Big|_{m^*=0} = \frac{\phi(0)}{\theta_i \sigma_m} \partial_y m^* < 0,$$

so the income sensitivity carries the same exact $1/\theta$ factor as the slope in (B.15): it is *dampened* under PRO (the dispersion form, with η held fixed, replaces the old temperature heuristic). Propagating through the pricing operator,

$$0 < \frac{\partial q_\theta}{\partial y}(B^*(y), y) < \frac{\partial q_1}{\partial y}(B^*(y), y), \quad \Rightarrow \quad F_y(B^*(y), y) = \frac{\partial q_\theta}{\partial y} - \frac{\partial q_1}{\partial y} < 0. \quad (\text{B.17})$$

Companion identity for the sovereign-indifference locus. The price pivot $B^*(y)$ above is defined by the *price wedge* $F(B^*(y), y) = q_\theta - q_1 = 0$, equivalently $\mathbb{E}_{y'|y} [P_\theta(y', B') - P_1(y', B')] = 0$, and is the object the Implicit Function Theorem is applied to below. A closely related, θ -independent threshold is the sovereign-indifference locus $\hat{B}(y)$ of Lemma 1, defined pointwise by $m^*(y, \hat{B}(y)) = 0$. Differentiating that identity,

$$\frac{d\hat{B}}{dy} = - \frac{\partial_y m^*(y, \hat{B}(y))}{\partial_{B'} m^*(y, \hat{B}(y))} > 0,$$

since $\partial_{B'} m^* > 0$ by (16) and $\partial_y m^* = -\partial_y \Delta V / \partial_m \Delta V < 0$, using $\partial_m \Delta V > 0$ and $\partial_y \Delta V = \partial_y V^R - \partial_y V^D > 0$ (a higher endowment raises the repayment value faster than the default value, the standard condition generating an upward-sloping default frontier). This monotone sovereign frontier underlies, but is not identical to, the price pivot.

Conclusion. By the Implicit Function Theorem applied to the price wedge F ,

$$\frac{dB^*}{dy} = -\frac{F_y(B^*(y), y)}{F_{B'}(B^*(y), y)} = -\frac{(-)}{(+)} > 0,$$

so the pivot threshold $B^*(y)$ is (strictly) increasing in y . □

B.7 Proof of Corollary 1

Proof. From $s_i(B', y) = \kappa / q_i(B', y) - \delta - r$ and $q_\theta, q_1 > 0$,

$$\Delta s(B', y) = \kappa \left(\frac{1}{q_\theta} - \frac{1}{q_1} \right) = -\frac{\kappa \Delta q(B', y)}{q_\theta(B', y) q_1(B', y)},$$

so $\text{sign}(\Delta s) = -\text{sign}(\Delta q)$ since $\kappa > 0$. Substituting the price-pivot pattern of Proposition 2 ($\Delta q \leq 0$ as $B' \leq B^*(y)$) flips the sign, yielding the stated spread pivot at the same threshold $B^*(y)$. □

B.8 Proof of Proposition 4

Proof. For a fixed endowment level y , define the repayment-default gap

$$G_i(B; y) \equiv V_i^R(y, B) - V^D(y), \quad i \in \{1, \theta\}.$$

By standard arguments (concavity of u , budget feasibility, and the fact that higher current debt tightens the budget set), $G_i(B; y)$ is strictly decreasing in B . The default threshold is the unique root $B_{D,i}^*(y)$ of $G_i(\cdot; y) = 0$.

Assume toward a contradiction that $B_{D,\theta}^*(y) \leq B_{D,1}^*(y)$. Monotonicity of $G_1(\cdot; y)$ then implies

$$G_1(B_{D,\theta}^*(y); y) = V_1^R(y, B_{D,\theta}^*(y)) - V^D(y) \geq V_1^R(y, B_{D,1}^*(y)) - V^D(y) = 0. \quad (\text{B.18})$$

Next, invoke Proposition 2 (“price pivot”): for the given y there exists $B^*(y)$ such that

$$q_\theta(y, B') \geq q_1(y, B') \quad \text{iff} \quad B' \geq B^*(y),$$

with strict inequality on a set of B' of positive measure. Near the brink of default (i.e., at $B = B_{D,\theta}^*(y)$), the sovereign’s optimal issuance under repayment places (by the Euler condition

and rollover incentives in long-term debt models) essentially all probability mass on future debt levels $B' \geq B^*(y)$.¹³ Hence, evaluating the choice-specific value

$$W_i(y, B, B') = u(y - \kappa B + [B' - (1 - \delta)B] q_i(y, B')) + \beta \mathbb{E}_{y'|y} [V_i(y', B')],$$

I have for all such relevant $B' \geq B^*(y)$,

$$q_\theta(y, B') \geq q_1(y, B') \implies u(\cdot \text{ with } q_\theta) \geq u(\cdot \text{ with } q_1),$$

and (by monotonicity of the Bellman operator with respect to the price schedule in those states) $\mathbb{E}[V_\theta(y', B')] \geq \mathbb{E}[V_1(y', B')]$. Therefore $W_\theta(y, B, B') \geq W_1(y, B, B')$ on the support of the (repayment) choice distribution, with strict inequality on a set of positive probability.

Aggregating with the log-sum-exp representation of the repayment value,

$$V_i^R(y, B) = \rho \log \left(\sum_{B'} \exp \{W_i(y, B, B') / \rho\} \right),$$

which is strictly increasing in each $W_i(y, B, B')$, I obtain

$$V_\theta^R(y, B_{D,\theta}^*(y)) > V_1^R(y, B_{D,\theta}^*(y)). \quad (\text{B.19})$$

Combining (B.18) and (B.19) yields

$$V_\theta^R(y, B_{D,\theta}^*(y)) > V^D(y),$$

which contradicts the definition of $B_{D,\theta}^*(y)$ as the point where $V_\theta^R = V^D$. Hence the assumed ordering is false, and I must have

$$B_{D,\theta}^*(y) > B_{D,1}^*(y).$$

□

B.9 Proof of Proposition 5

Proof. Setup. Fix a state (y, B) . For each lender type $i \in \{1, \theta\}$, let

$$B'_i(y, B) \in \arg \max_{B' \in \mathcal{B}} W(y, B, B'; q_i), \quad W(y, B, B'; q_i) \equiv u(c_i(B')) + \beta \mathbb{E}_{y'|y} [V_i(y', B')],$$

¹³Intuitively, when current debt is high, the sovereign rolls over rather than deleverages sharply; see, e.g., the rollover logic in long-term debt environments. Formally, with small taste-shock dispersion ρ , the logit policy concentrates around the deterministic maximizer, which lies weakly to the right of $B^*(y)$ at high current B .

with

$$c_i(B') = y - \kappa B + [B' - (1 - \delta)B] q_i(y, B').$$

By strict concavity of u and standard properties of the long-maturity debt problem, $B'_i(y, B)$ is unique.¹⁴ Define the marginal objective

$$\begin{aligned} g_i(B') \equiv \frac{\partial W(y, B, B'; q_i)}{\partial B'} &= u'(c_i(B')) \left[q_i(y, B') + (B' - (1 - \delta)B) q_{i, B'}(y, B') \right] \\ &\quad + \beta \frac{\partial}{\partial B'} \mathbb{E}_{y'|y} [V_i(y', B')]. \end{aligned}$$

FOC and uniqueness give

$$g_i(B'_i(y, B)) = 0, \quad i \in \{1, \theta\}. \quad (\text{B.20})$$

Region of comparison. By Proposition 2, there exists $B^*(y)$ such that

$$B' \leq B^*(y) \implies q_\theta(y, B') < q_1(y, B'), \quad B' \geq B^*(y) \implies q_\theta(y, B') > q_1(y, B'). \quad (\text{B.21})$$

I prove the claim on the empirically relevant *borrowing* side, i.e. for states with the baseline choice on (or to the left of) the pivot, where the PRO premium raises the cost of debt:

$$B'_1(y, B) \leq B^*(y). \quad (\text{B.22})$$

(The complementary case is analogous and yields the weak inequality; see the remark at the end.)

Two auxiliary inequalities.

(A1) Since $q_{i, B'}(y, B') \leq 0$ in equilibrium and $(B' - (1 - \delta)B) \geq 0$ whenever the sovereign issues (rolls over) debt,¹⁵

$$q_i(y, B') + (B' - (1 - \delta)B) q_{i, B'}(y, B') \leq q_i(y, B'). \quad (\text{B.23})$$

(A2) (Continuation marginal ordering.) Increasing next-period debt lowers the continuation value and it does so *more* under PRO lenders:

$$\frac{\partial}{\partial B'} \mathbb{E}_{y'|y} [V_\theta(y', B')] \leq \frac{\partial}{\partial B'} \mathbb{E}_{y'|y} [V_1(y', B')] \leq 0, \quad (\text{B.24})$$

¹⁴The usual argument is that the choice-specific value is strictly concave in B' because (i) u is strictly concave and $B' \mapsto c_i(B')$ is affine plus a (weakly) concave price-feedback term, and (ii) the continuation term $B' \mapsto \mathbb{E}[V_i(y', B')]$ is concave by the concavity of the Bellman operator.

¹⁵If net issuance is negative at the optimum, the conclusion below is even stronger because the self-impact term becomes favorable.

with strict inequality on a set of positive measure when $B' \leq B^*(y)$.¹⁶

Key evaluation at the baseline optimum. Evaluate g_θ at $B'_1 \equiv B'_1(y, B)$. Using (B.23) and (B.24),

$$\begin{aligned} g_\theta(B'_1) &= u'(c_\theta(B'_1)) \left[q_\theta(y, B'_1) + (B'_1 - (1 - \delta)B) q_{\theta, B'}(y, B'_1) \right] + \beta \frac{\partial}{\partial B'} \mathbb{E}[V_\theta(y', B')] \Big|_{B'=B'_1} \\ &\leq u'(c_\theta(B'_1)) q_\theta(y, B'_1) + \beta \frac{\partial}{\partial B'} \mathbb{E}[V_1(y', B')] \Big|_{B'=B'_1}. \end{aligned} \quad (\text{B.25})$$

By the baseline FOC (B.20) for $i = 1$,

$$0 = g_1(B'_1) = u'(c_1(B'_1)) \left[q_1 + (B'_1 - (1 - \delta)B) q_{1, B'} \right] + \beta \frac{\partial}{\partial B'} \mathbb{E}[V_1(y', B')] \Big|_{B'=B'_1},$$

and therefore, again using $q_{1, B'} \leq 0$,

$$\beta \frac{\partial}{\partial B'} \mathbb{E}[V_1(y', B')] \Big|_{B'=B'_1} = -u'(c_1(B'_1)) \left[q_1 + (B'_1 - (1 - \delta)B) q_{1, B'} \right] \leq -u'(c_1(B'_1)) q_1(y, B'_1). \quad (\text{B.26})$$

Substituting (B.26) into (B.25) yields

$$g_\theta(B'_1) \leq u'(c_\theta(B'_1)) q_\theta(y, B'_1) - u'(c_1(B'_1)) q_1(y, B'_1). \quad (\text{B.27})$$

Sign of the bound on the borrowing side. Under (B.22), (B.21) gives $q_\theta(y, B'_1) < q_1(y, B'_1)$. Moreover, $c_\theta(B'_1) < c_1(B'_1)$ because the only difference in the current-period budget is the price multiplying the same issuance $[B'_1 - (1 - \delta)B]$, hence by concavity $u'(c_\theta(B'_1)) > u'(c_1(B'_1))$. Combining these two facts in (B.27) implies

$$g_\theta(B'_1) < 0.$$

Since $B' \rightarrow W(y, B, B'; q_\theta)$ is strictly concave, g_θ is strictly decreasing, so $g_\theta(B'_1) < 0$ forces the θ -optimizer to lie strictly to the left of B'_1 :

$$B'_\theta(y, B) < B'_1(y, B).$$

From modes to distributions. Under the logit (Gumbel) regularization of the *borrowing* choice, the choice probabilities satisfy $\Pr_i(B' = b \mid y, B) \propto \exp\{W(y, B, b; q_i)/\rho\}$, governed by the borrowing-taste scale ρ (not the default-taste scale η); they are unimodal and

¹⁶This follows from monotonicity of the Bellman operator in the price schedule and Proposition 2. For $B' \leq B^*(y)$ (the borrowing region), PRO implies lower prices $q_\theta < q_1$ and (via the probability ordering embedded in the pricing operator) a higher perceived default likelihood, both of which tighten the next-period budget more when B' increases. Hence the marginal effect of B' on the continuation value is weakly more adverse under θ .

concentrate around the unique maximizer as $\rho \downarrow 0$. Because the mode shifts left from B'_1 to B'_θ , the θ -distribution first-order stochastically dominates (to the left) the baseline distribution, implying $\mathbb{E}_\theta[B' | y, B] < \mathbb{E}_1[B' | y, B]$.

Remark. If $B'_1(y, B) \geq B^*(y)$ (the softening-of-doom side, where $q_\theta > q_1$), the price effect runs the other way and the same argument yields only the weak inequality $B'_\theta(y, B) \leq B'_1(y, B)$; strictness may fail when prices and continuation effects offset on the relevant grid. The deleveraging is therefore strict precisely in the empirically relevant borrowing region $B'_1 \leq B^*(y)$. $\square$

B.10 Proof of Corollary 3

Proof. **Step 0: Bellman operators.** Let J_i be the Bellman operator for economy $i \in \{1, \theta\}$:

$$(J_i V_{in})(y, B) = \eta \ln \left(\exp \left(\frac{V^D(y; V_{in})}{\eta} \right) + \exp \left(\frac{V^R(y, B; q_i, V_{in})}{\eta} \right) \right),$$

where V^D and V^R are the default and repayment values respectively. The equilibrium value V_i is the unique fixed point of J_i on the space of bounded continuous functions on $\mathcal{Y} \times \mathcal{B}$ with the sup norm.

Step 1: Default value is identical across i . The default value

$$V^D(y; V_{in}) = u(h(y)) + \beta \left[\gamma \mathbb{E}_{y'|y} V_{in}(y', B_0) + (1 - \gamma) \mathbb{E}_{y'|y} V^D(y'; V_{in}) \right]$$

is independent of the current price schedule q_i ; hence

$$V^D(y; V_{in}) \quad \text{is the same for } i = 1 \text{ and } i = \theta. \quad (\text{B.28})$$

Step 2: Repayment value difference comes from prices. The repayment value is

$$V^R(y, B; q_i, V_{in}) = \rho \ln \left(\sum_{B' \in \mathcal{B}} \exp \left(\frac{W(y, B, B'; q_i, V_{in})}{\rho} \right) \right),$$

with

$$W(y, B, B'; q_i, V_{in}) = u(c_i(B')) + \beta \mathbb{E}_{y'|y} V_{in}(y', B'), \quad c_i(B') = y - \kappa B + [B' - (1 - \delta)B] q_i(y, B').$$

The continuation term $\mathbb{E}_{y'|y} V_{in}(y', B')$ is identical across i ; differences in W come solely from $c_i(B')$.

Step 3: Price ordering in the borrowing region. From Proposition 2, if $B' \leq B^*(y)$ then

$$q_\theta(y, B') < q_1(y, B'),$$

with strict inequality on a set of B' with positive measure: in the borrowing region below the pivot, PRO depresses prices and *raises the cost of debt*. If the equilibrium repayment policy places positive probability on $B' < B^*(y)$ (in particular if the baseline optimal choice $B'_1(y, B) \leq B^*(y)$), the relevant B' in the repayment maximization fall in this region with positive probability.

Step 4: Consumption and utility ordering. For any such B' , $c_\theta(B') < c_1(B')$, hence by strict concavity of u ,

$$u(c_\theta(B')) < u(c_1(B')).$$

Because the continuation term in W is identical, I have

$$W(y, B, B'; q_\theta, V_{in}) < W(y, B, B'; q_1, V_{in}) \quad \text{for such } B'. \quad (\text{B.29})$$

Step 5: Aggregating over choices. The log-sum-exp in V^R is strictly increasing in each W . From (B.29), and because the inequality is strict for some B' in the summation,

$$V^R(y, B; q_\theta, V_{in}) < V^R(y, B; q_1, V_{in})$$

whenever the repayment policy places positive probability on the borrowing region $B' < B^*(y)$ (in particular if $B'_1(y, B) \leq B^*(y)$). When all repayment mass sits weakly above the pivot, $B' \geq B^*(y)$, the softening of doom gives $q_\theta \geq q_1$ and the weak inequality $V^R(y, B; q_\theta, V_{in}) \geq V^R(y, B; q_1, V_{in})$ runs the other way pointwise; the global ordering then follows from the equilibrium fixed-point comparison below, in which the borrowing-region wedge dominates because issuance recurs there.

Step 6: Bellman operator dominance and fixed points. By (B.28) and the above repayment ordering,

$$(J_\theta V_{in})(y, B) \leq (J_1 V_{in})(y, B),$$

strictly when the policy places positive probability on $B' < B^*(y)$. The operators J_i are monotone and contractions (Blackwell conditions), hence the pointwise ordering is preserved at the unique fixed points:

$$V_\theta(y, B) \leq V_1(y, B),$$

with strict inequality whenever the equilibrium policy places positive probability on the borrowing region $B' < B^*(y)$, where PRO raises the cost of debt.

Remark (general case). The clean monotone-operator argument above is global when repayment issuance occurs in the borrowing region with positive probability, the empirically relevant case in which the sovereign borrows in $B' < B^*(y)$ with probability one along the ergodic path (Section 5). When some repayment mass falls on the softening

side $B' > B^*(y)$, the better terms there partially offset the borrowing-region tax; that the net effect remains a welfare loss is verified numerically (a uniformly lower value function under higher θ , Section 5) and we state the global dominance as a conjecture rather than a theorem. □

Assumptions for Proposition 6 Let $\alpha \equiv \psi_\beta a_\beta$ denote effective mean precision and let $w(s) \propto L'(-\Delta V(s)/\eta)$ be the margin kernel from the main text. Define

$$\mathcal{M}(\alpha, a_\sigma) := \mathbb{E}[w(s) \nabla f^\top \hat{\Sigma}_\beta(\alpha)^2 \nabla f], \quad \hat{\Sigma}_\beta(\alpha) \equiv (\Sigma_\beta^{-1} + \alpha I_k)^{-1},$$

$$\mathcal{S}(\alpha, a_\sigma) := \mathbb{E}[w(s) (-\widehat{\partial \sigma^2} / \partial a_\sigma)],$$

where expectations are taken under the stationary distribution of states. The first-order conditions in the main text (Eq. (35)) can be written compactly as

$$F_\beta(a_\beta, a_\sigma, \psi_\beta) := \kappa_\beta a_\beta - \frac{\psi_\beta}{\eta^2} \mathcal{M}(\alpha, a_\sigma) = 0, \quad F_\sigma(a_\beta, a_\sigma, \psi_\beta) := \kappa_\sigma a_\sigma - \frac{1}{\eta^2} \mathcal{S}(\alpha, a_\sigma) = 0.$$

Assume:

- (A1) Diminishing returns: $\mathcal{M}_\alpha \leq 0$ and $\mathcal{S}_{a_\sigma} \leq 0$ for all relevant arguments.
- (A2) Substitutability at the margin: $\mathcal{S}_\alpha \leq 0$ (greater mean precision lowers the marginal value of dispersion attention).
- (A3) Interior regularity: the Jacobian $J \equiv \partial(F_\beta, F_\sigma) / \partial(a_\beta, a_\sigma)$ is nonsingular at the optimum; sufficient conditions are

$$\kappa_\beta - \frac{\psi_\beta^2}{\eta^2} \mathcal{M}_\alpha > 0, \quad \kappa_\sigma - \frac{1}{\eta^2} \mathcal{S}_{a_\sigma} > 0, \quad \det J \neq 0.$$

- (A4) Parameter effect on mean attention: $\mathcal{M} + \alpha \mathcal{M}_\alpha > 0$.

B.11 Proof of Proposition 6

Proof. Suppress the state s for notational clarity. Let $F_\beta = 0$ and $F_\sigma = 0$ denote the two first-order conditions above. Write $\alpha \equiv \psi_\beta a_\beta$. The Jacobian with respect to (a_β, a_σ) is

$$J = \begin{bmatrix} \partial_{a_\beta} F_\beta & \partial_{a_\sigma} F_\beta \\ \partial_{a_\beta} F_\sigma & \partial_{a_\sigma} F_\sigma \end{bmatrix} = \begin{bmatrix} \kappa_\beta - \frac{\psi_\beta^2}{\eta^2} \mathcal{M}_\alpha & -\frac{\psi_\beta}{\eta^2} \mathcal{M}_{a_\sigma} \\ -\frac{\psi_\beta}{\eta^2} \mathcal{S}_\alpha & \kappa_\sigma - \frac{1}{\eta^2} \mathcal{S}_{a_\sigma} \end{bmatrix}.$$

By (A1)–(A3), the diagonal entries are positive and J is nonsingular. Differentiate $F_\beta = F_\sigma = 0$ with respect to ψ_β and apply the Implicit Function Theorem:

$$\begin{bmatrix} \partial a_\beta^* / \partial \psi_\beta \\ \partial a_\sigma^* / \partial \psi_\beta \end{bmatrix} = -J^{-1} \begin{bmatrix} \partial_{\psi_\beta} F_\beta \\ \partial_{\psi_\beta} F_\sigma \end{bmatrix}.$$

Parameter derivatives are

$$\partial_{\psi_\beta} F_\sigma = -\frac{1}{\eta^2} a_\beta \mathcal{S}_\alpha \geq 0 \quad \text{by (A2),}$$

$$\partial_{\psi_\beta} F_\beta = -\frac{1}{\eta^2} (\mathcal{M} + \alpha \mathcal{M}_\alpha) < 0 \quad \text{by (A4).}$$

Using the explicit inverse of a 2×2 matrix, one obtains

$$\frac{\partial a_\sigma^*}{\partial \psi_\beta} = \frac{1}{\det J} \left[\left(\kappa_\beta - \frac{\psi_\beta^2}{\eta^2} \mathcal{M}_\alpha \right) \partial_{\psi_\beta} F_\sigma + \frac{\psi_\beta}{\eta^2} \mathcal{S}_\alpha \partial_{\psi_\beta} F_\beta \right] \leq 0,$$

where the inequality uses $\det J \neq 0$ with positive diagonal terms, $\partial_{\psi_\beta} F_\sigma \geq 0$, $\partial_{\psi_\beta} F_\beta < 0$, and $\mathcal{S}_\alpha \leq 0$. Similarly,

$$\frac{\partial a_\beta^*}{\partial \psi_\beta} = \frac{1}{\det J} \left[\frac{\psi_\beta}{\eta^2} \mathcal{M}_{a_\sigma} \partial_{\psi_\beta} F_\sigma + \left(\kappa_\sigma - \frac{1}{\eta^2} \mathcal{S}_{a_\sigma} \right) (-\partial_{\psi_\beta} F_\beta) \right] \geq 0,$$

which is weakly positive under (A1)–(A3); it is strictly positive if at least one inequality in (A1)–(A2) is strict. This establishes that a fall in ψ_β (lower fundamental signal quality) weakly reduces a_β^* and weakly raises a_σ^* at an interior optimum. $\square$

B.12 Pricing Operator: Contraction and Differentiability

Setup. For each $\theta \geq 1$, define the bounded linear operator $\mathcal{T}_\theta$ on the Banach space $\mathcal{X}$ of bounded functions $q : \mathcal{Y} \times \mathcal{B} \rightarrow \mathbb{R}$ (with sup norm) by

$$(\mathcal{T}_\theta q)(y, B') := \frac{1}{1+r} \mathbb{E}_{y'|y} \left[(1 - \tilde{P}_\theta(y', B')) (1 - \delta) \mathbb{E}_{B''|y', B'} [q(y', B'')] \right], \quad (\text{B.30})$$

and the bounded function

$$F_\theta(y, B') := \frac{1}{1+r} \mathbb{E}_{y'|y} \left[(1 - \tilde{P}_\theta(y', B')) \kappa \right]. \quad (\text{B.31})$$

The equilibrium price q_θ solves the fixed point equation

$$q_\theta = F_\theta + \mathcal{T}_\theta q_\theta. \quad (\text{B.32})$$

Contraction and resolvent. Since $0 \leq 1 - \tilde{P}_\theta \leq 1$ and conditional expectation is a con-

traction under the sup norm,

$$\|\mathcal{T}_\theta q\|_\infty \leq \frac{1}{1+r} (1-\delta) \|q\|_\infty \quad \Rightarrow \quad \|\mathcal{T}_\theta\| \leq \frac{1-\delta}{1+r} < 1. \quad (\text{B.33})$$

Thus $\mathcal{T}_\theta$ is a contraction. By the Banach fixed-point theorem, (B.32) has a unique solution and $(I - \mathcal{T}_\theta)^{-1} = \sum_{n \geq 0} \mathcal{T}_\theta^n$ exists as a bounded linear operator. Moreover, $\mathcal{T}_\theta$ is *positive* (maps nonnegative functions to nonnegative functions), hence the resolvent $(I - \mathcal{T}_\theta)^{-1}$ is positive.

Differentiability in θ . Suppose $\tilde{P}_\theta(y', B')$ is C^1 in θ and $\sup_{y', B', \theta} |\partial_\theta \tilde{P}_\theta(y', B')| < \infty$. Then $\theta \mapsto F_\theta \in \mathcal{X}$ and $\theta \mapsto \mathcal{T}_\theta \in \mathcal{L}(\mathcal{X})$ are C^1 with

$$\partial_\theta F_\theta(y, B') = -\frac{1}{1+r} \mathbb{E}_{y'|y} [(\partial_\theta \tilde{P}_\theta(y', B')) \kappa], \quad (\text{B.34})$$

$$(\partial_\theta \mathcal{T}_\theta) q(y, B') = -\frac{1}{1+r} \mathbb{E}_{y'|y} [(\partial_\theta \tilde{P}_\theta(y', B')) (1-\delta) \mathbb{E}_{B''|y', B'} [q(y', B'')]]. \quad (\text{B.35})$$

Differentiate (B.32) and use $(I - \mathcal{T}_\theta)^{-1}$ to obtain the unique derivative $q_\theta^{(1)} := \partial_\theta q_\theta \in \mathcal{X}$:

$$(I - \mathcal{T}_\theta) q_\theta^{(1)} = \partial_\theta F_\theta + (\partial_\theta \mathcal{T}_\theta) q_\theta. \quad (\text{B.36})$$

Link to Equation (23). Combining (B.34)–(B.35) and defining

$$\Lambda(y', B') := \kappa + (1-\delta) \mathbb{E}_{B''|y', B'} [q_\theta(y', B'')], \quad (\text{B.37})$$

I can rewrite (B.36) pointwise as

$$(I - \mathcal{T}_\theta) \frac{\partial q_\theta(\cdot)}{\partial \theta}(y, B') = -\frac{1}{1+r} \mathbb{E}_{y'|y} [(\partial_\theta \tilde{P}_\theta(y', B')) \Lambda(y', B')], \quad (\text{B.38})$$

which coincides with Equation (23) used in the main text. Existence and uniqueness of $\partial_\theta q_\theta$ follow from invertibility of $I - \mathcal{T}_\theta$.

Rational Inattention (linear): Uniform Contraction and Convergence Assumption

RI.1 (Uniform contraction over $[1, \bar{\theta}]$). There exists $\alpha \in (0, 1)$ such that for every fixed $\theta \in [1, \bar{\theta}]$, the pricing operator $\mathcal{T}_\theta : \mathcal{Q} \rightarrow \mathcal{Q}$ satisfies $\|\mathcal{T}_\theta q - \mathcal{T}_\theta q'\|_\infty \leq \alpha \|q - q'\|_\infty$ for all $q, q' \in \mathcal{Q}$.

Proposition RI.1 (State-dependent θ with uniform modulus). Let $\theta_{\text{RI}} : S \rightarrow [1, \bar{\theta}]$ be continuous. Define $\mathcal{T}^{\text{RI}} q(s) := (\mathcal{T}_{\theta_{\text{RI}}(s)} q)(s)$. Under Assumption RI.1, $\mathcal{T}^{\text{RI}}$ is a contraction on $(\mathcal{Q}, \|\cdot\|_\infty)$ with modulus α . Hence there exists a unique fixed point $q^* \in \mathcal{Q}$ solving $q^* = \mathcal{T}^{\text{RI}} q^*$, and successive approximation converges at rate α .

Proof. For any $q, q' \in \mathcal{Q}$ and $s \in S$,

$$|(\mathcal{T}^{\text{RI}} q)(s) - (\mathcal{T}^{\text{RI}} q')(s)| = |(\mathcal{T}_{\theta_{\text{RI}}(s)} q)(s) - (\mathcal{T}_{\theta_{\text{RI}}(s)} q')(s)| \leq \alpha \|q - q'\|_\infty.$$

Taking sup over s delivers $\|\mathcal{T}^{\text{RI}}q - \mathcal{T}^{\text{RI}}q'\|_\infty \leq \alpha \|q - q'\|_\infty$.

B.13 Dispersion bridge (exact given the linearized mental model), and a local temperature reading

Dispersion bridge (exact given the linearized mental model). The rational-inattention block delivers a posterior perceived variance $\widehat{\sigma}_m^2(s)$ for the policy shock m (Section 6). Given $\widehat{\sigma}_m^2$, its mapping to the dispersion-form friction θ is exact: the same m -integral, with no logit-temperature approximation; the *only* approximation is the linearization of f at $\widehat{\beta}$, which enters solely in defining $\widehat{\sigma}_m^2$ itself. Pricing the default decision as if $m \sim \mathcal{N}(0, \widehat{\sigma}_m^2(s))$ gives the perceived default probability

$$\begin{aligned}\widehat{P}(y', B') &= \mathbb{E}_{m \sim \mathcal{N}(0, \widehat{\sigma}_m^2)}[L(-\Delta V(y', B', m)/\eta)] \\ &= \mathbb{E}_{m \sim \mathcal{N}(0, (\theta\sigma_m)^2)}[L(-\Delta V(y', B', m)/\eta)], \quad \theta = \frac{\widehat{\sigma}_m}{\sigma_m},\end{aligned}$$

because once $\theta\sigma_m = \widehat{\sigma}_m$ the two expectations integrate the *same* function against the *same* Gaussian. This is precisely the identity $\theta_{\text{RI}}(s) = \sqrt{\widehat{\sigma}_m^2(s)/\sigma_m^2}$ of (32): the perceived second moment maps one-for-one into the dispersion-form θ , with the true taste scale η held fixed and no logit-temperature step (exact given the linearized mental model). All of the pricing in the paper uses this bridge.

A local temperature reading (by-product, not used in pricing). The older effective-temperature intuition, that posterior dispersion in the *logit argument* $z = -\Delta V/\eta$ softens the choice probability, is a *separate*, purely local statement, recovered only to leading order and not invoked anywhere in the pricing operator. I record it for intuition.

Lemma 6 (Posterior mixtures and effective logit temperature). *Let L be the logistic CDF $P(z) = L(z)$, and let $z \equiv -\Delta V/\eta$. For a mean-zero perturbation ε with $\text{Var}(\varepsilon) = v$ and finite third moment, there exists $\tilde{\theta}(v) = 1 + \frac{v}{4} + o(v)$ such that*

$$\mathbb{E}[L(z + \varepsilon)] = L\left(\frac{z}{\tilde{\theta}(v)}\right) + O(v^{3/2})$$

uniformly on $|z| \leq \bar{z}$. In particular, at $z = 0$, $\mathbb{E}[L'(\varepsilon)] = \frac{1}{4}(1 - \frac{v}{4}) + o(v)$ and $L'_\theta(0) = \frac{1}{4\theta}$, so matching slopes yields $\tilde{\theta}(v) = 1 + \frac{v}{4} + o(v)$.

Proof. Let $f(z) = L(z)$, $f'(z) = f(z)(1 - f(z))$, $f''(z) = f'(z)(1 - 2f(z))$, and $f'''(z) = f''(z)(1 - 2f(z)) - 2(f'(z))^2$. At $z = 0$, $f(0) = \frac{1}{2}$, $f'(0) = \frac{1}{4}$, $f''(0) = 0$, $f'''(0) = -\frac{1}{8}$. A third-order Taylor around 0 gives $\mathbb{E}[f'(\varepsilon)] = f'(0) + \frac{v}{2}f'''(0) + o(v) = \frac{1}{4}(1 - \frac{v}{4}) + o(v)$. For any fixed $\bar{z} > 0$, f and its first three derivatives are bounded on $[-\bar{z}, \bar{z}]$; a standard remainder bound then yields the stated uniform approximation for $\mathbb{E}[f(z + \varepsilon)]$. For the

matching logit with temperature θ , $g(z) = L(z/\theta)$ has $g'(0) = \frac{1}{4\theta}$. Equating slopes gives $\tilde{\theta}(\nu) = 1 + \frac{\nu}{4} + o(\nu)$. Substituting back yields the uniform approximation. $\square$

Since $z = -\Delta V/\eta$, any small mean-zero dispersion in the posterior of ΔV (variance $\text{Var}(\xi)$) translates into $\nu = \text{Var}(\xi)/\eta^2$ and inflates the effective temperature by $\tilde{\theta} \approx 1 + \text{Var}(\xi)/(4\eta^2)$. This formalizes the main-text claim that posterior dispersion around the default margin softens the logit in a way that is locally equivalent to evaluating the same pricing operator at a higher θ .

B.14 Posterior of ℓ_σ and the sensitivity of $\widehat{\sigma^2}$

Setup and posterior. Recall the prior and signal for $\ell_\sigma \equiv \log \sigma^2$:

$$\ell_\sigma \sim \mathcal{N}(\bar{\ell}, \tau^2), \quad s_\ell | \ell_\sigma \sim \mathcal{N}(\ell_\sigma, (\psi_\sigma a_\sigma)^{-1}).$$

Bayesian updating gives the Gaussian posterior $\ell_\sigma | s_\ell \sim \mathcal{N}(\hat{\ell}, \hat{\tau}^2)$ with

$$\hat{\tau}^2(a_\sigma) = (\tau^{-2} + \psi_\sigma a_\sigma)^{-1}, \quad \hat{\ell}(a_\sigma, s_\ell) = \frac{\tau^{-2} \bar{\ell} + \psi_\sigma a_\sigma s_\ell}{\tau^{-2} + \psi_\sigma a_\sigma}. \quad (\text{B.39})$$

Since $\sigma^2 = \exp(\ell_\sigma)$, the posterior mean is

$$\widehat{\sigma^2}(a_\sigma, s_\ell) = \mathbb{E}[\exp(\ell_\sigma) | s_\ell] = \exp\left(\hat{\ell}(a_\sigma, s_\ell) + \frac{1}{2} \hat{\tau}^2(a_\sigma)\right), \quad (\text{B.40})$$

which coincides with the main-text expression (36).

Sensitivity with respect to a_σ . Differentiating (B.40) and using (B.39),

$$\partial_{a_\sigma} \widehat{\sigma^2} = \widehat{\sigma^2} \left(\partial_{a_\sigma} \hat{\ell} + \frac{1}{2} \partial_{a_\sigma} \hat{\tau}^2 \right), \quad \partial_{a_\sigma} \hat{\tau}^2 = -\psi_\sigma \hat{\tau}^4.$$

A quotient rule gives $\partial_{a_\sigma} \hat{\ell} = \psi_\sigma (s_\ell - \bar{\ell}) \hat{\tau}^4 / \tau^2$. Therefore

$$\frac{\partial \widehat{\sigma^2}}{\partial a_\sigma} = \widehat{\sigma^2} \psi_\sigma \hat{\tau}^4 \left(\frac{s_\ell - \bar{\ell}}{\tau^2} - \frac{1}{2} \right), \quad (\text{B.41})$$

matching Equation (37) in the main text. The term $-(1/2)$ captures the certainty effect (posterior concentration), while the surprise term $\frac{s_\ell - \bar{\ell}}{\tau^2}$ captures the information content of the instability signal.

C Policy and Information Extensions: Formal Statements and Proofs

This section remains within the same operator framework: I vary information and policy mappings that enter *through* the pricing operator $\mathcal{T}_\theta$, while holding primitives and the sovereign's choice aggregator J_ρ fixed.

Ramsey Planning Under PRO Corollary 3 shows that PRO reduces welfare; can an optimal fiscal authority undo the loss? I study a Ramsey planner who chooses lump-sum transfers but takes bond pricing as given: the planner's instruments enter only the resource constraint, leaving the choice aggregator J_ρ and the operator $\mathcal{T}_\theta$ unchanged. PRO acts as a persistent wedge in the intertemporal price of borrowing. Transfers reallocate resources subject to a zero present-value constraint but cannot change the shadow price at which the government trades across dates, so the PRO schedule distorts debt choices even under optimal fiscal policy.

The planner chooses $\{c_t, B_{t+1}, \tau_t\}_{t \geq 0}$ to maximize expected utility subject to the per-period resource constraint and a zero present-value (PV) condition for transfers:

$$c_t + \kappa B_t + \tau_t = y_t + (B_{t+1} - (1 - \delta)B_t)q_\theta(y_t, B_{t+1}), \quad t \geq 0, \text{ a.s.} \quad (\text{C.42})$$

and

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \tau_t \right] = 0, \quad c_t \geq 0, \quad B_{t+1} \in \mathcal{B}. \quad (\text{C.43})$$

Proposition 7. Let $r > -1$ satisfy $\beta(1 + r) = 1$. For $i \in \{1, \theta\}$ define the Ramsey value

$$W_i^R = \sup_{\{c_t, B_{t+1}, \tau_t\}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t) \right]$$

subject to (C.42)–(C.43). Assume:

(A1) u is strictly increasing and strictly concave; $\mathcal{B}$ is compact.

(A2) (Price dominance by trade sign, consistent with Propositions 2 and 5). Let $\Delta_t \equiv B_{t+1} - (1 - \delta)B_t$. For every history,

$$\Delta_t \geq 0 \Rightarrow q_1(y_t, B_{t+1}) \geq q_\theta(y_t, B_{t+1}), \quad \Delta_t \leq 0 \Rightarrow q_1(y_t, B_{t+1}) \leq q_\theta(y_t, B_{t+1}),$$

with strict inequality on a set of positive probability and $\Pr(|\Delta_t| > 0) > 0$.

Then $W_\theta^R < W_1^R$.

Proof. See Appendix C.1. □

Transfers ensure first-best *conditional* consumption smoothing but cannot change the intertemporal proceeds term $\mathbb{E}_0[\sum \beta^t \Delta_t q_i(y_t, B_{t+1})]$ that appears in the implementability constraint; therefore the wedge in q_θ generates a genuine efficiency loss. See also [Aiyagari et al. \[2002\]](#).

Endogenous Belief Formation I next let lenders learn about the taste-shock dispersion parameter θ . This is a state augmentation of the baseline economy: I extend the state to $S_t = (y_t, B_t, \theta_t)$ and keep the pricing operator unchanged except that it is evaluated at the current belief, i.e., prices satisfy the same fixed-point condition with $\mathcal{T}_{\theta_t}$.

Even with Bayesian updating, negativity bias (overweighting unexpected defaults relative to unexpected repayments) can make PRO persistent and self-reinforcing. Rare but salient defaults move beliefs more than frequent, modestly good realizations, creating a drift toward PRO.

Let $\theta_t \in [\underline{\theta}, \bar{\theta}]$ with $\underline{\theta} = 1 < \bar{\theta}$, and suppose beliefs evolve according to

$$\theta_{t+1} = \lambda \theta_t + (1 - \lambda) \hat{\theta}(\{d_s\}_{s=0}^t), \quad \lambda \in (0, 1), \quad (\text{C.44})$$

where $d_t \in \{0, 1\}$ indicates default and $\hat{\theta}(\cdot)$ is a (possibly biased) estimator that places extra weight on *surprising* defaults. For instance, with

$$\xi(y, B) = \max \left\{ 0, \frac{P_1(y, B) - P_{\theta_t}(y, B)}{P_1(y, B)} \right\}, \quad (\text{C.45})$$

negativity bias means that $\hat{\theta}$ responds more to larger ξ when $d_t = 1$ than to the analogous repayment surprise when $d_t = 0$ [Baumeister et al. \[2001\]](#), [Bordalo et al. \[2018\]](#).

Proposition 8. *Under a stable, Feller belief-update kernel induced by (C.44) with negativity bias as above and mild drift/minorization conditions, the Markov process $\{\theta_t\}$ admits a unique invariant distribution Θ^* such that:*

- (i) $\mathbb{E}[\Theta^*] > 1$ (persistent PRO);
- (ii) Θ^* first-order stochastically dominates the rational benchmark (history dependence through rare defaults);
- (iii) $\|\mathbb{E}[\theta_t] - \mathbb{E}[\Theta^*]\| = O(\lambda^t)$, with slow convergence when defaults are rare (large λ).

Proof. See Appendix C.2. □

Optimal Policy Communication This extension likewise preserves the baseline structure: transparency affects only the *effective* belief input into the same pricing operator, while the sovereign's choice aggregator J_ρ and other primitives are unchanged. Now allow

the government to choose a transparency level $\alpha \in [0, 1]$ that affects lenders' effective dispersion:

$$\theta_{\text{eff}}(\alpha, \theta) = \alpha \cdot 1 + (1 - \alpha) \cdot \theta, \quad (\text{C.46})$$

and let period utility be $u(c, \alpha) = c^{1-\sigma}/(1-\sigma) - \phi(\alpha)$ with $\phi(\alpha) = \gamma \alpha^2/2$.

Proposition 9. *Let $W(\alpha)$ be the sovereign's value when the price schedule is $q_{\theta_{\text{eff}}(\alpha, \theta)}$ and the planner otherwise solves the same Ramsey problem. Suppose $W(\alpha)$ is strictly concave in α (e.g., q_θ is smooth and ϕ is strictly convex). Then:*

(i) *There exists a unique $\alpha^* \in [0, 1]$ satisfying the FOC*

$$\frac{d}{d\alpha} W(\alpha^*) = \phi'(\alpha^*) = \gamma \alpha^*. \quad (\text{C.47})$$

(ii) *If $\partial W / \partial \theta < 0$ and $\partial \theta_{\text{eff}} / \partial \alpha = 1 - \theta < 0$, then $\partial \alpha^* / \partial \theta > 0$: greater PRO raises optimal transparency.*

(iii) *There exists $\theta_c > 1$ such that $W(\alpha^*) > W(0)$ for all $\theta > \theta_c$.*

Proof. See Appendix C.3. □

Transparency improves welfare by steepening lenders' price schedule toward the rational benchmark, but convex disclosure costs imply an interior optimum. The comparative static in (ii) formalizes that transparency is most valuable when PRO is severe.

Testable Predictions and Empirical Hooks As brief guidance for future empirical work, the framework yields concise, testable implications: (i) a single-crossing pivot in price/spread schedules with $B^*(y)$ increasing in y ; (ii) higher default thresholds despite higher average spreads; (iii) deleveraging alongside higher mean spreads and lower volatility ("stability illusion"); and (iv) cross-maturity and peer "decoupling" patterns consistent with second-moment (dispersion) beliefs. I view these as light-touch hooks for calibration and event studies (e.g., Argentina), not the focus here.

C.1 Proof of Proposition 7

Proof. Step 1 (Intertemporal implementability). Multiply (C.42) by β^t and sum expectations over t , and use $\beta(1+r) = 1$ together with (C.43). Writing $\Delta_t \equiv B_{t+1} - (1-\delta)B_t$, feasibility under schedule q_i is equivalent to the *implementability constraint*

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t c_t \right] = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t (y_t - \kappa B_t) \right] + \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \Delta_t q_i(y_t, B_{t+1}) \right], \quad (\text{C.48})$$

together with $c_t \geq 0$ and $B_{t+1} \in \mathcal{B}$. In particular, the transfer sequence $\{\tau_t\}$ only enforces period constraints and has zero present value, so it does not affect (C.48).

Step 2 (Set inclusion of feasible (c, B)). Fix any feasible pair (c, B) under q_θ ; it satisfies (C.48) with $i = \theta$. Consider the same (c, B) under q_1 . Subtracting the two versions of (C.48) gives the slack generated by switching from q_θ to q_1 :

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \Delta_t (q_1(y_t, B_{t+1}) - q_\theta(y_t, B_{t+1})) \right] = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t c_t \right]_{q_1} - \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t c_t \right]_{q_\theta}. \quad (\text{C.49})$$

By (A2), each summand in $\mathcal{S}(c, B)$ is nonnegative and strictly positive with positive probability whenever $|\Delta_t| > 0$. Hence

$$\mathcal{S}(c, B) \geq 0 \quad \text{and} \quad \mathcal{S}(c, B) > 0 \quad \text{if} \quad \Pr(|\Delta_t| > 0 \text{ and } q_1 \neq q_\theta) > 0.$$

Therefore, every (c, B) feasible under q_θ is also feasible under q_1 , and generically with strictly more present-value resources available for consumption under q_1 .

Step 3 (Value comparison). Let (c^θ, B^θ) be Ramsey-optimal under q_θ . By Step 2, (c^θ, B^θ) is feasible under q_1 and produces a weakly larger right-hand side in (C.48); because u is strictly increasing, the planner can raise consumption at (at least) one date by an $\varepsilon > 0$ while keeping feasibility under q_1 , yielding strictly higher utility. Hence

$$W_1^R \geq \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t^\theta) \right], \quad \text{with strict inequality under the strict part of (A2).}$$

Taking the supremum over all feasible (c, B) under q_1 gives $W_1^R > W_\theta^R$.

Remark (Why transfers cannot undo the loss). Condition (C.43) fixes the present value of transfers at zero, so transfers only reshuffle consumption across dates without changing the right-hand side of (C.48). The loss stems from the lower proceeds from intertemporal trade encoded in q_θ whenever Δ_t and the price dominance have the same sign; no transfer sequence can increase the present value on the right-hand side of (C.48). $\square$

C.2 Proof of Proposition 8

Proof. I establish each part through rigorous analysis of the belief updating dynamics.

Part (i): Persistent PRO. Define the conditional maximum likelihood estimators:

$$\hat{\theta}_D(\{d_s\}_{s=0}^t) = \arg \max_{\theta \in [\underline{\theta}, \bar{\theta}]} \prod_{s: d_s=1} P_\theta(y_s, B_s) \prod_{s: d_s=0} (1 - P_\theta(y_s, B_s)) \quad (\text{C.50})$$

$$\hat{\theta}_R(\{d_s\}_{s=0}^t) = \arg \max_{\theta \in [\underline{\theta}, \bar{\theta}]} \prod_{s: d_s=0} (1 - P_\theta(y_s, B_s)) \prod_{s: d_s=1} P_\theta(y_s, B_s) \quad (\text{C.51})$$

Under negativity bias, the belief updating mechanism assigns different weights $w_D >$

w_R to default versus repayment observations:¹⁷

$$\hat{\theta}(\{d_s\}_{s=0}^t) = \frac{w_D \sum_{s:d_s=1} \ln P_\theta(y_s, B_s) + w_R \sum_{s:d_s=0} \ln(1 - P_\theta(y_s, B_s))}{w_D N_D + w_R N_R} \quad (\text{C.52})$$

where N_D and N_R are the numbers of defaults and repayments, respectively.

Taking expectations over the ergodic distribution $\mu(y, B)$ of state variables:

$$\mathbb{E}[\hat{\theta}] = \int \left[w_D P_1(y, B) \frac{\partial \ln P_\theta(y, B)}{\partial \theta} + w_R (1 - P_1(y, B)) \frac{\partial \ln(1 - P_\theta(y, B))}{\partial \theta} \right] d\mu(y, B) \quad (\text{C.53})$$

Since, to leading order in η , $P_\theta(y, B) = \Phi(m^*(y, B)/(\theta\sigma_m))$ with m^* the default threshold of Lemma 1 (the m -integrated logit at the perceived dispersion $\theta\sigma_m$):¹⁸

$$\frac{\partial \ln P_\theta(y, B)}{\partial \theta} = - \frac{\phi(m^*(y, B)/(\theta\sigma_m))}{\Phi(m^*(y, B)/(\theta\sigma_m))} \frac{m^*(y, B)}{\theta^2 \sigma_m} < 0 \quad \text{whenever } m^*(y, B) > 0. \quad (\text{C.54})$$

Every realized-default observation entering the weighted MLE (C.52) occurs in a state where default was perceived likely, i.e. in the risky region $m^*(y, B) > 0$; restricting the expectation (C.53) to the default-conditional distribution μ_D (the events that actually enter the w_D sum), the integrand sign is governed by (C.54) with $m^* > 0$, so $|m^*| = -m^*$ and $\partial_\theta \ln P_\theta < 0$ throughout the relevant domain.

Substituting (C.54) into (C.53) and using the negativity bias $w_D > w_R$:

$$\mathbb{E}[\hat{\theta}] > 1 + \frac{(w_D - w_R)}{w_D + w_R} \int_{m^*(y, B) > 0} P_1(y, B) \frac{\phi(m^*(y, B)/(\theta\sigma_m)) |m^*(y, B)|}{\theta^2 \sigma_m \Phi(m^*(y, B)/(\theta\sigma_m))} d\mu_D(y, B) > 1 \quad (\text{C.55})$$

By the ergodic theorem and (C.44), $\lim_{t \rightarrow \infty} \mathbb{E}[\theta_t] = \mathbb{E}[\hat{\theta}] > 1$.

Part (ii): History Dependence. Let $\mathcal{H}_t = \{(y_s, B_s, d_s)\}_{s=0}^{t-1}$ denote the history up to time t . The belief updating process creates a Markov chain on the augmented state space $(y, B, \theta, \mathcal{H})$.¹⁹

For any two histories $\mathcal{H}^A$ and $\mathcal{H}^B$ with different default patterns but identical fundamentals, define the corresponding stationary distributions Θ^A and Θ^B . The key insight is that the transition operator:

$$T_{\mathcal{H}}(\theta, \theta') = \lambda \delta(\theta' - \theta) + (1 - \lambda) \delta(\theta' - \hat{\theta}(\mathcal{H})) \quad (\text{C.56})$$

depends explicitly on the history $\mathcal{H}$.

¹⁷The negativity bias reflects well-documented cognitive biases where agents overweight negative information relative to positive information, consistent with prospect theory and related behavioral findings.

¹⁸This is the leading-order form (15); PRO enters only through the perceived dispersion of the policy shock m .

¹⁹The history dependence arises because the MLE $\hat{\theta}(\mathcal{H}_t)$ depends on the entire sequence of past defaults, not just the current state.

Since $\hat{\theta}(\mathcal{H}^A) \neq \hat{\theta}(\mathcal{H}^B)$ for histories with different default intensities, I have $\Theta^A \neq \Theta^B$. Moreover, if $\mathcal{H}^A$ contains more surprising defaults than $\mathcal{H}^B$, then Θ^A first-order stochastically dominates Θ^B :

$$\int_{\underline{\theta}}^{\theta} d\Theta^A(\tilde{\theta}) \leq \int_{\underline{\theta}}^{\theta} d\Theta^B(\tilde{\theta}) \quad \forall \theta \in [\underline{\theta}, \bar{\theta}] \quad (\text{C.57})$$

Part (iii): Slow Convergence. Let π_t denote the default probability in period t under the current belief θ_t . The frequency of belief updates follows a Poisson process with intensity π_t .²⁰

Between updates, beliefs evolve according to:

$$\theta_{t+1} = \lambda \theta_t \quad (\text{no default}) \quad (\text{C.58})$$

The time between defaults follows an exponential distribution with rate $\pi_t \approx \pi^*$ in steady state. The expected duration between updates is $1/\pi^*$, and during each interval of length τ , beliefs decay by factor λ^τ .

For the convergence rate, consider the deviation from the stationary mean: $\epsilon_t = \theta_t - \mathbb{E}[\Theta^*]$. The dynamics satisfy:

$$\mathbb{E}[\epsilon_{t+1}] = \lambda \mathbb{E}[\epsilon_t] + (1 - \lambda)(\mathbb{E}[\hat{\theta}] - \mathbb{E}[\Theta^*]) \quad (\text{C.59})$$

In steady state, $\mathbb{E}[\hat{\theta}] = \mathbb{E}[\Theta^*]$, so:

$$\mathbb{E}[\epsilon_{t+1}] = \lambda \mathbb{E}[\epsilon_t] \quad (\text{C.60})$$

Therefore, $\|\theta_t - \mathbb{E}[\Theta^*]\| = O(\lambda^t)$. Since empirical persistence in sovereign spreads requires $\lambda \approx 0.95 - 0.99$, convergence is indeed slow with half-life approximately $\ln(2)/\ln(1/\lambda) \approx 14 - 69$ periods. $\square$

C.3 Proof of Proposition 9

Proof. I establish each part through detailed analysis of the sovereign's optimization problem under strategic communication.

Setup: The Communication Problem. Let $\mathcal{W}(\alpha; \theta)$ denote the sovereign's value function under transparency level α and baseline PRO θ . This satisfies the Bellman equation:

$$\mathcal{W}(\alpha; \theta) = \max_{\{c_t, B_{t+1}\}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t (u(c_t) - \phi(\alpha)) \right] \quad (\text{C.61})$$

²⁰This approximation is valid when defaults are rare events, which is the empirically relevant case for most sovereigns.

subject to:

$$c_t = y_t - \kappa B_t + [B_{t+1} - (1 - \delta)B_t]q_{\theta_{\text{eff}}}(y_t, B_{t+1}) \quad (\text{C.62})$$

$$\theta_{\text{eff}}(\alpha, \theta) = \alpha \cdot 1 + (1 - \alpha) \cdot \theta \quad (\text{C.63})$$

Part (i): Interior Solution. Define the value function net of communication costs: $V(\alpha) = \mathcal{W}(\alpha; \theta) - \mathbb{E}[\phi(\alpha)]$. The first-order condition for optimality is:

$$\left. \frac{\partial V(\alpha)}{\partial \alpha} \right|_{\alpha=\alpha^*} = \left. \frac{\partial \mathcal{W}(\alpha; \theta)}{\partial \alpha} \right|_{\alpha=\alpha^*} - \gamma \alpha^* = 0 \quad (\text{C.64})$$

To characterize $\frac{\partial \mathcal{W}}{\partial \alpha}$, note that transparency affects welfare through the bond pricing channel. Using the envelope theorem:²¹

$$\frac{\partial \mathcal{W}}{\partial \alpha} = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u'(c_t) \frac{\partial c_t}{\partial \alpha} \right] \quad (\text{C.65})$$

From the budget constraint (C.62):

$$\frac{\partial c_t}{\partial \alpha} = [B_{t+1} - (1 - \delta)B_t] \frac{\partial q_{\theta_{\text{eff}}}}{\partial \theta_{\text{eff}}} \frac{\partial \theta_{\text{eff}}}{\partial \alpha} = [B_{t+1} - (1 - \delta)B_t] \frac{\partial q_{\theta_{\text{eff}}}}{\partial \theta_{\text{eff}}} (1 - \theta) \quad (\text{C.66})$$

Since $\frac{\partial q_{\theta_{\text{eff}}}}{\partial \theta_{\text{eff}}} < 0$ (higher perceived volatility reduces bond prices) and $(1 - \theta) < 0$ for $\theta > 1$, I have:

$$\frac{\partial \mathcal{W}}{\partial \alpha} = (\theta - 1) \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u'(c_t) [B_{t+1} - (1 - \delta)B_t] \left| \frac{\partial q_{\theta_{\text{eff}}}}{\partial \theta_{\text{eff}}} \right| \right] > 0 \quad (\text{C.67})$$

For an interior solution, I need both concavity and appropriate boundary conditions. The second-order condition requires:

$$\left. \frac{\partial^2 V(\alpha)}{\partial \alpha^2} \right|_{\alpha=\alpha^*} = \left. \frac{\partial^2 \mathcal{W}}{\partial \alpha^2} \right|_{\alpha=\alpha^*} - \gamma < 0 \quad (\text{C.68})$$

Since $\frac{\partial^2 \mathcal{W}}{\partial \alpha^2} < 0$ (diminishing returns to transparency) and $\gamma > 0$, condition (C.68) is satisfied.²²

The boundary conditions require $\frac{\partial V}{\partial \alpha}|_{\alpha=0} > 0$ and $\frac{\partial V}{\partial \alpha}|_{\alpha=1} < 0$, which hold for intermediate values of γ satisfying:

$$\left. \frac{\partial \mathcal{W}}{\partial \alpha} \right|_{\alpha=0} > \gamma > \left. \frac{\partial \mathcal{W}}{\partial \alpha} \right|_{\alpha=1} \quad (\text{C.69})$$

Part (ii): PRO Amplifies Transparency. Differentiating the first-order condition

²¹The envelope theorem applies because the sovereign optimally chooses consumption and debt policies for any given price schedule.

²²The diminishing returns arise because transparency has the largest impact when moving from very opaque to moderately transparent, with smaller gains from further increases.

(C.64) with respect to θ using the implicit function theorem:

$$\left. \frac{\partial^2 \mathcal{W}}{\partial \alpha \partial \theta} \right|_{\alpha=\alpha^*} + \left. \frac{\partial^2 \mathcal{W}}{\partial \alpha^2} \right|_{\alpha=\alpha^*} \frac{\partial \alpha^*}{\partial \theta} - \gamma \frac{\partial \alpha^*}{\partial \theta} = 0 \quad (\text{C.70})$$

Rearranging:

$$\frac{\partial \alpha^*}{\partial \theta} = \frac{\left. \frac{\partial^2 \mathcal{W}}{\partial \alpha \partial \theta} \right|_{\alpha=\alpha^*}}{\gamma - \left. \frac{\partial^2 \mathcal{W}}{\partial \alpha^2} \right|_{\alpha=\alpha^*}} \quad (\text{C.71})$$

To sign the numerator, differentiate (C.67) with respect to θ :

$$\frac{\partial^2 \mathcal{W}}{\partial \alpha \partial \theta} = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u'(c_t) [B_{t+1} - (1 - \delta)B_t] \left| \frac{\partial q_{\theta, \text{eff}}}{\partial \theta} \right| \right] + (\theta - 1) \frac{\partial}{\partial \theta} \mathbb{E}[\dots] \quad (\text{C.72})$$

The first term is positive (direct effect), and the second term captures the indirect effect through equilibrium adjustments. Under reasonable parameter values, the direct effect dominates, ensuring $\frac{\partial^2 \mathcal{W}}{\partial \alpha \partial \theta} > 0$.²³

Since the denominator in (C.71) is positive by the second-order condition (C.68), I conclude:

$$\frac{\partial \alpha^*}{\partial \theta} > 0 \quad (\text{C.73})$$

Part (iii): Welfare Dominance. Define the net welfare gain from communication as:

$$\Delta W(\theta) = \mathcal{W}(\alpha^*(\theta); \theta) - \mathcal{W}(0; \theta) - \mathbb{E}[\phi(\alpha^*(\theta))] \quad (\text{C.74})$$

For small $\theta - 1$, using a second-order Taylor expansion around $\theta = 1$:

$$\Delta W(\theta) \approx \frac{1}{2} \left. \frac{\partial^2 \mathcal{W}}{\partial \alpha^2} \right|_{\alpha=\alpha^*(1)} [\alpha^*(\theta)]^2 - \frac{\gamma [\alpha^*(\theta)]^2}{2} \quad (\text{C.75})$$

Using the first-order condition and part (ii), $\alpha^*(\theta) \approx C(\theta - 1)$ for some constant $C > 0$. Substituting:

$$\Delta W(\theta) \approx \frac{C^2(\theta - 1)^2}{2} \left[\left. \frac{\partial^2 \mathcal{W}}{\partial \alpha^2} \right|_{\alpha=\alpha^*(1)} - \gamma \right] \quad (\text{C.76})$$

Since the second-order condition requires $\left. \frac{\partial^2 \mathcal{W}}{\partial \alpha^2} \right|_{\alpha=\alpha^*(1)} - \gamma < 0$, I have $\Delta W(\theta) < 0$ for θ close to 1. However, as θ increases, the benefits of transparency grow faster than the costs.²⁴

The critical threshold θ_c is defined by $\Delta W(\theta_c) = 0$. For $\theta > \theta_c$, the welfare gains from reduced borrowing costs exceed the communication costs, establishing welfare dominance. $\square$

²³The direct effect dominates because the immediate benefit of transparency increases linearly with the degree of baseline PRO, while the indirect effects through equilibrium adjustments are second-order.

²⁴This occurs because the marginal benefit of transparency is proportional to $(\theta - 1)$, while the marginal cost is constant in θ .

D Computational Algorithm

I solve the model numerically using value function iteration on a discretized state space, following the computational strategies developed by [Mihalache \[2024\]](#). The algorithm is implemented in Fortran and parallelized using OpenMP to accelerate computation. The process is as follows:

D.1 Discretization and Initialization

The state space is discretized first.

1. I discretize the exogenous endowment process, an AR(1) in logarithms defined in equation (1), into a $N_y = 201$ state Markov chain using the [Tauchen \[1986\]](#) method, as implemented in my `discretizeAR1` subroutine. This yields a grid for income levels $\{y_i\}_{i=1}^{N_y}$ and a transition matrix $\Pi(y_i, y_j)$.
2. I construct the space for bond holdings, $\mathcal{B}$, as a grid of $N_B = 600$ points, uniformly spaced between $B_{\min} = 0$ and $B_{\max} = 1.5$ of mean output, created using a `linspace` function.
3. I initialize the algorithm with guesses for the value functions and the bond price schedule. I set the price of bonds, $q_0(y, B')$, initially to 1 for all states. I initialize the value function, $V_0(y, B)$, to the utility of consuming the endowment net of coupon payments, and the value of default, $Vd_0(y)$, to the utility of consuming the post-default output $h(y)$.

D.2 Value Function and Price Iteration

The core of my solution method is iterating on the value functions (V, V^D) and the bond price function q until they converge to a joint fixed point. This process solves the system of equations defined in Section 3. My main DO WHILE loop in the Fortran code continues until the maximum absolute difference between successive iterates for both the value function and the bond prices falls below a specified tolerance ($\epsilon_V = 10^{-6}, \epsilon_q = 10^{-6}$). A single iteration, from step $k - 1$ to k , is a carefully ordered sequence of updates, parallelized with OpenMP where possible.

1. **Update the Value of Default.** I begin each iteration by updating the value of default, $V_k^D(y)$. This calculation uses the total value function V_{k-1} and default value V_{k-1}^D from the *previous* iteration. For each income state y_i , I follow the discrete version of equation (8):

$$V_k^D(y_i) = u(h(y_i)) + \beta \sum_{j=1}^{N_y} \Pi(y_i, y_j) \left[\gamma V_{k-1}(y_j, B_1) + (1 - \gamma) V_{k-1}^D(y_j) \right],$$

where B_1 is the grid point corresponding to zero debt.

2. **Update the Value of Repayment and Sovereign Policies.** This is a multi-step process that I compute inside a large parallel loop over the state space (y_i, B_j) .

- (a) First, for each state (y_i, B_j) , I calculate the **choice-specific value of borrowing**, $W(y_i, B_j, B'_l)$, for *every* possible next-period debt level B'_l . This uses the bond price q_{k-1} and value function V_{k-1} from the prior iteration:

$$W(y_i, B_j, B'_l) = u(y_i - \kappa B_j + [B'_l - (1 - \delta)B_j]q_{k-1}(y_i, B'_l)) + \beta \sum_{m=1}^{N_y} \Pi(y_i, y_m) V_{k-1}(y_m, B'_l).$$

- (b) With the full vector of W values, I compute the **value of repayment**, $V_k^R(y_i, B_j)$, using the "log-sum-exp" formulation following Mihalache [2024]. This numerically stable technique, which uses the borrowing taste shock parameter ρ (or rhoB in my code), replaces a non-differentiable 'max' operation over the borrowing grid:

$$V_k^R(y_i, B_j) = \rho \ln \left(\sum_{l=1}^{N_B} \exp \frac{W(y_i, B_j, B'_l)}{\rho} \right).$$

- (c) Simultaneously, I compute the state-contingent **borrowing policy**, $\Pr(B'_l | y_i, B_j)$, as the softmax of the choice-specific values:

$$\Pr(B'_l | y_i, B_j) = \frac{\exp(W(y_i, B_j, B'_l) / \rho)}{\sum_{s=1}^{N_B} \exp(W(y_i, B_j, B'_s) / \rho)}.$$

- (d) Finally, using the *just-updated* values $V_k^R(y_i, B_j)$ and $V_k^D(y_i)$, I calculate the new **ex-ante value function** $V_k(y_i, B_j)$, again using the log-sum-exp formula but with the default taste shock parameter η (rhoD in my code):

$$V_k(y_i, B_j) = \eta \ln \left(\exp \frac{V_k^D(y_i)}{\eta} + \exp \frac{V_k^R(y_i, B_j)}{\eta} \right).$$

- (e) I also determine the sovereign's true **default probability**, $\Pr\{d = 1 | y_i, B_j\}$, at this point from the logistic choice formula:

$$\Pr\{d = 1 | y_i, B_j\} = \frac{\exp(V_k^D(y_i) / \eta)}{\exp(V_k^D(y_i) / \eta) + \exp(V_k^R(y_i, B_j) / \eta)}.$$

3. **Update Bond Prices.** I update the bond price schedule $q_k(y, B')$ based on the lenders' zero-profit condition. This is the crucial step where I introduce the behavioral PRO wedge, θ (thetaD). For each state (y_i, B'_l) :

- (a) I calculate the **lender's perceived default probability**, $\tilde{P}(y_m, B'_l)$. It uses the

most recently updated value functions, V_k^R and V_k^D , with the *true* taste scale η , and applies PRO *only* through the perceived dispersion $\theta\sigma_m$ of the policy shock. Concretely, I integrate the logit default rule over $m \sim \mathcal{N}(0, (\theta\sigma_m)^2)$ by N_m -point Gauss–Hermite quadrature with nodes $\{\zeta_n\}$ and weights $\{\omega_n\}$ (so that $m_n = \sqrt{2}\theta\sigma_m\zeta_n$ and the Gaussian weights are $\omega_n/\sqrt{\pi}$):

$$\tilde{P}(y_m, B'_l) = \sum_{n=1}^{N_m} \frac{\omega_n}{\sqrt{\pi}} \frac{\exp(V_k^D(y_m)/\eta)}{\exp(V_k^D(y_m)/\eta) + \exp(V_k^R(y_m, B'_l, m_n)/\eta)},$$

where $V_k^R(y_m, B'_l, m_n)$ is the repayment value at shock realization m_n (the budget shifted by $+m_n$). This is the discretization of the m -integrated logit $\tilde{P}_\theta$ in (12); at $\theta = 1$ it collapses to the lender pricing the default decision at the true dispersion σ_m .

- (b) The new price, $q_k(y_i, B'_l)$, is the expected discounted payoff from the lender's perspective. This expectation is taken over future income states y_m and future bond choices B_s'' . It uses the lender's perceived default probability $\tilde{P}$, the sovereign's *true* borrowing policy $\Pr(B_s''|y_m, B'_l)$ (computed in step 2), and the bond prices from the *previous* iteration, q_{k-1} . The full pricing equation implemented in the code is:

$$q_k(y_i, B'_l) = \frac{1}{1+r} \sum_{m=1}^{N_y} \Pi(y_i, y_m) (1 - \tilde{P}(y_m, B'_l)) \times \left(\kappa + (1 - \delta) \sum_{s=1}^{N_B} \Pr(B_s''|y_m, B'_l) \cdot q_{k-1}(y_m, B_s'') \right).$$

4. **Check for Convergence.** The iteration concludes by calculating the supremum norm of the change in the value functions and price functions: $\text{errV} = \max |V_k - V_{k-1}|$ and $\text{errQ} = \max |q_k - q_{k-1}|$. If the errors are within tolerance, the loop terminates. Otherwise, the algorithm sets $V_0 \leftarrow V_k$, $V_d \leftarrow V_k^D$, and $q_0 \leftarrow q_k$, and proceeds to the next iteration.

D.3 Simulation

Once the functions have converged, the `simulate` subroutine is called to generate the business cycle moments. For reproducibility, the random number generator is initialized with a fixed seed. The simulation generates a long time series of 100,000 periods, with the first 299 periods discarded as a burn-in phase.

The simulation starts from a deterministic state: zero initial debt and the median level of the endowment grid. For each subsequent period, the simulation proceeds as follows:

1. The new endowment level is determined by drawing from the discretized Markov

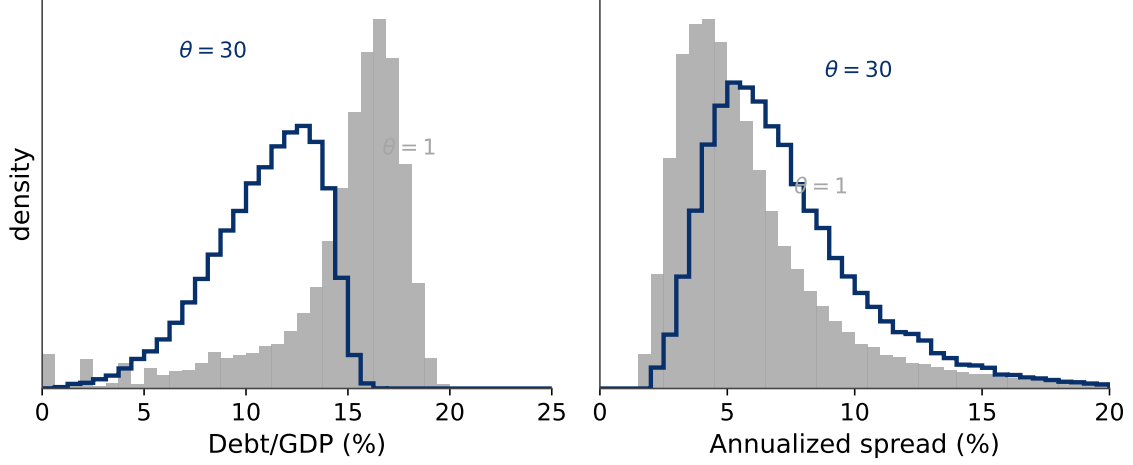


Figure 8: **Stationary distributions: rational ($\theta = 1$, shaded) vs. PRO ($\theta = 30$, outline).** Under PRO the debt/GDP distribution shifts left (deleveraging) while the spread distribution does not, so spreads persist on a safer balance sheet.

transition matrix, Π .

2. If the economy was in default in the previous period, a random draw determines if it regains market access (with probability γ). If not, it remains in default.
3. If the economy has market access, a uniform random draw is compared against the converged default probability function, $\Pr\{d = 1|y, B\}$, to determine if a default occurs.
4. If the economy does not default, a second uniform random draw is used to select the next period's debt level, B' , from the converged borrowing probability distribution, $\Pr\{B'|y, B\}$.
5. Key economic variables (consumption, trade balance, GDP, credit spreads) for the current period are calculated based on the state and choices, and then stored.

The business cycle moments reported in Table 3 are calculated from this simulated data series. The code consistently uses the standard numerical stability technique of subtracting the maximum value within any exp calculation in the log-sum-exp formulas to prevent floating-point overflow.

D.4 Numerical Stability Techniques

The log-sum-exp formulation requires care to prevent floating-point overflow and underflow [Mihalache, 2024]. This subsection details the techniques used in evaluating the choice probabilities and expected values of the main text.

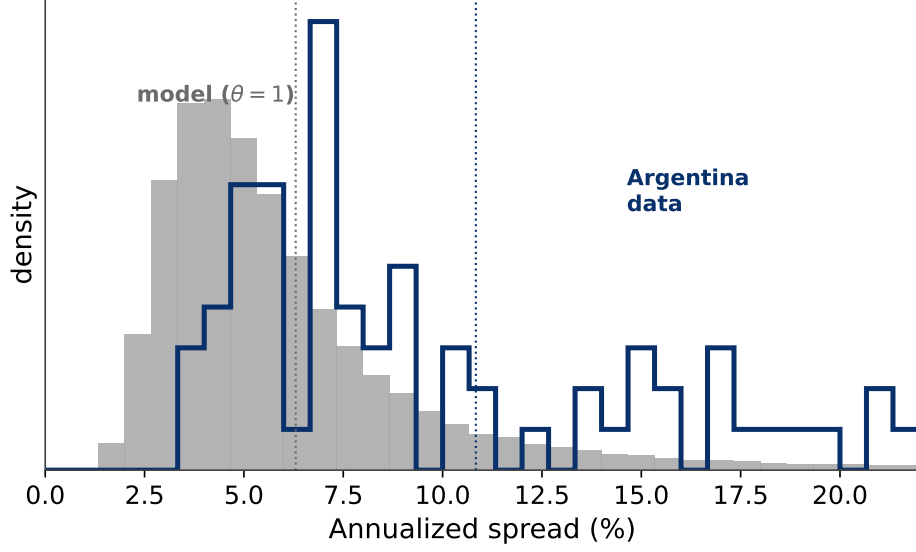


Figure 9: **Model vs data.** The simulated spread distribution of the rational model ($\theta = 1$) against Argentina's realized EMBI distribution; the model captures the location but under-shoots the level and the fat right tail (the modest-mean-spread feature, standard for risk-neutral default models).

D.4.1 The Log-Sum-Exp Numerical Stability Problem

The naive implementation of the log-sum-exp function

$$\text{LSE}(x_1, \dots, x_n) = \ln \left(\sum_{i=1}^n \exp(x_i) \right) \quad (\text{D.77})$$

is prone to numerical instability when the values x_i are large in magnitude. For large positive values, $\exp(x_i)$ may overflow, returning infinity. For large negative values, $\exp(x_i)$ may underflow to zero, causing a loss of precision in the sum. In my sovereign default model, these issues arise when the choice-specific values $W(y, B, B')$ span a wide range due to varying consumption levels across borrowing choices, when the value functions $V^R(y, B)$ and $V^D(y)$ differ substantially near the default boundary, and when the small scale parameters $\rho \approx 10^{-5}$ and $\eta \approx 10^{-4}$ amplify the ratios W/ρ or V/η to extreme magnitudes.

D.4.2 Stabilized Log-Sum-Exp Implementation

To address these issues, I employ the standard “max subtraction” technique following [Mihalache \[2024\]](#). For any set of values $\{x_1, \dots, x_n\}$, I first compute their maximum:

$$x_{\max} = \max\{x_1, \dots, x_n\} \quad (\text{D.78})$$

Then I rewrite the log-sum-exp as:

$$\text{LSE}(x_1, \dots, x_n) = x_{\max} + \ln \left(\sum_{i=1}^n \exp(x_i - x_{\max}) \right) \quad (\text{D.79})$$

This transformation ensures that all terms satisfy $(x_i - x_{\max}) \leq 0$, yielding $\exp(x_i - x_{\max}) \in (0, 1]$, where the largest exponential term equals exactly 1 to prevent overflow, while smaller terms decay exponentially but remain numerically representable to preserve precision.

D.4.3 Application to Borrowing Choice Probabilities

In my computation of borrowing choice probabilities corresponding to equation (6), I use the stabilized implementation:

$$\overline{W}(y, B) = \max_{B'} W(y, B, B') \quad (\text{D.80})$$

$$\Pr(B' = B_i | y, B) = \frac{\exp \frac{W(y, B, B_i) - \overline{W}(y, B)}{\rho}}{\sum_{j \in \mathcal{B}} \exp \frac{W(y, B, B_j) - \overline{W}(y, B)}{\rho}} \quad (\text{D.81})$$

Similarly, the expected value of repayment corresponding to equation (10) becomes:

$$V^R(y, B) = \overline{W}(y, B) + \rho \ln \left(\sum_{j \in \mathcal{B}} \exp \frac{W(y, B, B_j) - \overline{W}(y, B)}{\rho} \right) \quad (\text{D.82})$$

D.4.4 Application to Default Choice Probabilities

For the default decision corresponding to equation (5), I apply the same stabilization technique:

$$\overline{V}(y, B) = \max\{V^D(y), V^R(y, B)\} \quad (\text{D.83})$$

$$\Pr(d = 1 | y, B) = \frac{\exp \frac{V^D(y) - \overline{V}(y, B)}{\eta}}{\exp \frac{V^D(y) - \overline{V}(y, B)}{\eta} + \exp \frac{V^R(y, B) - \overline{V}(y, B)}{\eta}} \quad (\text{D.84})$$

The corresponding expected value function is:

$$V(y, B) = \overline{V}(y, B) + \eta \ln \left[\exp \frac{V^D(y) - \overline{V}(y, B)}{\eta} + \exp \frac{V^R(y, B) - \overline{V}(y, B)}{\eta} \right] \quad (\text{D.85})$$

D.4.5 PRO and Numerical Stability

The behavioral extension introducing PRO as in equation (12) requires computing the lender's perceived default probability as the m -integrated logit at the *true* taste scale

η , with PRO entering *only* through the perceived dispersion $\theta\sigma_m$ of the policy shock. I evaluate the expectation by N_m -point Gauss–Hermite quadrature (nodes ζ_n , weights ω_n ; $m_n = \sqrt{2}\theta\sigma_m\zeta_n$) and apply the same log-sum-exp stabilization as in the default choice probabilities, separately at each node:

$$\tilde{V}(y, B, m_n) = \max\{V^D(y), V^R(y, B, m_n)\}, \quad (\text{D.86})$$

$$\tilde{P}(y, B) = \sum_{n=1}^{N_m} \frac{\omega_n}{\sqrt{\pi}} \frac{\exp \frac{V^D(y) - \tilde{V}(y, B, m_n)}{\eta}}{\exp \frac{V^D(y) - \tilde{V}(y, B, m_n)}{\eta} + \exp \frac{V^R(y, B, m_n) - \tilde{V}(y, B, m_n)}{\eta}} \quad (\text{D.87})$$

where $V^R(y, B, m_n)$ is the repayment value with the budget shifted by the node m_n , and $\tilde{V}(y, B, m_n)$ serves as the per-node stabilizing maximum. An additional safeguard prevents division by zero when both exponential terms become numerically zero at a node, setting that node’s contribution to 0.5. At $\theta = 1$ the quadrature integrates over the true dispersion σ_m , recovering the rational benchmark.

D.4.6 Additional Numerical Safeguards

Beyond the log-sum-exp stabilization, my implementation includes several other numerical safeguards. For consumption bounds, I prevent evaluation of the utility function at non-positive consumption levels by setting $W(y, B, B') = -10^6$ whenever the implied consumption is non-positive, which effectively removes infeasible choices from consideration. I set the convergence tolerances to $\epsilon_V = \epsilon_q = 10^{-6}$, balancing computational accuracy with reasonable iteration counts. All my computations use double precision arithmetic to minimize accumulated rounding errors.

These numerical stability techniques, following the approach of [Mihalache \[2024\]](#), are essential for ensuring convergence of my value function iteration algorithm, particularly in the presence of the small taste shock parameters that characterize the discrete choice approach.

E Notation

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Table 4: Notation

Symbol	Meaning
<i>Preferences and technology</i>	
β, σ	discount factor; CRRA risk aversion
y, B, B'	endowment; current and next-period debt
δ, κ	principal decay rate; coupon
γ	re-entry probability after default
$m \sim N(0, \sigma_m^2)$	i.i.d. transitory policy shock
η, ρ	Gumbel taste scales (default; borrowing)
<i>The PRO friction</i>	
$\theta \geq 1$	PRO coefficient (perceived-dispersion inflation)
$\theta \sigma_m$	lenders' perceived policy-shock dispersion
$\theta_{RI}, \widehat{\sigma}_m^2$	RI-implied θ ; perceived variance
<i>Equilibrium objects</i>	
q_θ, s_θ	bond price and spread schedules
V, V^R, V^D	value; repayment value; default value
$\tilde{p}_\theta, m^*$	perceived default prob.; indifference shock
$B^*(y)$	price-pivot threshold
Φ, ϕ	standard normal CDF and density

Note: A few symbols are reused in secondary material and disambiguated by context: in the rational-inattention microfoundation (Section 6), β also denotes the fundamentals parameter vector and Ψ the attention-cost function; the operator-analysis appendix (Appendix B.12) writes the one-period discount $1/(1+r)$ explicitly to avoid reusing β ; the transparency cost in Appendix C reuses ϕ . The Euler–Mascheroni constant in the Gumbel footnote and derivation is written $\gamma_E \approx 0.5772$ to keep γ for the re-entry probability.

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